

Azure Administrator (AZ-104) | Lab Guide

Version 22.9

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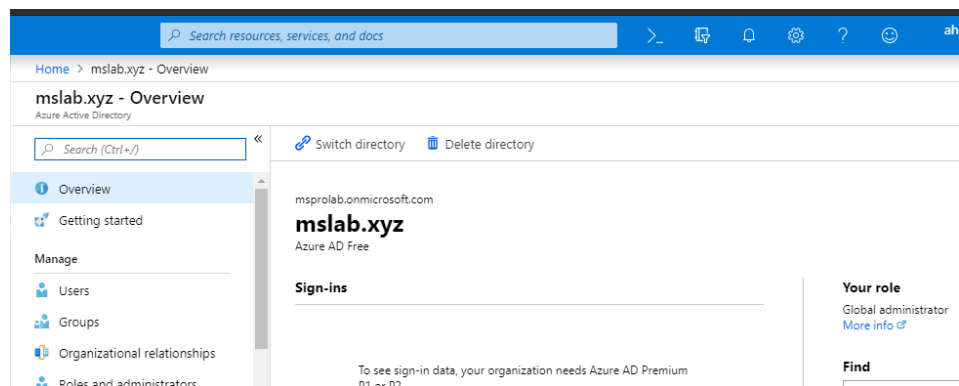
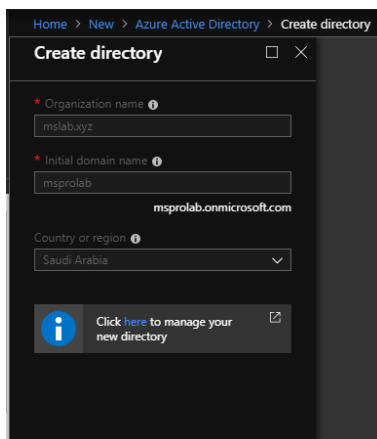
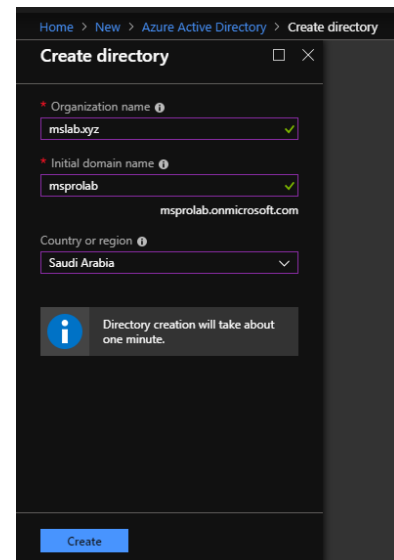
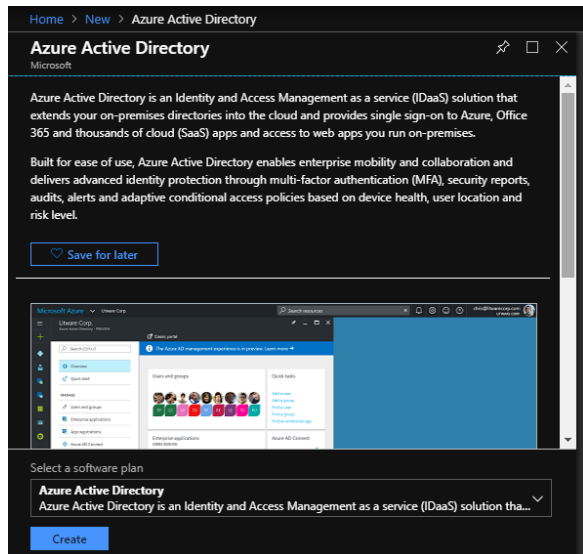
Identity in the Cloud

Important Note: As of now, everything related to Azure AD has been updated to Microsoft Entra ID. The configurations and functionalities remain the same.

AAD is a cloud-based directory and identity management service that provides application access management and identity protection. It's often referred to as **IaaS**.

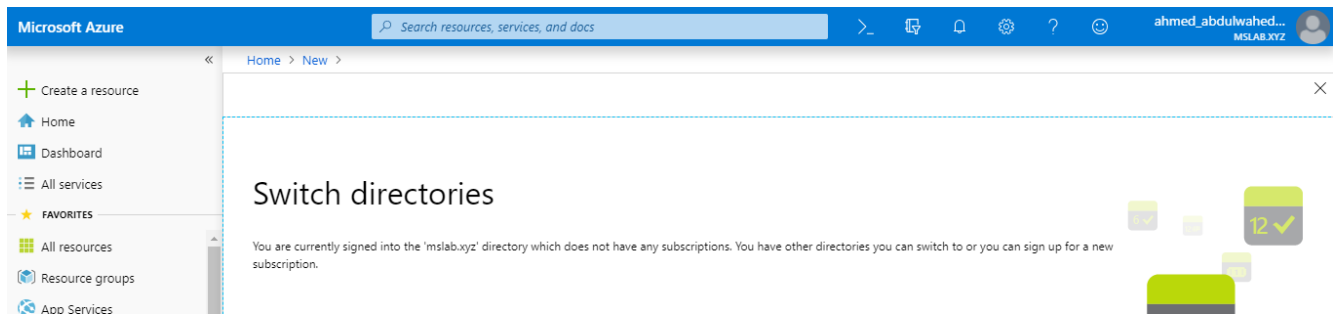
- **Azure Active Directory Free:** Basic identity and access management features for users and groups.
- **Azure Active Directory Basic:** Enhanced user productivity with application access and self-service capabilities.
- **Azure Active Directory Premium P1:** Advanced identity protection, self-service, and access management features.
- **Azure Active Directory Premium P2:** Comprehensive identity protection, advanced monitoring, and security governance.

Creating a new directory (Tenant)

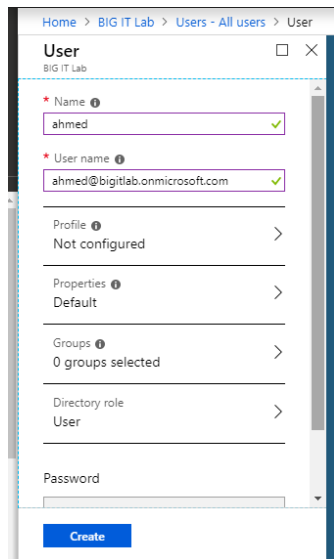
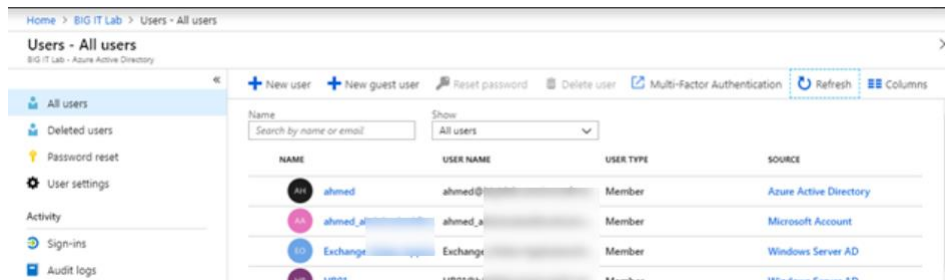


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Now you can switch to the new tenant

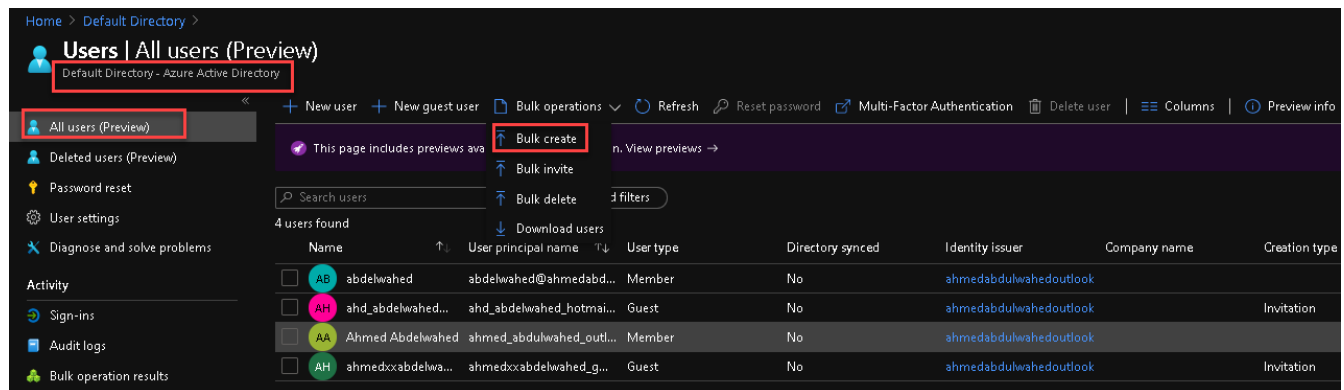


Creating a new user



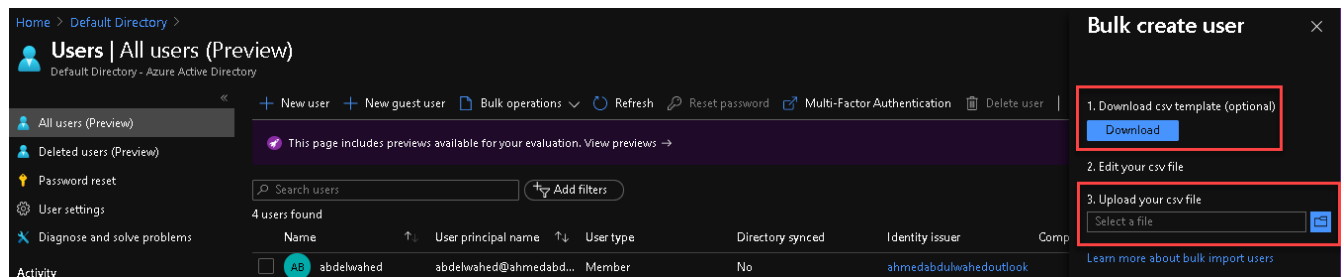
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Bulk User Creation



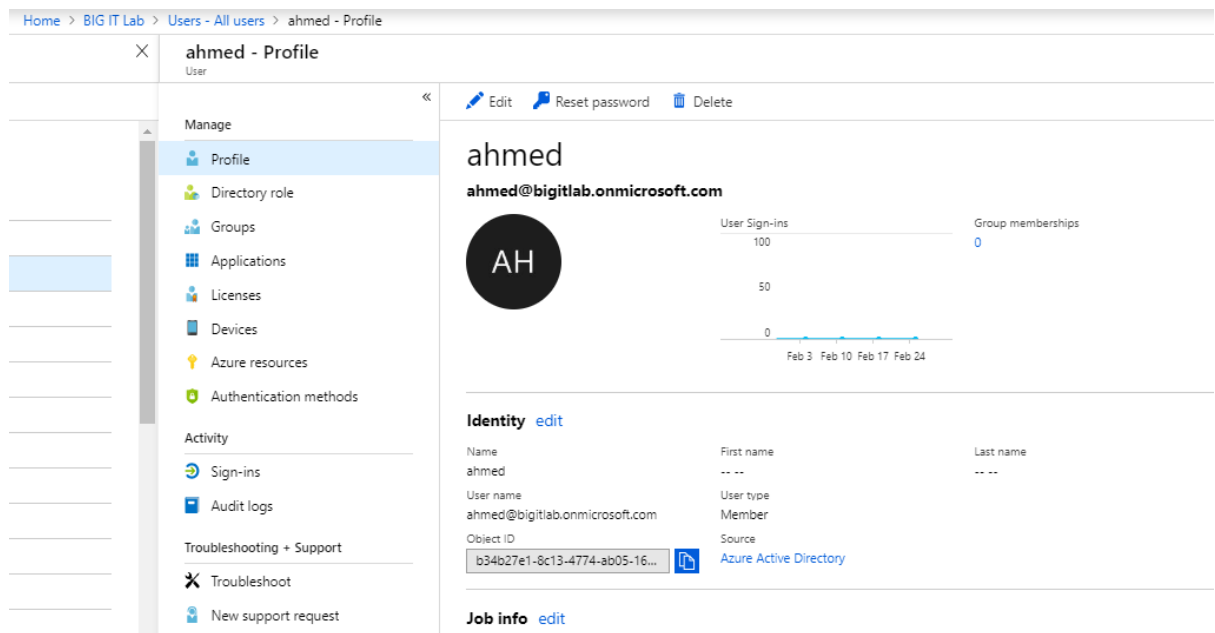
The screenshot shows the 'Users | All users (Preview)' page in the Azure Active Directory portal. The 'Bulk create' button is highlighted with a red box. Below the button, there is a table of users with columns: Name, User principal name, User type, Directory synced, Identity issuer, Company name, and Creation type. The table contains 4 users found.

Name	User principal name	User type	Directory synced	Identity issuer	Company name	Creation type
AB abdelwahed	abdelwahed@ahmedabd...	Member	No	ahmedabdulwahedoutlook		
AH ahd_abdelwahed...	ahd_abdelwahed_hotmai...	Guest	No	ahmedabdulwahedoutlook		Invitation
AA Ahmed Abdelwahed	ahmed_abdulwahed_outl...	Member	No	ahmedabdulwahedoutlook		
AH ahmedxxabdelwa...	ahmedxxabdelwahed_g...	Guest	No	ahmedabdulwahedoutlook		Invitation



The screenshot shows the 'Bulk create user' dialog box. It contains three steps: 1. Download csv template (optional) with a 'Download' button. 2. Edit your csv file. 3. Upload your csv file with a 'Select a file' button and a file upload icon. The 'Download' button is highlighted with a red box.

Managing user options and permissions



The screenshot shows the 'ahmed - Profile' page in the Azure Active Directory portal. The page displays user information, including the user's name, email address, and a profile picture. It also shows a 'User Sign-ins' graph and a 'Group memberships' section. The 'Profile' tab is selected in the left-hand navigation pane.

ahmed
ahmed@bigitlab.onmicrosoft.com

User Sign-ins
100
50
0
Feb 3 Feb 10 Feb 17 Feb 24

Group memberships
0

Identity [edit](#)

Name	First name	Last name
ahmed	---	---

User name	User type	Source
ahmed@bigitlab.onmicrosoft.com	Member	Azure Active Directory

Job info [edit](#)

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Home > BIG IT Lab > Users - All users > ahmed - Groups

ahmed - Groups

User

Manage

Profile

Directory role

Groups

Applications

Licenses

Devices

Azure resources

Authentication methods

Activity

<<

+ Add

Refresh

NAME	GROUP TYPE
Not a member of any groups	

Select Group

Select

Search by name or email address

DN

DnsAdmins

DN

DnsUpdateProxy

EI

Exchange Install Domain Servers

ES

Exchange Servers

Selected group:

No group selected

Home > BIG IT Lab > Users - All users > ahmed - Directory role

ahmed - Directory role

User

Manage

Profile

Directory role

Groups

Applications

Licenses

Devices

Azure resources

Authentication methods

Activity

Sign-ins

Audit logs

Troubleshooting + Support

Troubleshoot

New support request

<<

+ Add role

Remove

ROLE
No directory roles assigned

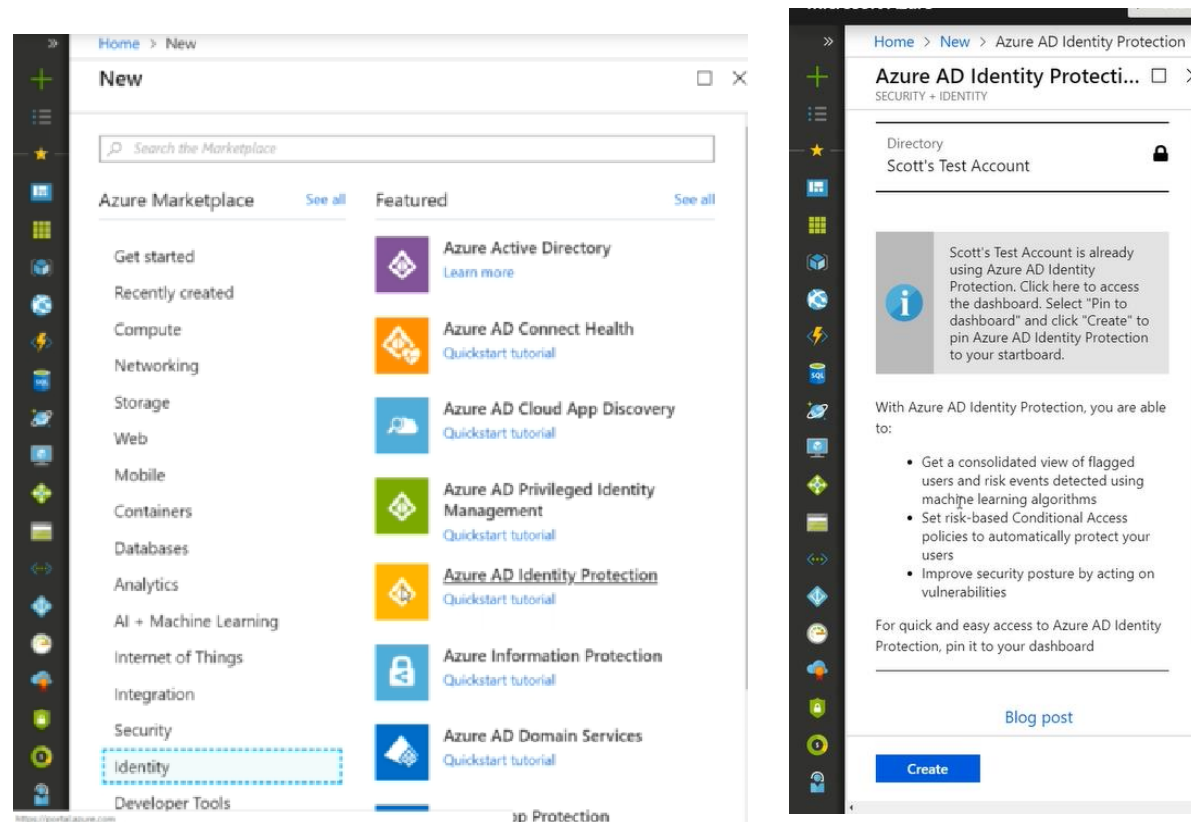
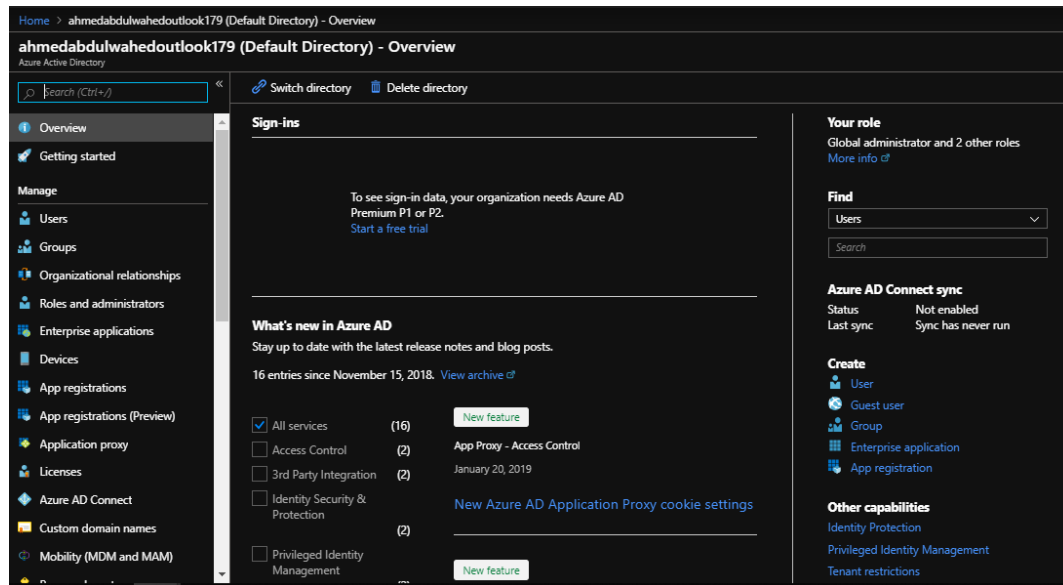
Directory roles

ROLE	DESCRIPTION
Application administrator	Can create and manage all aspects of app registrations and enterprise apps.
Application developer	Can create application registrations independent of the 'Users can register applications' sett...
Authentication administrator	Can access to view, set and reset authentication method information for any non-admin user.
Billing administrator	Can perform common billing related tasks like updating payment information.
Cloud application administrator	Can create and manage all aspects of app registrations and enterprise apps except App Pro...
Cloud device administrator	Full access to manage devices in Azure AD.
Compliance administrator	Can read and manage compliance configuration and reports in Azure AD and Office 365.
Conditional Access administrator	Can manage conditional access capabilities.
Customer LockBox access approver	Can approve Microsoft support requests to access customer organizational data.
Desktop Analytics administrator	Can access and manage Desktop management tools and services.
Dynamics 365 administrator	Can manage all aspects of the Dynamics 365 product.
Exchange administrator	Can manage all aspects of the Exchange product.
Global administrator	Can manage all aspects of Azure AD and Microsoft services that use Azure AD identities.
Guest inviter	Can invite guest users independent of the 'members can invite guests' setting.
Information Protection administrator	Can manage all aspects of the Azure Information Protection product.

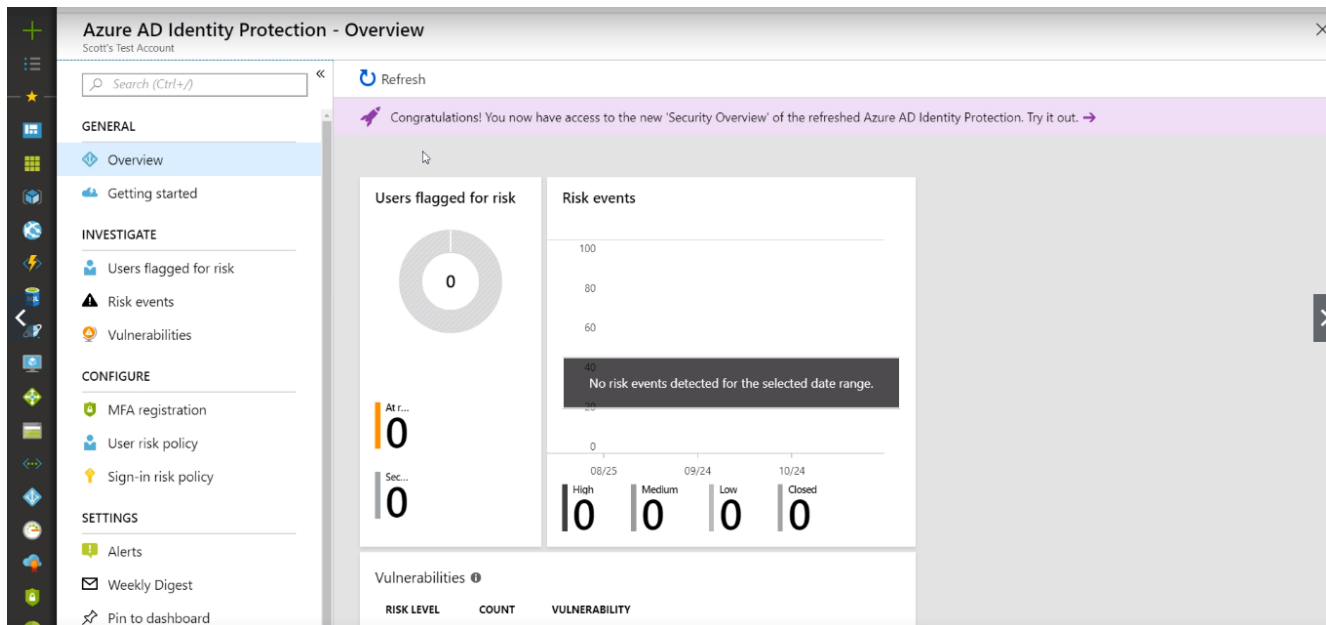
Select

www.abdelwahed.me

Upgrade Azure AD to Premium P2



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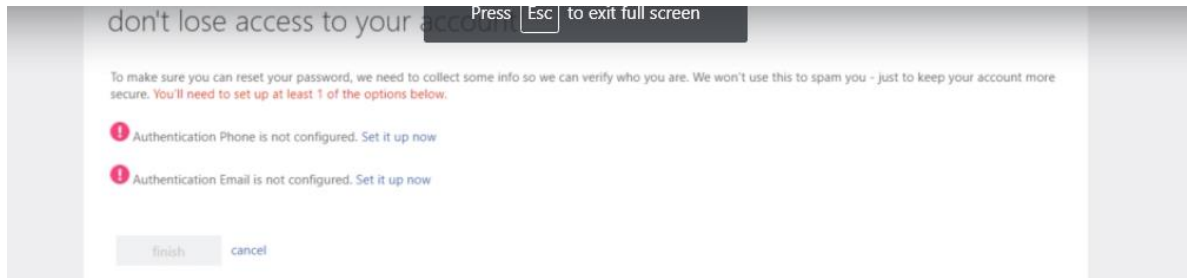


The screenshot shows the 'Password reset - Authentication methods' configuration page. The left sidebar contains a navigation menu with sections: Manage (Properties, Authentication methods, Registration, Notifications, Customization, On-premises integration), Activity (Audit logs), and Troubleshooting + Support (Troubleshoot). The main content area displays a 'Save' button and a 'Discard' button. A tooltip states: 'Defines the number of alternate methods of identification a user in this directory must have to reset their password'. Below this, there is a 'Number of methods required to reset' field with a value of 2. A section titled 'Methods available to users' contains a list of checkboxes: Mobile app notification (preview), Mobile app code (preview), Email (checked), Mobile phone (checked), Office phone, and Security questions.

This screenshot is identical to the one above, showing the 'Password reset - Authentication methods' configuration page. It displays the 'Save' and 'Discard' buttons, the tooltip about the number of alternate methods required to reset a password, the 'Number of methods required to reset' field set to 2, and the 'Methods available to users' section with checkboxes for Mobile app notification (preview), Mobile app code (preview), Email (checked), Mobile phone (checked), Office phone, and Security questions.

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First time you try to access with that user , you asked to setup MFA

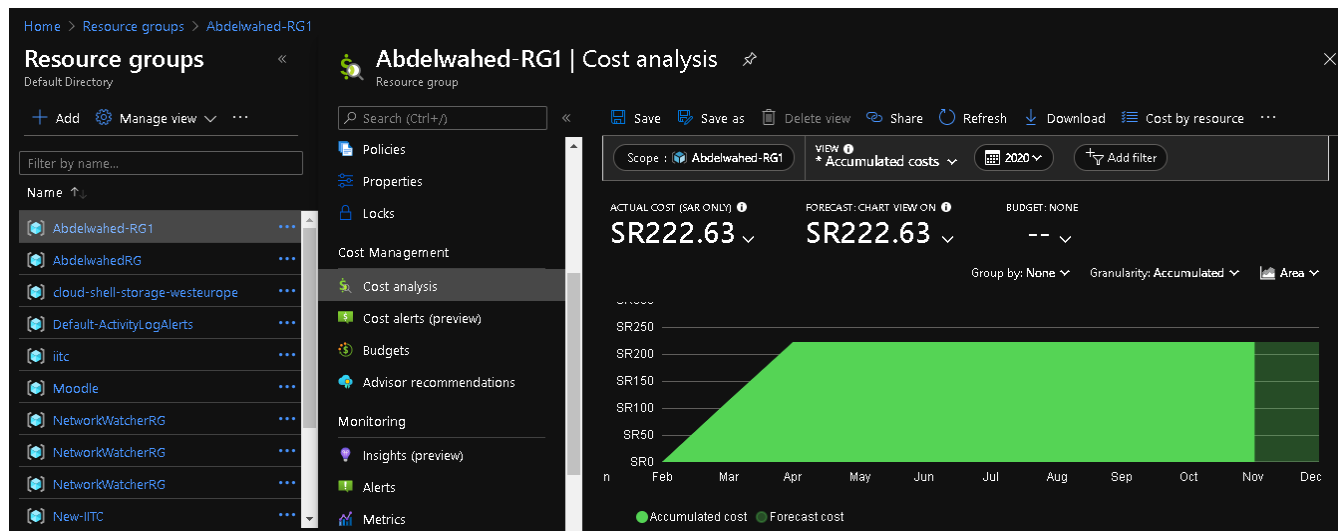


Governance and Compliance

Azure subscription types

- Sponsored subscriptions
- Pay as you go
- Enterprise subscriptions
- Azure for Students subscription (100\$ per month for 12 months)

Cost Management



Resource Tags (Azure resources to logically organize)

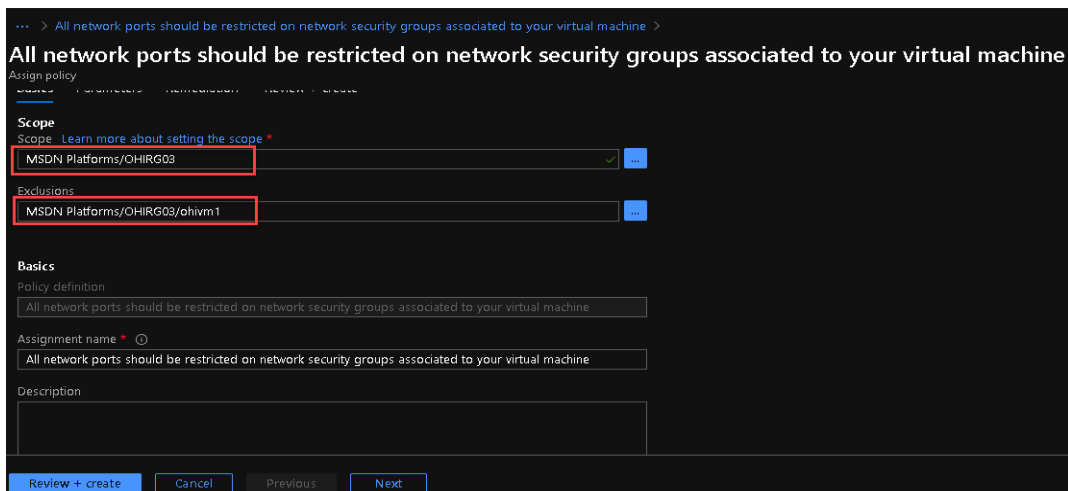
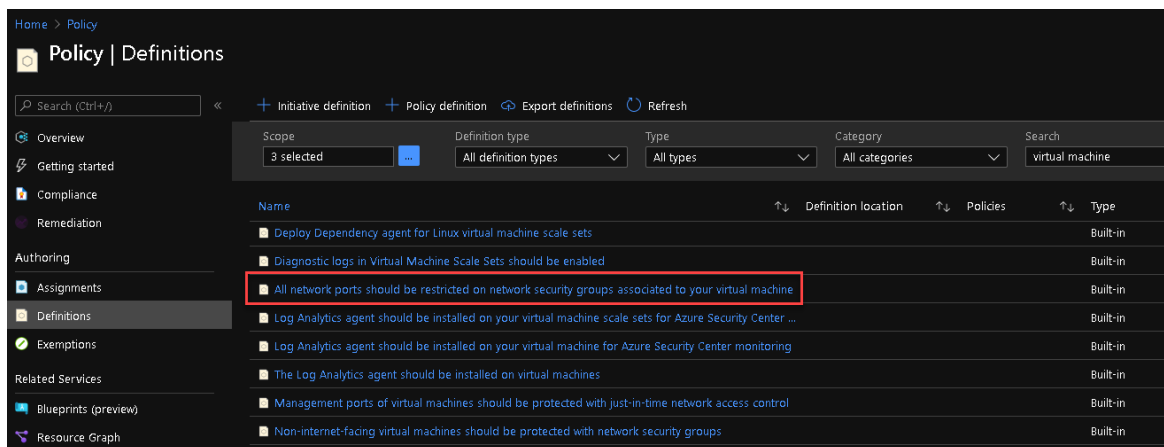
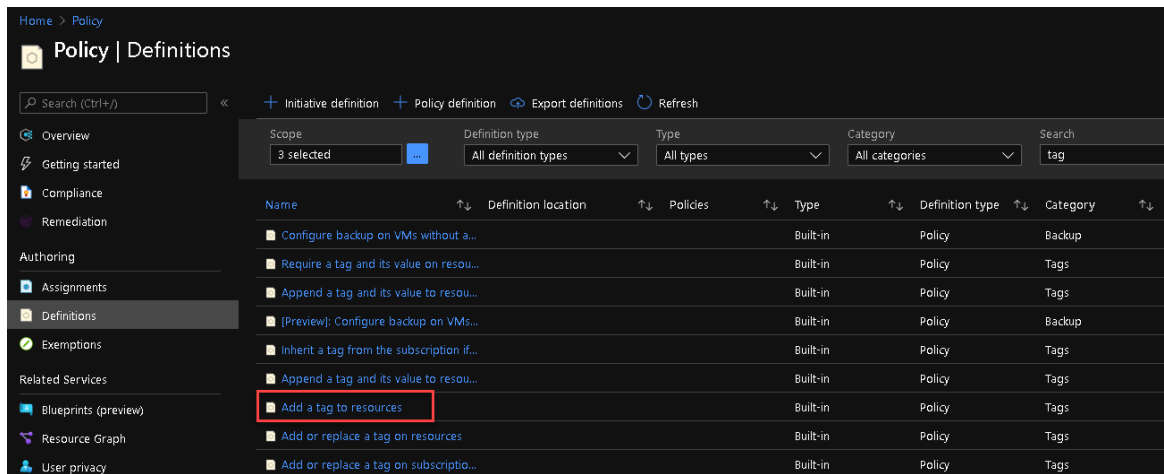
The screenshot shows the Azure Tags interface. The left sidebar lists tags, including 'applicationType : web', 'Environment : Test', and 'ms-resource-usage : azure-cloud-shell'. The main panel displays the 'Resources with tag Environment : Test' view. It shows a list of resources with columns for Name, Type, Resource group, Location, and Subscription. The resources are filtered by 'Subscription == all' and 'Type == all'. The table shows 6 records, including 'abdelwahed', 'AbdelwahedCert', 'AbdelwahedCertVault', 'AbdelwahedHub (abdelwahednoti/Abdelwahed...', 'abdelwahednoti', and 'ServicePlan10a1fc46-a829'.

Name	Type	Resource group	Location	Subscription
abdelwahed	App Service	RG	Central US	MSDN Platforms
AbdelwahedCert	App Service Certificate	RG	Global	MSDN Platforms
AbdelwahedCertVault	Key vault	RG	Central US	MSDN Platforms
AbdelwahedHub (abdelwahednoti/Abdelwahed...	Notification Hub	RG	Central US	MSDN Platforms
abdelwahednoti	Notification Hub Namesp...	RG	Central US	MSDN Platforms
ServicePlan10a1fc46-a829	App Service plan	RG	Central US	MSDN Platforms

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Azure Policy

add your company governance through azure policies so you can detect any non-compliant resource easily.



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Now you can see policy stat from resource portal

Home > Virtual machines > ohivm1

Virtual machines
Default Directory

+ Add Reservations

Try the new virtual machine resource browser! This experience is faster and has improved sorting and filtering capabilities. Please note that the new experience will not show classic virtual machines and does not include support for some columns such as maintenance status.

Filter by name...

Name ↑↓

ohivm1

ohivm1 | Policies
Virtual machine

Search (Ctrl+J)

Assign policy Assign initiative Refresh

Scope: MSDN Platform Type: All definition types Compliance state: All compliance st... Search: Filter by name or ID...

Overall resource compliance 0%
0 out of 1

Resources by compliance state

- 0 - Compliant
- 0 - Exempt
- 1 - Non-compliant
- 0 - Conflicting

Non-compliant initiatives 1 out of 1

Non-compliant policies 16 out of 140

Name	Scope	Compliance state	Resource compl...	Non-Compliant	No
ASC Default (subs...	MSDN Platforms	Non-compliant	0% (0 out of 1)	1	16

Also, from policy overview you can see summary for your all policies.

Home > **Policy**

Search (Ctrl+J)

Overview

Getting started

Compliance

Remediation

Authoring

- Assignments
- Definitions
- Exemptions

Related Services

- Blueprints (preview)
- Resource Graph
- User privacy

Announcing Policy Exemption! Learn more <https://aka.ms/AzPolicyExemption>

Overall resource compliance 0%
0 out of 14

Resources by compliance state

- 0 - Compliant
- 0 - Exempt
- 14 - Non-compliant
- 0 - Conflicting

Non-compliant initiatives 3 out of 3

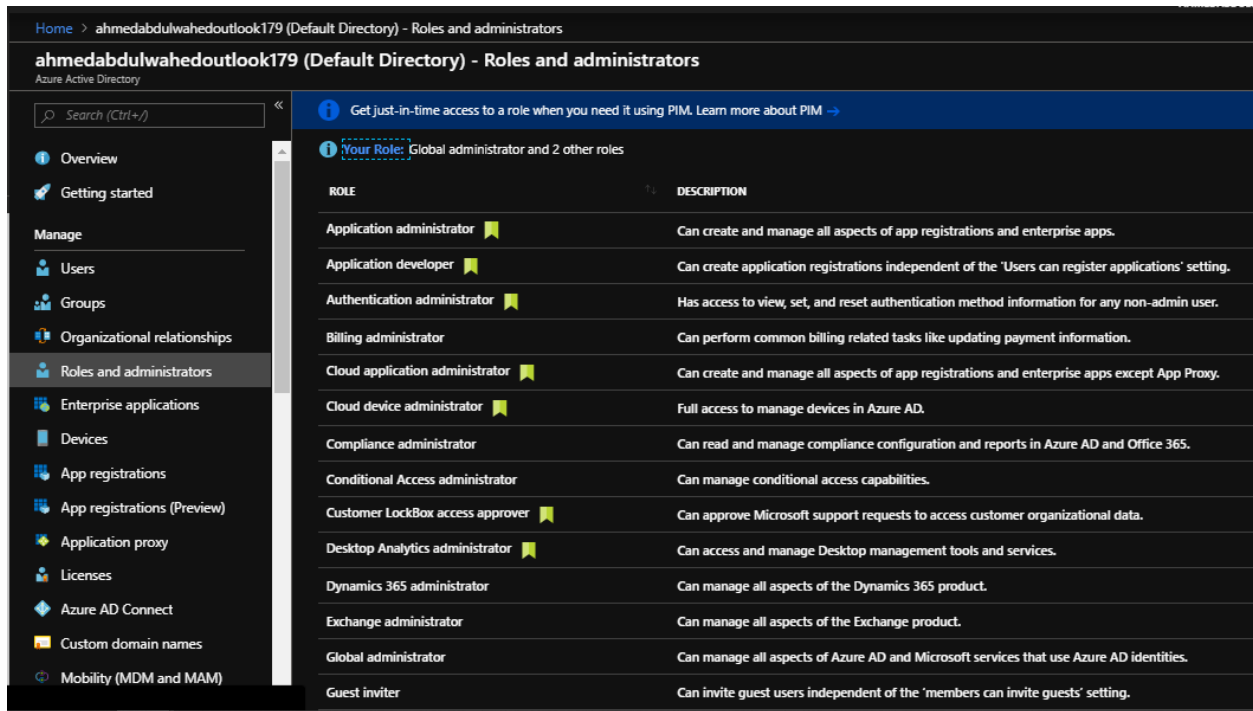
Non-compliant policies 61 out of 421

LEARN MORE
[Learn about Policy](#)
[Onboarding tutorial](#)

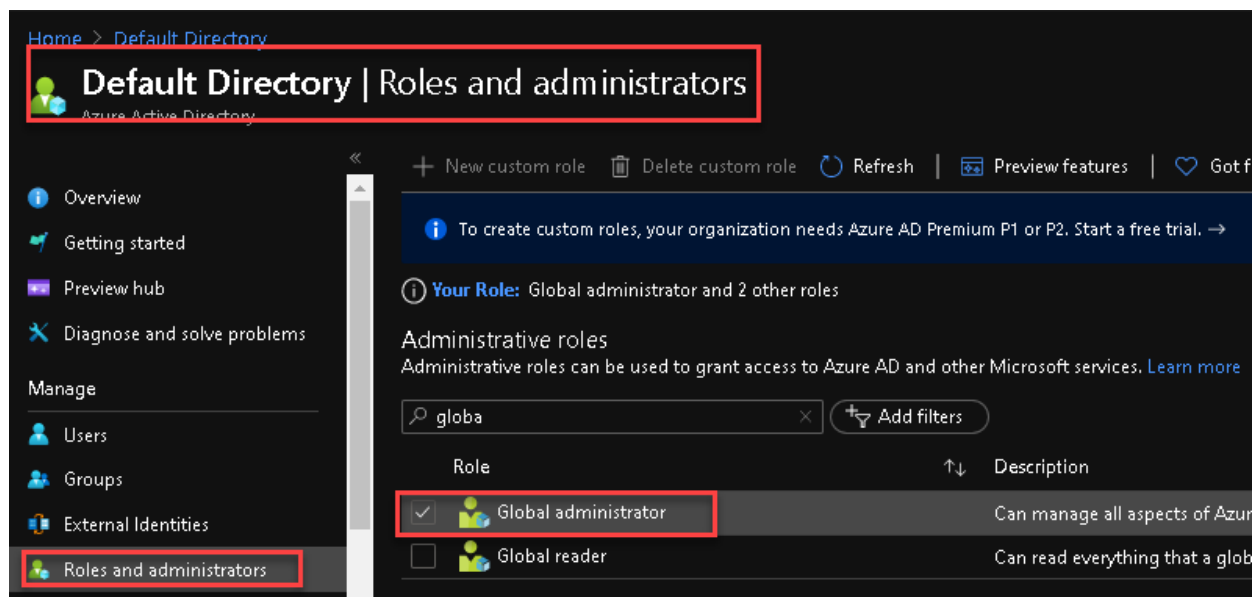
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Role-based access control (RBAC)

is a system that provides fine-grained access management of Azure resources. Using Azure RBAC, you can segregate duties within your team and grant only the amount of access to users that they need to perform their jobs.

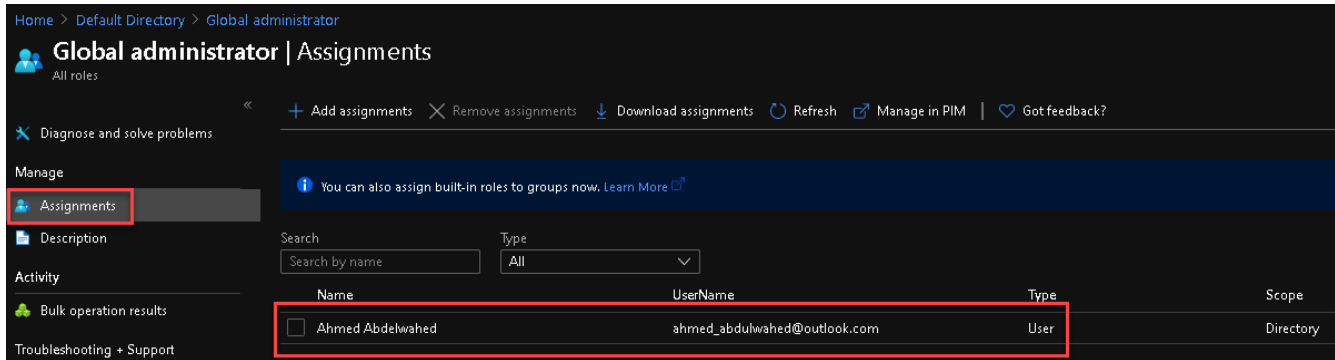


All members of Global admins



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through assignments tab you can see which user has this role.



Assign RBAC using Azure PowerShell and CLI

When you have large numbers of role assignments, you may prefer to use Azure PowerShell or the CLI.

```
#Role assignment properties
$roleName = "Contributor"
$assigneeName = ahmed@abdelwahed.me
$resourceGroupName = "DBRG"
```

Azure PowerShell

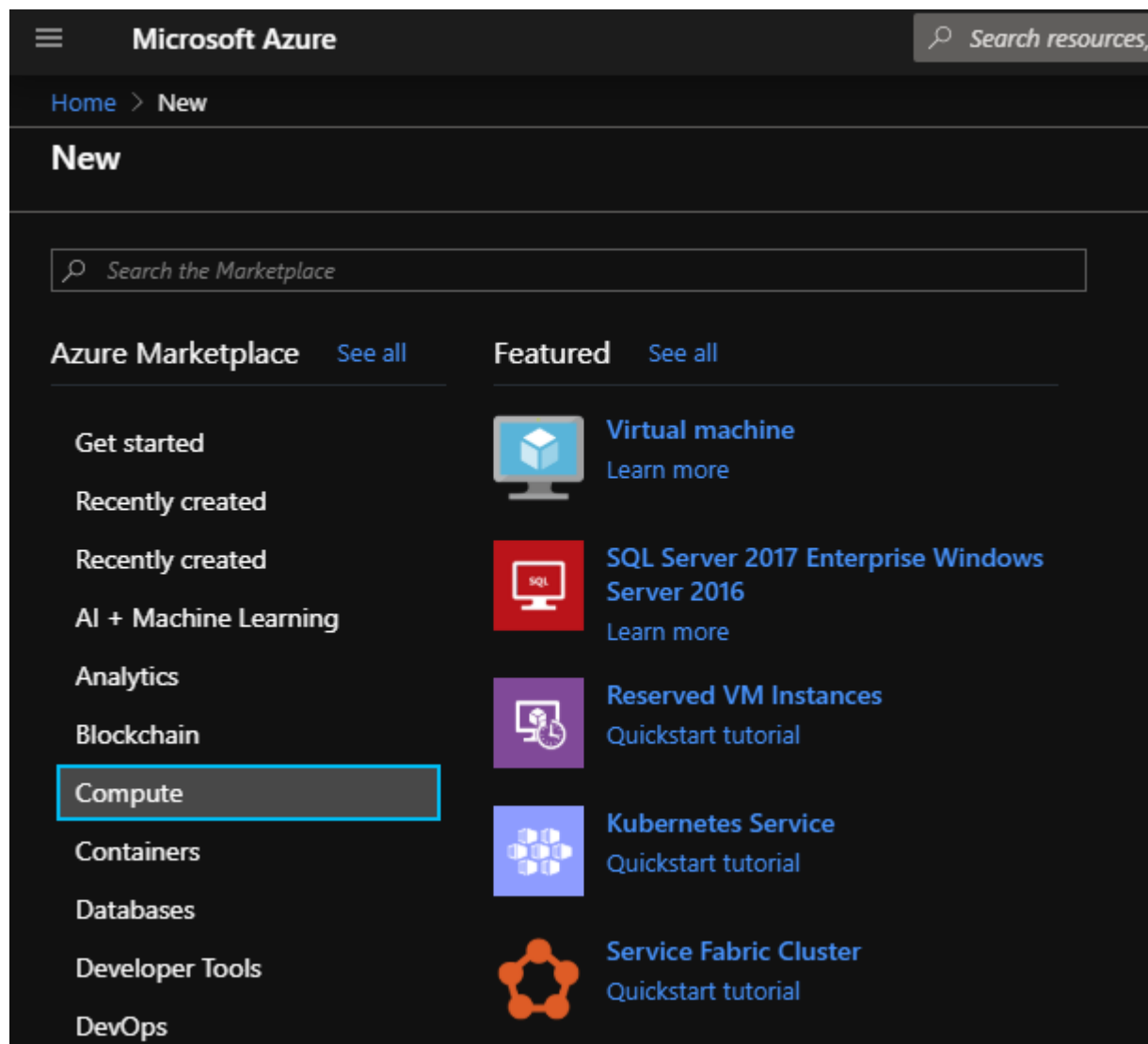
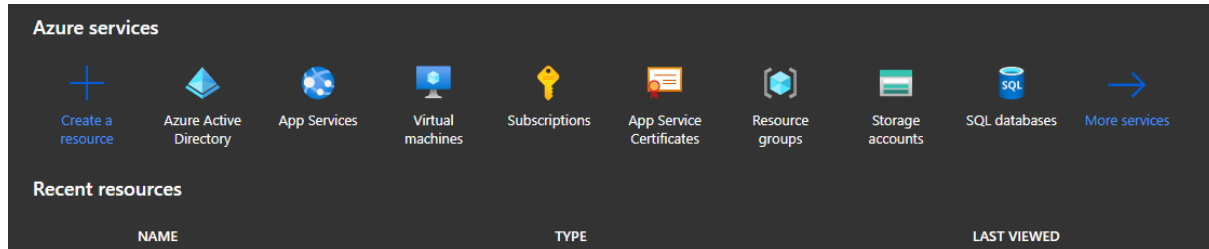
```
New-AzureRmRoleAssignment -RoleDefinitionName $roleName -SignInName $assigneeName -
ResourceGroupName $resourceGroupName
```

CLI

```
az role assignment create --role $roleName --assignee $assigneeName --resource-group $resourceGroupName
```

Azure Virtual Machines

Creating Virtual Machines in the Portal



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in first page fill the requested info starting with resource group, VM name, Image source, user name and password to access and VM size

The screenshot shows the 'Create a virtual machine' page in the Microsoft Azure portal. The left sidebar contains 'Instance details' and 'Administrator account' sections. The main area is titled 'Select an image' and shows the 'Marketplace' tab. A search bar at the top of the marketplace lists categories: Analytics, Blockchain, Compute, Containers, Databases, Developer Tools, DevOps, Identity, Integration, Internet of Things, and IT & Management Tools. The 'AI + Machine Learning' category is selected and highlighted. Below the search bar, several images are listed, including 'OpenVPN Access Server 2.7.5', 'Windows Server 2019 Datacenter with Containers', 'Windows Server 2019 Datacenter Server Core with Containers', and 'Windows Server 2019 Datacenter Server Core'.

The screenshot shows the 'Select a VM size' section of the 'Create a virtual machine' page. It displays a table of available VM sizes with filters for Size, Generation, Family, and Premium disk. The table shows 11 of 166 VM sizes. The selected size is B1ms.

VM Si...	Offering	Family	vCP...	RAM (...)	Data disks	Max IOPS	Temporary stor...	Premium disk s...	Cost/month (es...
B1ls	Standard	General purpose	1	0.5	2	200	1	Yes	SAR 17.41
B1ms	Standard	General purpose	1	2	2	640	4	Yes	SAR 69.75
B1s	Standard	General purpose	1	1	2	3200	2	Yes	SAR 34.88
B2ms	Standard	General purpose	2	8	4	1920	16	Yes	SAR 278.44
B2s	Standard	General purpose	2	4	4	1280	8	Yes	SAR 139.22

Home > New > Create a virtual machine

Create a virtual machine

Virtual machine name *	Abdelwahed-VM01	✓
Region *	(US) Central US	▼
Availability options	No infrastructure redundancy required	▼
Image *	Windows Server 2019 Datacenter	▼
	Browse all public and private images	
Size *	Standard B1ms 1 vcpu, 2 GiB memory Change size	
Administrator account		
Username *	aabdelwahed	✓
Password *	*****	✓
Confirm password *	*****	✓

Home > New > Create a virtual machine

Create a virtual machine

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ☐ None ☒ Allow selected ports

Select inbound ports * HTTP (80), HTTPS (443), SSH (22), RDP (3389) ▼

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Save money

Save up to 49% with a license you already own using Azure Hybrid Benefit. [Learn more](#)

Already have a Windows Server license? * ☐ Yes ☒ No

ⓘ

[Review + create](#) [< Previous](#) [Next : Disks >](#)

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Next Page its related for Disks options, you can add data disk from here or you can add it later

Create a virtual machine

Disk options

OS disk type * ⓘ

Premium SSD^

Enable Ultra Disk compatibility ⓘ

Standard HDD

Standard SSD

Premium SSD

Data disks

You can add and configure additional data disks for your virtual machine or attach existing disks. This VM also comes with a temporary disk.

LUN	Name	Size (GiB)	Disk type	Host caching
Create and attach a new disk Attach an existing disk				

Advanced

Use managed disks ⓘ

☐ No ☒ Yes

Use ephemeral OS disk ⓘ

☒ No ☐ Yes

Next page for network options

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ

(new) AZ-103-vnet^

Create new

Subnet * ⓘ

(new) default (10.0.1.0/24)^

Public IP ⓘ

(new) Abdelwahed-VM01-ip^

Create new

NIC network security group ⓘ

☐ None ☒ Basic ☐ Advanced

Public inbound ports * ⓘ

☐ None ☒ Allow selected ports

Select inbound ports *

HTTP (80), HTTPS (443), SSH (22), RDP (3389)^

⚠ This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Accelerated networking ⓘ ☐ On ☒ Off

The selected VM size does not support accelerated networking.

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Place this virtual machine behind an existing load balancing solution? ☐ Yes ☒ No

Next page is to add some management options like automatic shutdown machine

Home > New > Create a virtual machine

Create a virtual machine

Basics Disks Networking Management Advanced Tags Review + create

Configure monitoring and management options for your VM.

Azure Security Center

Azure Security Center provides unified security management and advanced threat protection across hybrid cloud workloads. [Learn more](#)

✔ Your subscription is protected by Azure Security Center basic plan.

Monitoring

Boot diagnostics ⓘ ☒ On ☐ Off

OS guest diagnostics ⓘ ☐ On ☒ Off

Diagnostics storage account * ⓘ [Create new](#)

Identity

System assigned managed identity ⓘ ☐ On ☒ Off

Identity

System assigned managed identity [ⓘ] ☐ On ☒ Off

Auto-shutdown

Enable auto-shutdown [ⓘ] ☒ On ☐ Off

Shutdown time [ⓘ]

Time zone [ⓘ] [▼]

Notification before shutdown [ⓘ] ☒ On ☐ Off

Email * [ⓘ] [✓]

Backup

Enable backup [ⓘ] ☐ On ☒ Off

Now you can go direct to review your configuration and start deployment

Home > New > Create a virtual machine

Create a virtual machine

✓ Validation passed

Basics

Subscription	MSDN Platforms
Resource group	(new) AZ-103
Virtual machine name	Abdelwahed-VM01
Region	(US) Central US
Availability options	No infrastructure redundancy required
Username	aabelwahed
Public inbound ports	RDP, HTTP, HTTPS, SSH
Already have a Windows Server license?	No

Disks

OS disk type	Premium SSD
Use managed disks	Yes
Use ephemeral OS disk	No

Networking

Virtual network	(new) AZ-103-vnet
Subnet	(new) default (10.0.1.0/24)

Create < Previous Next > Download a template for automation

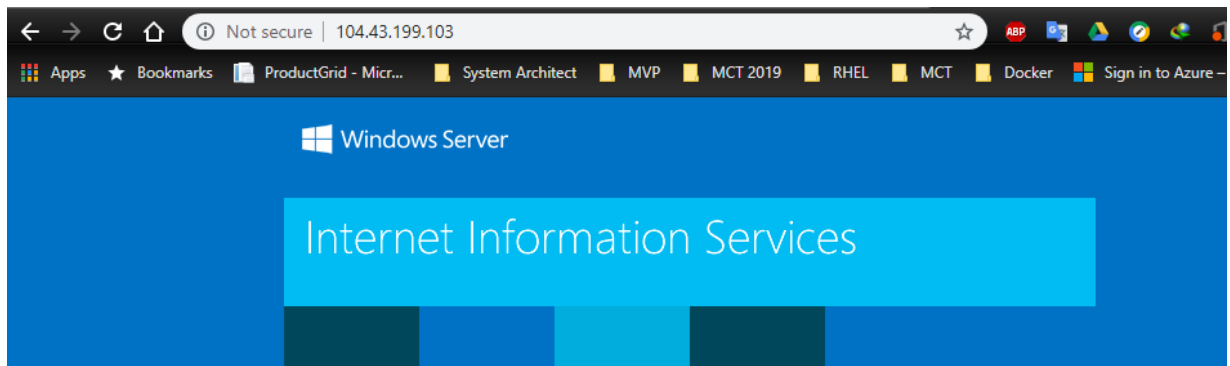
The screenshot displays the Azure portal interface. The top section, titled "CreateVm-MicrosoftWindowsServer.WindowsServer-201-20191123150855 - Overview", shows the deployment status as "Your deployment is underway". It lists the deployment name, subscription, resource group, start time, and correlation ID. Below this, a table provides details for each resource in the deployment.

Resource	Type	Status	Operation details
✓ abdelwahed-vm01442	Microsoft.Network/netw...	Created	Operation details
✓ Abdelwahed-VM01-ip	Microsoft.Network/publ...	OK	Operation details
✓ AZ-103-vnet	Microsoft.Network/virtu...	OK	Operation details
✓ Abdelwahed-VM01-nsg	Microsoft.Network/netw...	OK	Operation details
⌚ az103diag591	Microsoft.Storage/stora...	Accepted	Operation details

The bottom section, titled "Virtual machines", shows the "Abdelwahed-VM01" virtual machine. It lists various settings such as resource group, status, location, subscription, and operating system. The operating system is "Windows (Windows Server 2019 Datacenter)".

now you can connect to it by RDP and install different services you need, in the following example will add web server role

```
Install-WindowsFeature -name Web-Server -IncludeManagementTools
```



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Creating a Virtual Machine with PowerShell

First connect to your account using `Connect-AzAccount` and select specific subscription

```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\WINDOWS\system32> Connect-AzAccount

Account                               SubscriptionName TenantId                               Environment
-----
ahmed_abdulwahed@outlook.com MSDN Platforms                               AzureCloud

PS C:\WINDOWS\system32> Get-AzResourceGroup

ResourceGroupName : AbdelwahedRG
Location           : centralus
ProvisioningState   : Succeeded
Tags               :
ResourceId          : /subscriptions/ /resourceGroups/AbdelwahedRG

PS C:\WINDOWS\system32> Get-AzSubscription

Name            Id                               TenantId                               State
-----
MSDN Platforms 61                               0                               Enabled
MSDN Platforms a1                               0                               Enabled
MSDN Platforms b1                               0                               Enabled

PS C:\WINDOWS\system32> Connect-AzAccount -Subscription b15

Account                               SubscriptionName TenantId                               Environment
-----
ahmed_ah MSDN Platforms 0610                               AzureCloud

PS C:\WINDOWS\system32> Connect-AzAccount -Tenant
```

```
$cred = Get-Credential
$vm = New-AzVMConfig -VMName az103vm -VMSize Standard_D1
$vm = Set-AzVMOperatingSystem `
>> -VM $vm `
>> -Windows `
>> -ComputerName az103vm `
>> -Credential $cred `
>> -ProvisionVMAgent -EnableAutoUpdate
$vm = Set-AzVMSourceImage `
>> -VM $vm `
>> -PublisherName MicrosoftWindowsServer `
>> -Offer WindowsServer `
>> -Skus 2016-Datacenter `
>> -Version latest

New-AzVm `
>> -ResourceGroupName "az103RG" `
>> -Name "az103vm" `
>> -Location "East US" `
>> -VirtualNetworkName "az103vnet" `
>> -SubnetName "az103subnet" `
>> -SecurityGroupName "az103SG" `
>> -PublicIpAddressName "az103PublicAddress" `
```

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>> -OpenPorts 80,3389

Finally, you will ask to provide login credential

```
Administrator: Windows PowerShell
PS C:\WINDOWS\system32> $um = New-AzVMConfig -VMName az103vm -UMSize Standard_D1

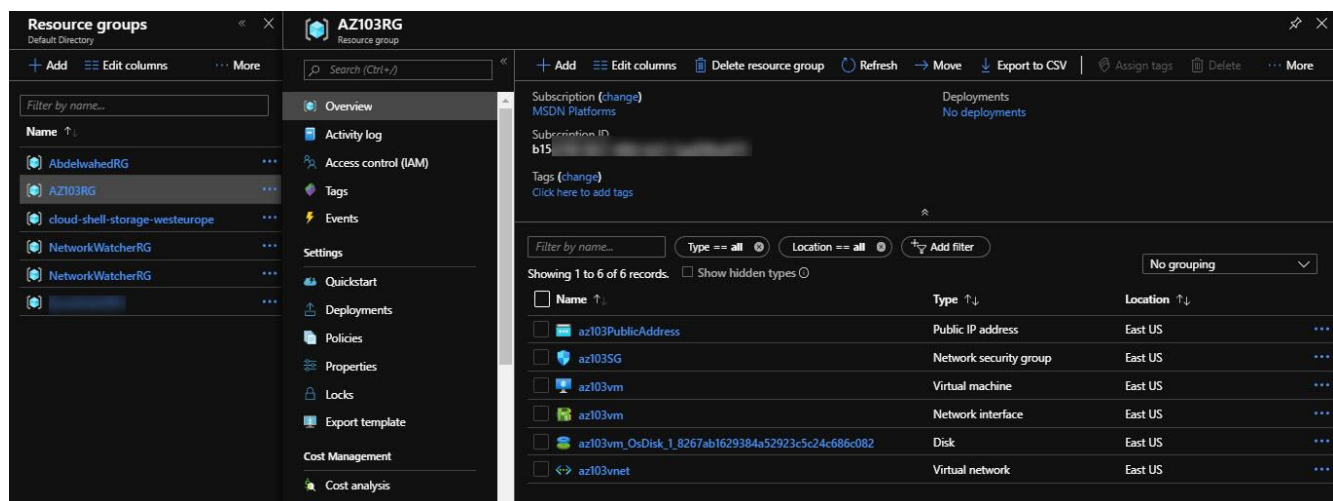
Creating Azure resources
26%
1
Creating virtualMachines/az103vm.

>> -UM $um
>> -PublisherName MicrosoftWindowsServer
>> -Offer WindowsServer
>> -Skus 2019-Datacenter
>> -Version latest
PS C:\WINDOWS\system32> New-AzVM -ResourceGroupName "az103RG" -Name "az103vm" -Location "East US" -VirtualNetworkName "az103vnet" -SubnetName "az103subnet" -SecurityGroupName "az103SG" -PublicIpAddressName "az103PublicAddress" -OpenPorts 80,3389

cmdlet New-AzVM at command pipeline position 1
Supply values for the following parameters:
Credential

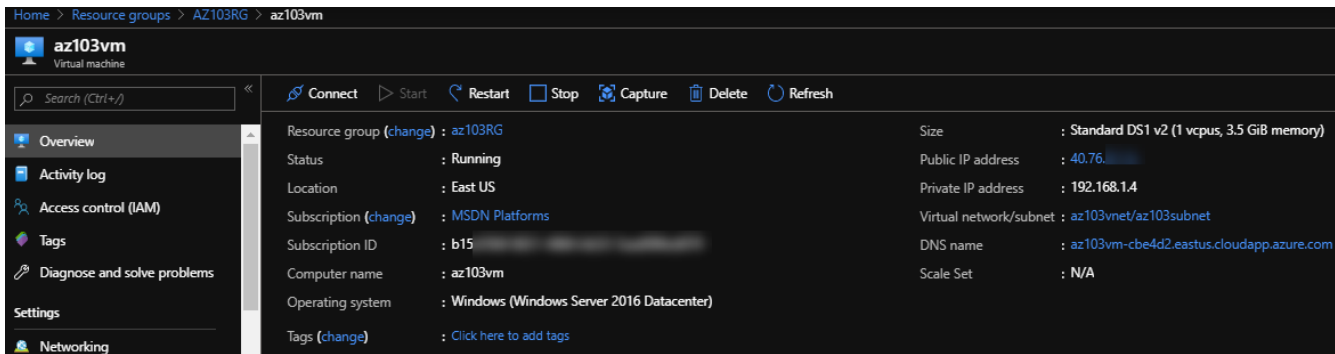
cmdlet New-AzVM at command pipeline position 1
Supply values for the following parameters:
Credential

ResourceGroupName      : az103RG
Id                    : /subscriptions/b15d766f-8021-4866-bb33-5aad096ed079/resourceGroups/az103RG/providers/Microsoft.Compute/virtualMachines/az103vm
VmId                  : 552f8344-5793-492a-9a8b-5363ae90bd98
Name                  : az103vm
Type                  : Microsoft.Compute/virtualMachines
Location              : eastus
Tags                  : {}
HardwareProfile        : {UmSize}
NetworkProfile         : {NetworkInterfaces}
OSProfile              : {ComputerName, AdminUsername, WindowsConfiguration, Secrets, AllowExtensionOperations}
ProvisioningState      : Succeeded
StorageProfile         : {ImageReference, OsDisk, DataDisks}
FullyQualifiedDomainName : az103vm-685fe9.East US.cloudapp.azure.com
```



Azure Administrator (AZ-104) | Lab Guide

try to connect vm using `mstsc /v:PublicIP`



The screenshot shows the Azure portal interface for a virtual machine named 'az103vm'. The left sidebar contains navigation options: Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Settings, and Networking. The main area displays the VM's status and configuration details.

Property	Value
Resource group (change)	az103RG
Status	Running
Location	East US
Subscription (change)	MSDN Platforms
Subscription ID	b15
Computer name	az103vm
Operating system	Windows (Windows Server 2016 Datacenter)
Tags (change)	Click here to add tags
Size	Standard DS1 v2 (1 vcpu, 3.5 GiB memory)
Public IP address	40.76
Private IP address	192.168.1.4
Virtual network/subnet	az103vnet/az103subnet
DNS name	az103vm-cbe4d2.eastus.cloudapp.azure.com
Scale Set	N/A

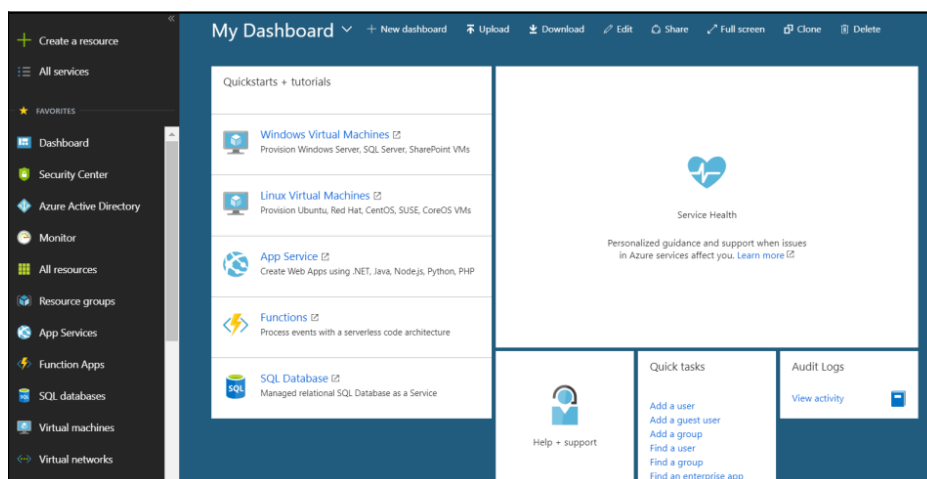
to clean up your resources use `Remove-AzResourceGroup -Name AZ103RG`

Azure Administration

Azure Resource Manager

Azure Resource Manager enables you to work with the resources in your solution as a group. You can deploy, update, or delete all the resources for your solution in a single, coordinated operation. You use a template for deployment and that template can work for different environments such as testing, staging, and production. Resource Manager provides security, auditing, and tagging features to help you manage your resources after deployment.

<https://portal.azure.com>



ARM templates, Microsoft stepped up and really changed cloud business. In the cloud and in DevOps, the **Infrastructure as code (IaC)** concept is very important and that was exactly what ARM templates were. You are able to create an ARM template and **reuse it multiple times to create similar environments**. By doing so, you automated your infrastructure deployment steps and removed possible mistakes in the deployment and configuration process.

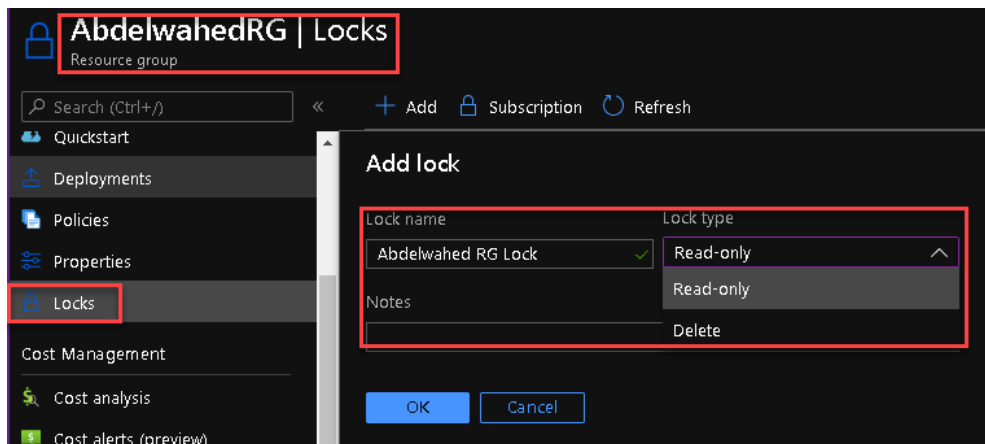
Managing Resources from ARM

Resource Manager Locks

Lock types

There are two types of resource locks.

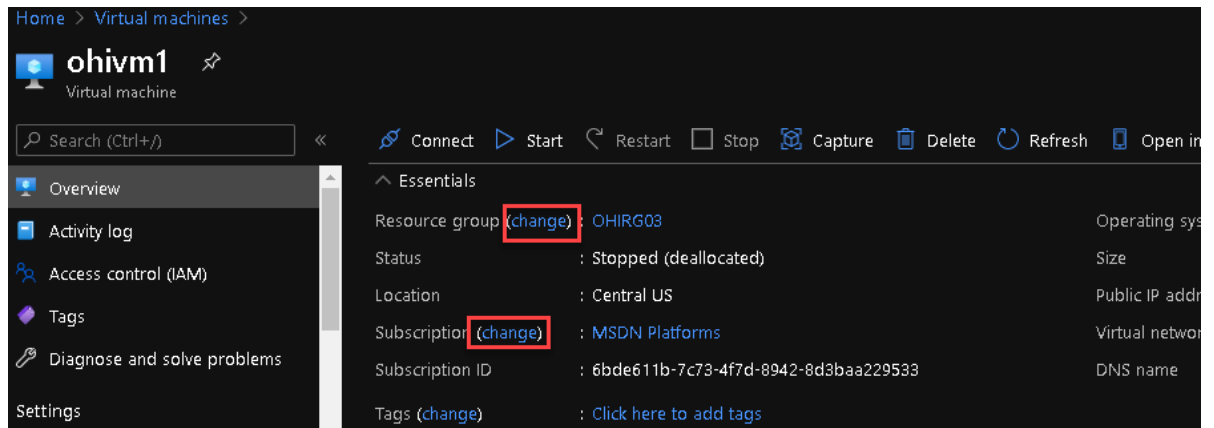
- **Read-Only locks**, which prevent any changes to the resource.
 - **Delete locks**, which prevent deletion.
- ✓ Only the Owner and User Access Administrator roles can create or delete management locks.



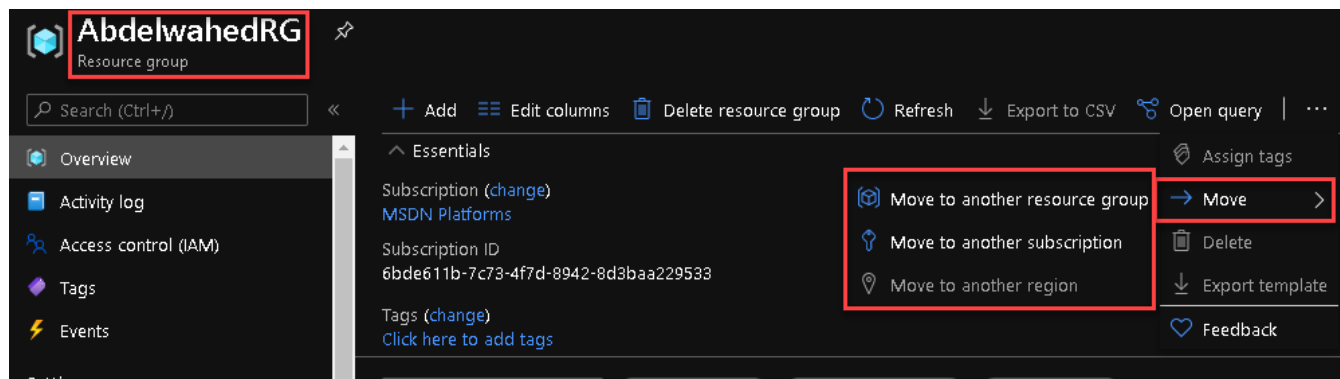
Moving Resources

Sometimes you may need to move resources to either a new subscription or a new resource group in the same subscription.

The following way from resource.



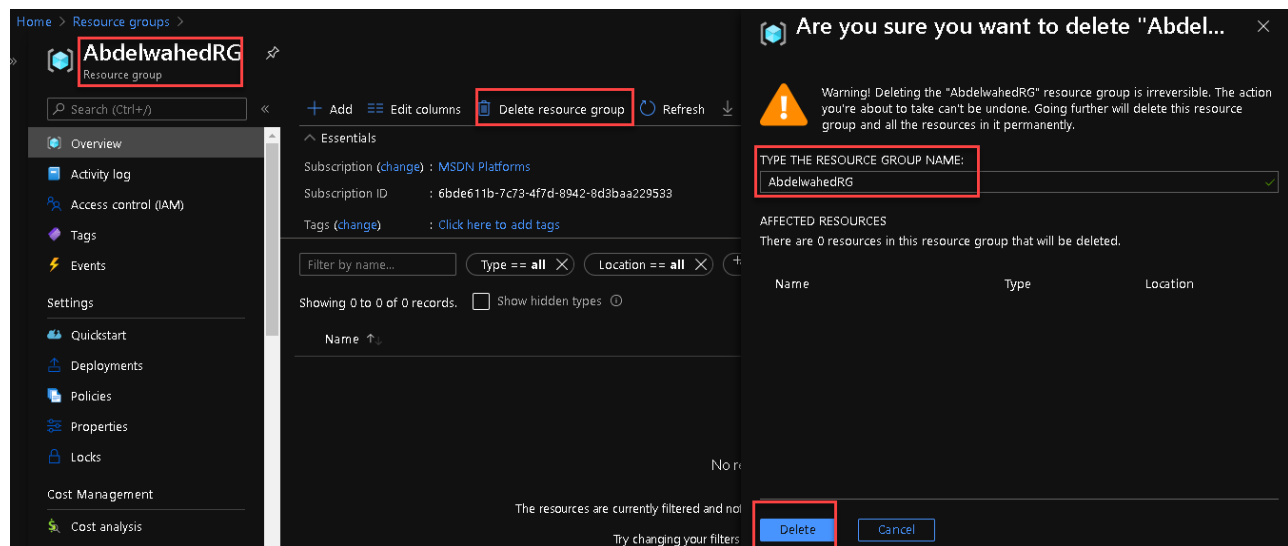
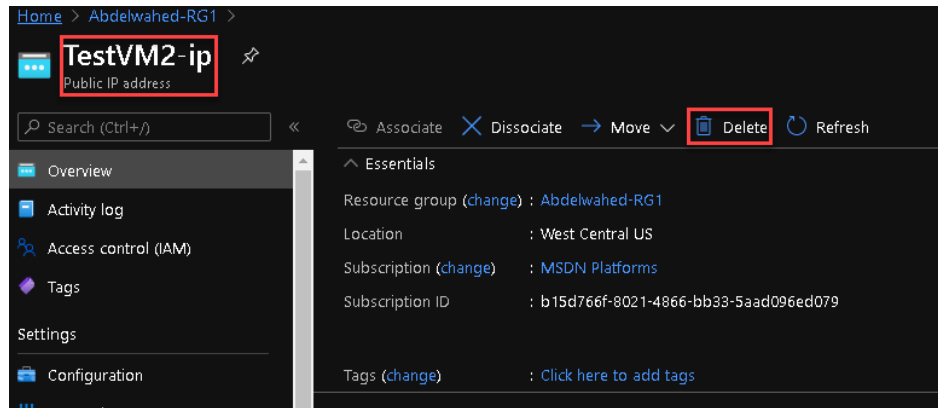
Another way from resource group as shown down.



Azure Administrator (AZ-104) | Lab Guide

Removing Resources and Resource Groups

If you delete RG all resources included will removed also. Or you can delete some resources inside resource group.



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Resource Limits

You can use each Microsoft Azure resource up to its quota. Each subscription has separate quotas and usage is tracked per subscription. If you reach a quota cap, you can request an increase via [Help + Support](#).

MSDN Platforms | Usage + quotas

Subscription

Search (Ctrl+/)

Request Quota Increase Refresh

You can use each Microsoft Azure resource up to its quota. Each subscription has separate quotas and usage is tracked per subscription. If you reach a quota cap, you can request an increase via [Help + Support](#). [Request increase](#)

8 services All providers Central US Show all

Filter items...

Quota	Provider	Location	Usage
Total Regional vCPUs	Microsoft.Compute	Central US	0 % 0 of 20
Virtual Machine Scale Sets	Microsoft.Compute	Central US	0 % 0 of 2500
Premium Storage Managed Disks	Microsoft.Compute	Central US	0 % 0 of 50000
Static Public IP Addresses	Microsoft.Network	Central US	0 % 0 of 10
Public IP Addresses	Microsoft.Network	Central US	0 % 0 of 10
Subnets per Virtual Network	Microsoft.Network	Central US	0 % 0 of 3000
Peering per Virtual Network	Microsoft.Network	Central US	0 % 0 of 500
Storage Accounts	Microsoft.Storage	Central US	0 % 0 of 250

Manage Azure Resources using cloud shell

connect from local PowerShell

Administrator: Windows PowerShell ISE

File Edit View Tools Debug Add-ons Help

Untitled1.ps1 X

```
1 Install-Module -name azuread
2
3 Get-Module azuread
4
5 Connect-AzureAD
```

PS C:\WINDOWS\system32> Install-Module -name azuread

Get-Module azuread

PS C:\WINDOWS\system32> Get-Module azuread

PS C:\WINDOWS\system32> Connect-AzureAD

Environment: TenantId TenantDomain AccountType

1 Get-AzureADUser

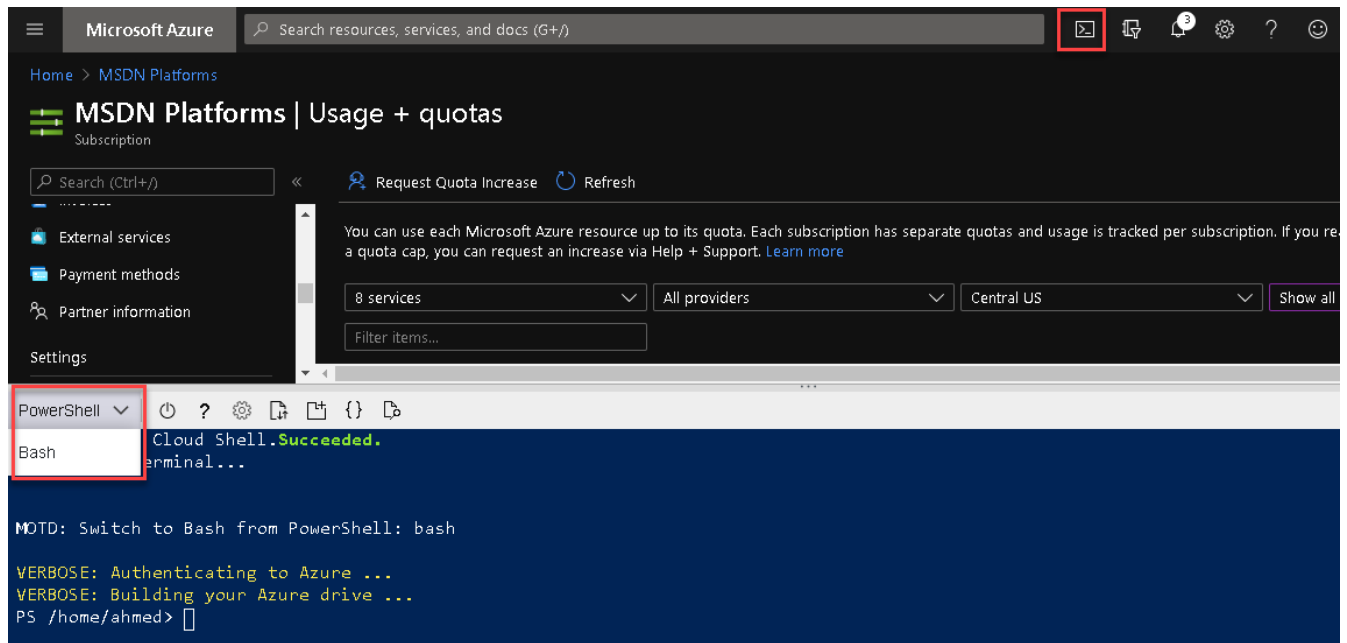
```
PS C:\WINDOWS\system32> Get-AzureADUser

ObjectID          DisplayName          UserPrincipalName
-----
ca351d00-8ca5-4b96-a431-95f40a77ca56 Ahmed Abdelwahed    ahmed@IIITCLab10.onmicrosoft.com
00d9b105-24fb-47f1-989b-74d423f7cd09 Ali Mohamed          ali@IIITCLab10.onmicrosoft.com
584875fc-38a2-4e88-bea3-0c251164d46c HR01                 hr01@IIITCLab10.onmicrosoft.com
498d9341-1f0b-40fa-898f-1a06dc73d8a8 HR02                 hr02@IIITCLab10.onmicrosoft.com
a7243505-6756-43ce-b76a-57d9e89c6054 HR03                 hr03@IIITCLab10.onmicrosoft.com
a7ff0ce1-c34c-4a17-8b3a-ce267777f402 HR04                 hr04@IIITCLab10.onmicrosoft.com
8039effb-4b54-45c1-89b4-c78f4831393f HR05                 hr05@IIITCLab10.onmicrosoft.com
a5e45c1a-57e6-4a63-96df-e6b0854ee2ba IT01                 it01@IIITCLab10.onmicrosoft.com
3d90cb5c-a650-4dd0-8237-a84a95a96939 IT02                 it02@IIITCLab10.onmicrosoft.com
b86f1ba9-f514-49d5-9f4a-f53c87946fb2 IT03                 it03@IIITCLab10.onmicrosoft.com
da051d88-2cba-4628-bc5a-b9195336f20c IT04                 it04@IIITCLab10.onmicrosoft.com
5a09c00d-d16a-4b3f-a017-01e54d6972a5 IT05                 it05@IIITCLab10.onmicrosoft.com
```

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- Get-AzureRmSubscription | gm
- help Get-AzureRmTenant -examples
- Get-AzureRmTenant
- Get-AzureRmNetworkUsage
- Get-Module -ListAvailable
- **Get-azurerms, commands start with azureRm**
- Install-Module az, install azure module
- Login-azureRmAccount

Through cloud shell



- az account list

Azure CLI and Working with Azure CLI Locally

Azure CLI is a command-line program to connect to Azure and execute administrative commands on Azure resources. It runs on Linux, macOS, and Windows, and allows administrators and developers to execute their commands through a terminal or a command-line prompt, (or script!) instead of a web browser. For example, to restart a VM, you would use a command such as the following:

You can download this tool <https://aka.ms/installazurecliwindows> and install it.

- Az login
- Az account list
- az group list
- az vm list
- Az storage list
- Az vm create
- Az group create

Azure Administrator (AZ-104) | Lab Guide

ARM Templates

There are two ways to deploy resource using ARM template

- 1- Download resource template and follow instruction below

Install-Module AzureRM

Install-Module Azure

Download and install Azure Cli

Set-ExecutionPolicy -Scope process -ExecutionPolicy Bypass

cd E:\ExportedTemplate

.\deploy.ps1 -subscriptionId a7632f95-7f0d-4342-adda-dbc286dbe45c -resourceGroupName AZ300RG -deploymentName az300deploy1

```
PS E:\ExportedTemplate-CreateVm-MicrosoftWindowsServer.WindowsServer-200-20190317161336> .\deploy.ps1 -subscriptionId a7632f95-7f0d-4342-adda-dbc286dbe45c -resourceGroupName AZ300RG -deploymentName az300deploy1
Logging in...

Selecting subscription 'a7632f95-7f0d-4342-adda-dbc286dbe45c'
Account          SubscriptionName TenantId          Environment
-----
ahmed_abdulwahed@outlook.com MSDN Platforms 06163712-12c2-4ae9-8e69-73d879a0e896 AzureCloud

Name      : MSDN Platforms (a7632f95-7f0d-4342-adda-dbc286dbe45c) - ahmed_abdulwahed@outlook.com
Account   : ahmed_abdulwahed@outlook.com
Environment : AzureCloud
Subscription : a7632f95-7f0d-4342-adda-dbc286dbe45c
Tenant    : 06163712-12c2-4ae9-8e69-73d879a0e896
TokenCache : Microsoft.Azure.Commands.Common.Authentication.ProtectedFileTokenCache
VersionProfile :
ExtendedProperties : {}

Registering resource providers
Registering resource provider 'microsoft.network'

ProviderNamespace : Microsoft.Network
RegistrationState  : Registered
ResourceTypes     : {virtualNetworks, natGateways, publicIPAddresses, networkInterfaces...}
Locations         : {West US, East US, North Europe, West Europe...}
ZoneMappings      :

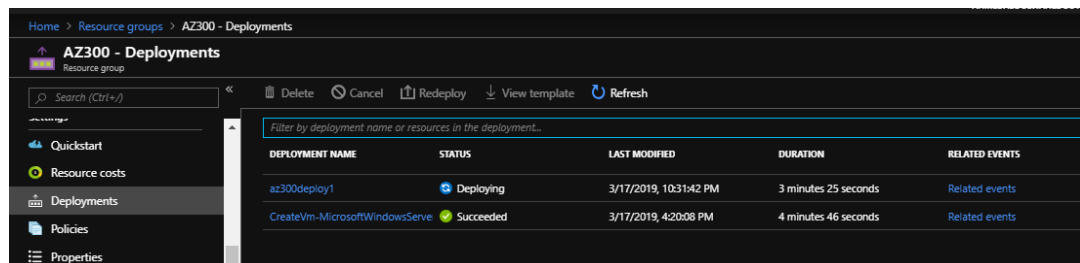
Registering resource provider 'microsoft.compute'

ProviderNamespace : Microsoft.Compute
RegistrationState  : Registered
ResourceTypes     : {availabilitySets, virtualMachines, virtualMachines/extensions, virtualMachineScaleSets...}
Locations         : {East US, East US 2, West US, Central US...}
ZoneMappings      :

Registering resource provider 'microsoft.devtestlab'

ProviderNamespace : Microsoft.DevTestLab
RegistrationState  : Registered
ResourceTypes     : {labs, schedules, labs/virtualMachines, labs/serviceRunners...}
Locations         : {West Central US, Japan East, West US, Australia Central...}
ZoneMappings      :

Using existing resource group 'AZ300RG'
Starting deployment...
```



DEPLOYMENT NAME	STATUS	LAST MODIFIED	DURATION	RELATED EVENTS
az300deploy1	Deploying	3/17/2019, 10:31:42 PM	3 minutes 25 seconds	Related events
CreateVm-MicrosoftWindowsServe	Succeeded	3/17/2019, 4:20:08 PM	4 minutes 46 seconds	Related events

```
"name": "RDP",
"properties": {
  "priority": 360,
  "protocol": "TCP",
  "access": "Allow",
  "direction": "Inbound",
  "sourceAddressPrefix": "**",
  "sourcePortRange": "**",
  "destinationAddressPrefix": "**",
  "destinationPortRange": "3389"
}
}
]
subnetName      String      default
virtualNetworkName String    AZ300-vnet
addressPrefixes Array      [
  "10.0.0/24"
]
subnets        Array      [
  {
    "name": "default",
    "properties": {
      "addressPrefix": "10.0.0/24"
    }
  }
]
publicIpAddressName String    AZ300-VM02-ip
publicIpAddressType String    Dynamic
publicIpAddressSku String    Basic
virtualMachineName String    AZ300-VM02
virtualMachineRG String    AZ300
osDiskType String    Standard_LRS
virtualMachineSize String    Standard_B1ms
adminUsername String    aabelwahed
adminPassword SecureString
autoShutdownStatus String    Enabled
autoShutdownTime String    19:00
autoShutdownTimeZone String    UTC
autoShutdownNotificationStatus String    Disabled
autoShutdownNotificationLocale String    en

Outputs
: {[adminUsername,
Microsoft.Azure.Commands.ResourceManager.Cmdlets.SdkModels.DeploymentVariable]}
OutputsString
:
  Name      Type      Value
  =====
  adminUsername String    aabelwahed
```

Home > Resource groups > AZ300 - Deployments

AZ300 - Deployments
Resource group

Search (Ctrl+/) [Delete] [Cancel] [Redeploy] [View template] [Refresh]

Filter by deployment name or resources in the deployment...

DEPLOYMENT NAME	STATUS	LAST MODIFIED	DURATION	RELATED EVENTS
az300deploy1	Succeeded	3/17/2019, 10:33:05 PM	4 minutes 42 seconds	Related events
CreateVm-MicrosoftWindowsServe	Succeeded	3/17/2019, 4:20:08 PM	4 minutes 46 seconds	Related events

az vm image list --output table

Home > Virtual machines

Virtual machines
ahmedabdulwahedoutlook179 (Default Directory)

+ Add [Reservations] [Edit columns] [Refresh] [Assign tags] [Start] [Restart] [Stop] [Delete] [Services]

Subscriptions: All 3 selected - Don't see a subscription? Open Directory + Subscription settings

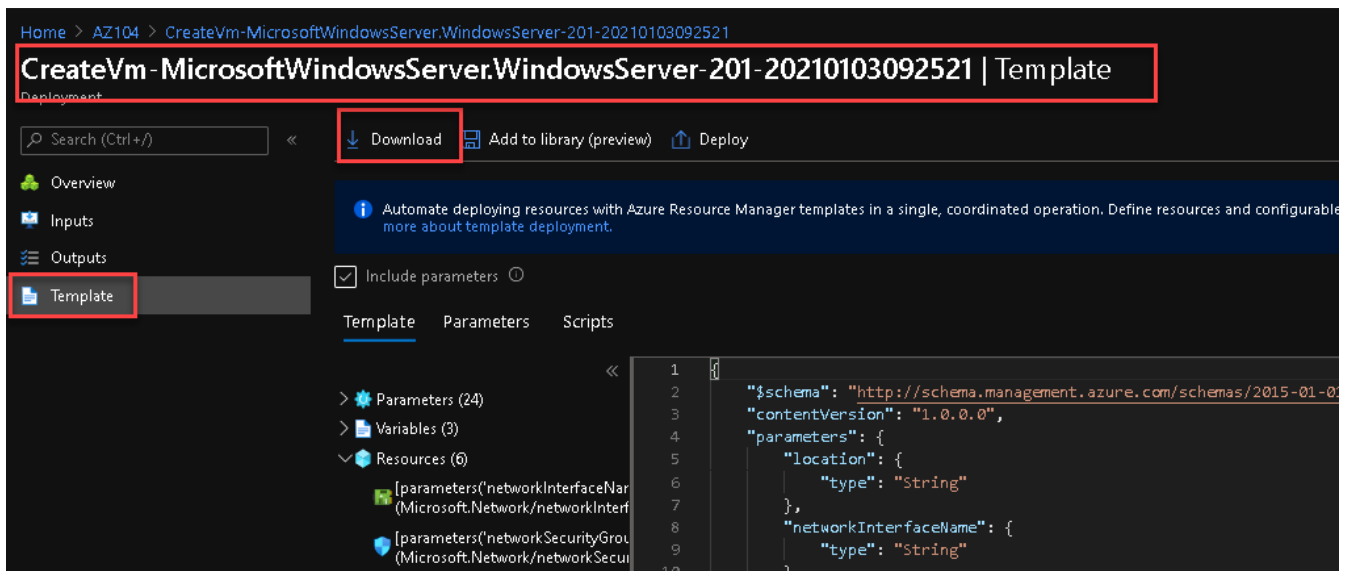
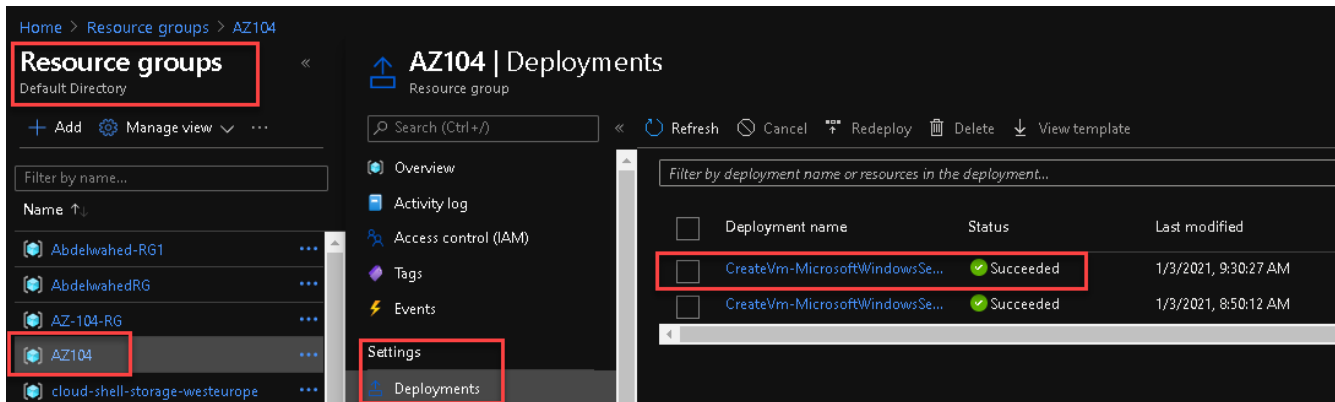
Filter by name... All subscriptions All resource groups All types All locations All tags No grouping

2 items

NAME	TYPE	STATUS	RESOURCE GROUP	LOCATION	MAINTENANCE STATUS	SUBSCRIPTION
AZ300-VM01	Virtual machine	Stopped (deallocated)	AZ300	Central US	-	MSDN Platforms (a7...)
AZ300-VM02	Virtual machine	Running	AZ300	Central US	-	MSDN Platforms (a7...)

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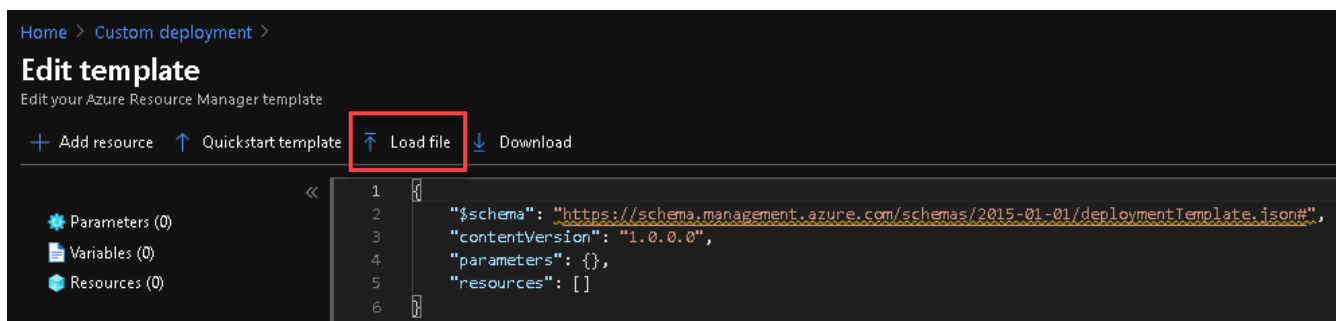
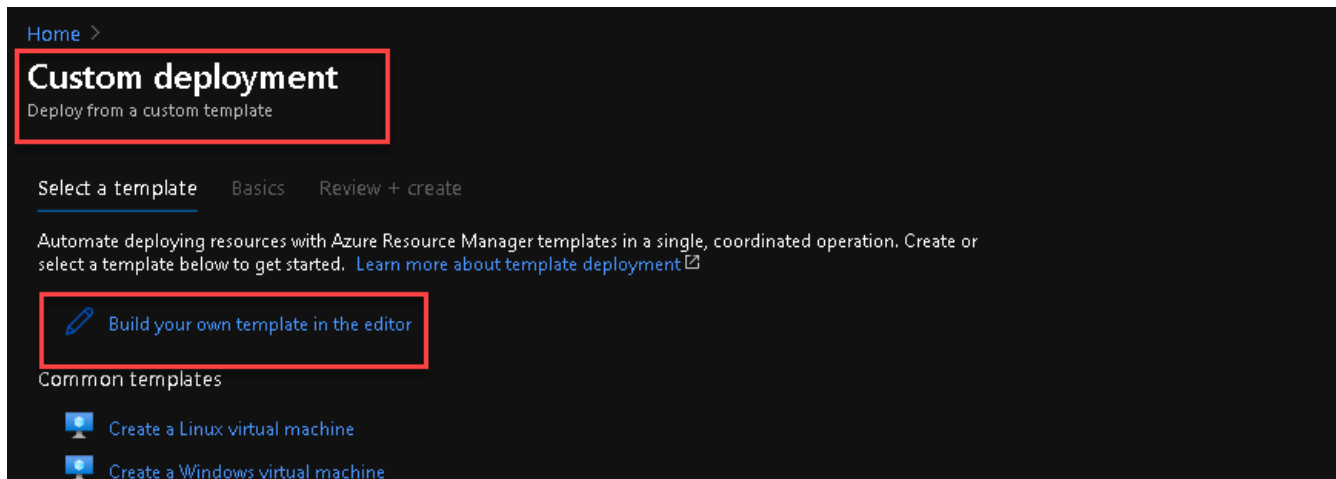
2- after deploy resource follow the instruction below



edit your Json file

Azure Administrator (AZ-104) | Lab Guide

search for **Custom deployment** in azure portal



And upload the .Json file and edit it

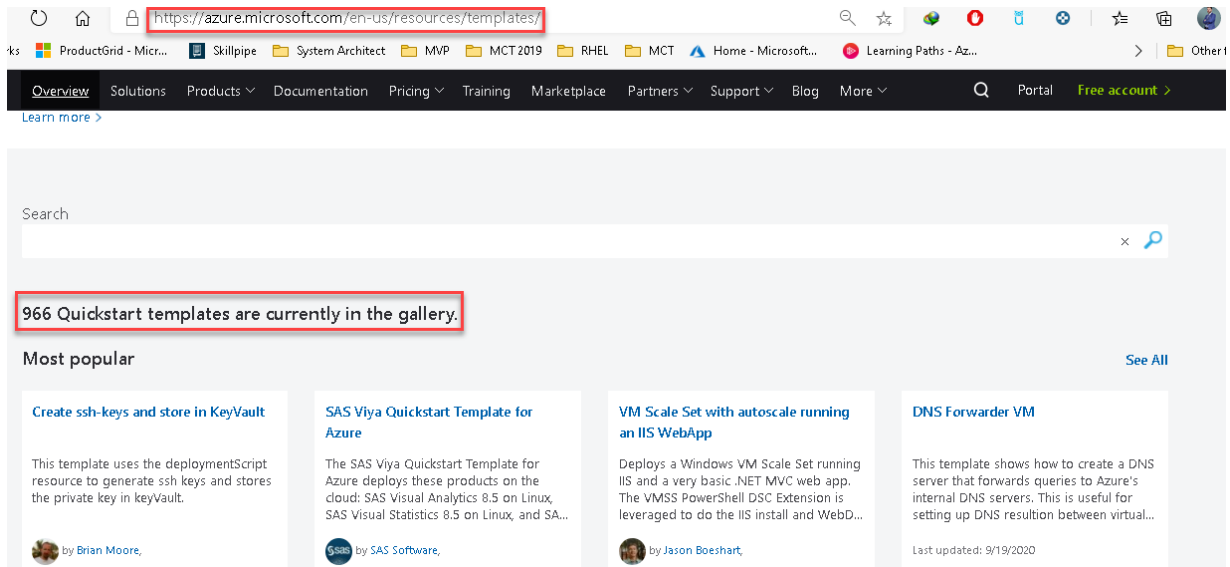
Azure Administrator (AZ-104) | Lab Guide

3- Using QuickStart Templates

are Resource Manager templates provided by the Azure community?

[Azure Quickstart Templates \(microsoft.com\)](https://azure.microsoft.com/en-us/resources/templates/)

Select template



So,
you
can

```
PS C:\Users\Anand> New-AzResourceGroupDeployment -Name NewVmDeployment -ResourceGroupName Prod_IT -TemplateUri "C:\Users\Anand\Desktop\ARM Templates\NewVM.json"

cmdlet New-AzResourceGroupDeployment at command pipeline position 1
Supply values for the following parameters:
(Type !? for Help.)
adminUsername:
```

deploy directly to azure

[Templates](#) / [Create a VM from Image Version](#)

Create a VM from Image Version

by Akshay Joshi
Last updated: 3/2/2020

[Deploy to Azure](#)

[Browse on GitHub](#)

This template allows you to create a Virtual Machines from an Image Version in a Shared Image Gallery. This template also deploys a Virtual Network, Public IP address, and an Interface. Please ensure that you have created an Image Version using Image Version 101 Template first.

This Azure Resource Manager template was created by a member of the community and not by Microsoft. Each Resource Manager template is licensed to you under a license agreement by its owner. Microsoft is not responsible for Resource Manager templates provided and licensed by community members and does not screen for security, compatibility, or performance. Community Resource Manager templates are made available AS IS without warranty of any kind.

Parameters

PARAMETER NAME	DESCRIPTION
----------------	-------------

www.abdelwahed.me

[Templates](#) / Create a VM from Image Version

Create a VM from Image Version



by Akshay Joshi

Last updated: 3/2/2020

Deploy to Azure

Browse on GitHub

This template allows you to create a Virtual Machines from an Image Version in a Shared Image Gallery. This template also deploys a Interface. Please ensure that you have created an Image Version using Image Version 101 Template first.

This Azure Resource Manager template was created by a member of the community and not by Microsoft. Each Resource Manager template is licensed to you not responsible for Resource Manager templates provided and licensed by community members and does not screen for security, compatibility, or performance under any Microsoft support program or service, and are made available AS IS without warranty of any kind.

Parameters

PARAMETER NAME	DESCRIPTION
adminUsername	Username for the Virtual Machine.

Also, you can edit the selected template to meet your goal.

[Home](#) >

Create a VM from Image Version

Azure quickstart template

[Basics](#) [Review + create](#)

Template

101-vm-from-sig 5 resources

Edit template Edit paramet...

Deployment scope

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ MSDN Platforms (6bde611b-7c73-4f7d-8942-8d3baa229533) ▼

Resource group * ⓘ

Create new

 ▼

Parameters

Region * ⓘ West US 2 ▼

[Review + create](#) < Previous Next : Review + create >

Intersite Connectivity

Virtual Network Peering – VNet Peering

You can connect virtual networks to each other with virtual network peering. Once virtual networks are peered, resources in both virtual networks can communicate with each other, with the same latency and bandwidth as if the resources were in the same virtual network.

Peering lab

Create 2 subnets inside one virtual network

Virtual network: 192.168.0.0/16

Subnet1: 192.168.10.0/24

Subnet2: 192.168.20.0/24

Now create 2 VMs:

VM1: is webserver1 connected to subnet1

VM2: is webserver2 connected to subnet2

Now test connectivity between both VMs (connected), this because the default NSG rule (check NSG rules)

There is default routing between all subnets in same VNet

There are two types of VNet peering.

- **Regional VNet peering** connects Azure virtual networks in the same region.
- **Global VNet peering** connects Azure virtual networks in different regions.

VNet1: 192.168.0.0/24

Subnet1: 192.168.10.0/24

Location: CentralUS

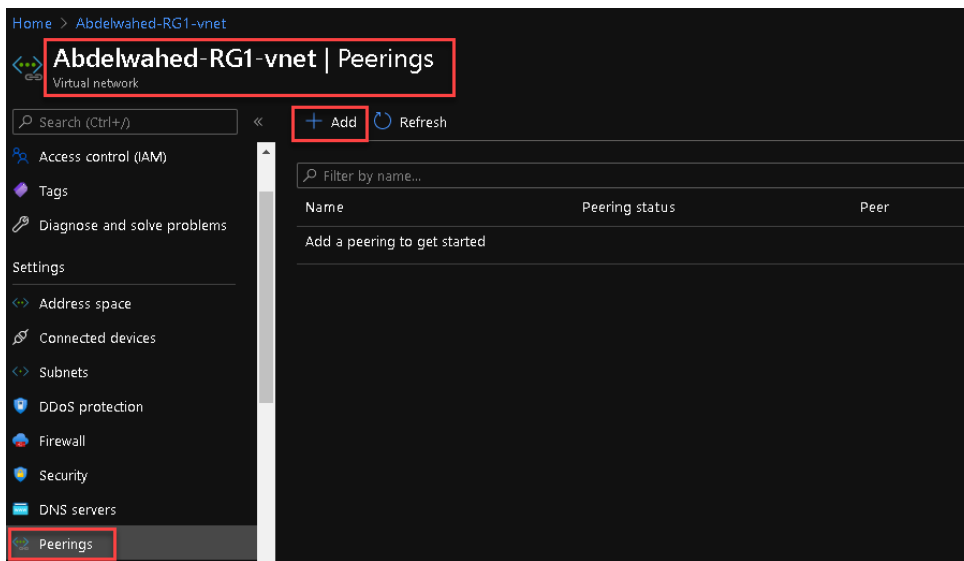
VNet2: 10.0.0.0/8

Subnet1: 10.0.0.0/8

Location: South Africa

You can connect these VNets using VNet peering as explained below

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The screenshot shows the 'Add peering' form in the Azure portal. The form is titled 'Add peering' and contains several sections for configuring a new peering link. The 'Add' button at the bottom is highlighted with a red box. The form includes the following fields and options:

- Peering link name ***: A text input field containing 'VNet1-VNet2-Peer'.
- Traffic to remote virtual network**: A radio button group with 'Allow (default)' selected.
- Traffic forwarded from remote virtual network**: A radio button group with 'Allow (default)' selected.
- Virtual network gateway or Route Server**: A radio button group with 'None (default)' selected.
- Remote virtual network**: A section containing:
 - Peering link name ***: A text input field containing 'VNet1-VNet2-Peer'.
 - Virtual network deployment model**: A radio button group with 'Resource manager' selected.
 - I know my resource ID**: An unchecked checkbox.
 - Subscription ***: A dropdown menu showing 'MSDN 5C'.
 - Virtual network ***: A dropdown menu showing 'PeeringVnet2'.
 - Traffic to remote virtual network**: A radio button group with 'Allow (default)' selected.

Network Traffic Management

Network Routing

First, we will create firewall as a virtual appliance then we will learn how to route all traffic to it

Azure Firewall is a managed, cloud-based network security service that protects Azure Virtual Network resources. It is **Firewall as a Service** with built-in high availability and scalability.

With Azure Firewall, you can create, enforce, and log application and network connectivity policies. Static public IP addresses for your virtual network resources allow outside firewalls to identify traffic originating from your virtual network. The service is integrated with Azure Monitor for logging and analytics.

Configuring Azure Firewall

- Set up a network environment (Create Resource group, VNet and VM)
- Deploy a firewall
- Create a default route
- Configure RDP and internet through the firewall rules

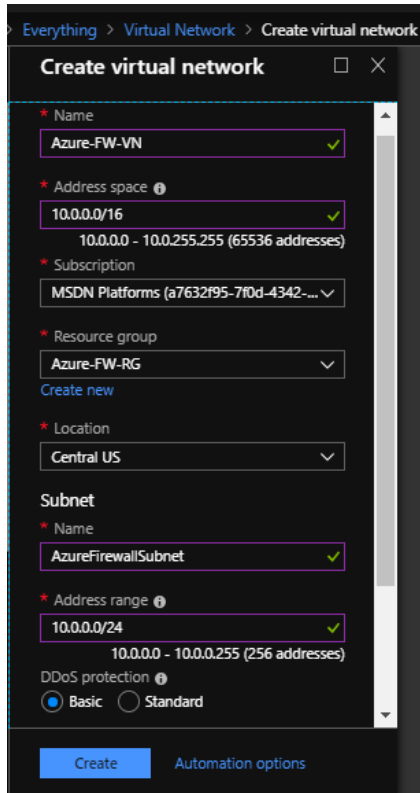
Create a resource group

The screenshot shows the 'Create a resource group' page in the Azure portal. The page has a dark theme. At the top, there's a title bar with navigation icons and the title 'Create a resource group'. Below the title bar, there's a sidebar on the left with a search bar and a list of resource groups. The main content area has a tabbed interface with 'Basics', 'Tags', and 'Review + Create'. The 'Basics' tab is active. Below the tabs, there's a description of a resource group. Then, there are two sections: 'PROJECT DETAILS' and 'RESOURCE DETAILS'. Under 'PROJECT DETAILS', there's a 'Subscription' dropdown menu showing 'MSDN Platforms (a7632f95-7f0d-4342-adda-dbc286dbe45c)' and a 'Resource group' dropdown menu showing 'Azure-FW-RG' with a green checkmark. Under 'RESOURCE DETAILS', there's a 'Region' dropdown menu showing 'Central US'.

Azure Administrator (AZ-104) | Lab Guide

Create a VNet

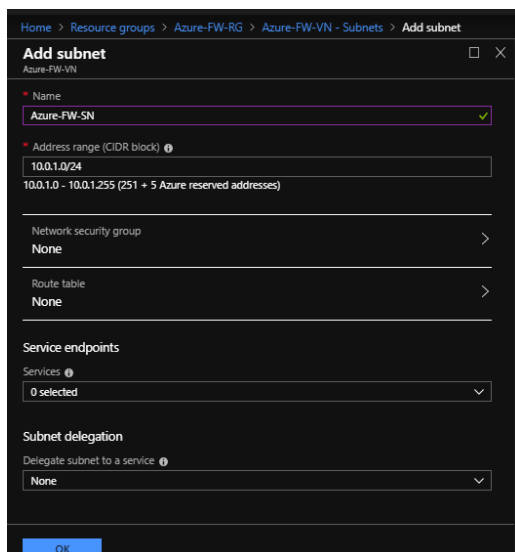
The minimum size of the AzureFirewallSubnet subnet is /26.



The screenshot shows the 'Create virtual network' dialog box in the Azure portal. The dialog is titled 'Create virtual network' and has a close button (X) in the top right corner. The breadcrumb navigation at the top reads 'Everything > Virtual Network > Create virtual network'. The form contains the following fields and options:

- Name:** 'Azure-FW-VN' with a green checkmark.
- Address space:** '10.0.0.0/16' with a green checkmark. Below it, it says '10.0.0.0 - 10.0.255.255 (65536 addresses)'.
- Subscription:** 'MSDN Platforms (a7632f95-7f0d-4342-....)'.
- Resource group:** 'Azure-FW-RG' with a dropdown arrow. Below it is a link 'Create new'.
- Location:** 'Central US' with a dropdown arrow.
- Subnet:**
 - Name:** 'AzureFirewallSubnet' with a green checkmark.
 - Address range:** '10.0.0.0/24' with a green checkmark. Below it, it says '10.0.0.0 - 10.0.0.255 (256 addresses)'.
- DDoS protection:** Two radio buttons: 'Basic' (selected) and 'Standard'.
- Buttons:** 'Create' (blue) and 'Automation options' (grey).

Create additional subnets to associate it with firewall



The screenshot shows the 'Add subnet' dialog box in the Azure portal. The breadcrumb navigation at the top reads 'Home > Resource groups > Azure-FW-RG > Azure-FW-VN - Subnets > Add subnet'. The dialog is titled 'Add subnet' and has a close button (X) in the top right corner. The form contains the following fields and options:

- Name:** 'Azure-FW-SN' with a green checkmark.
- Address range (CIDR block):** '10.0.1.0/24'. Below it, it says '10.0.1.0 - 10.0.1.255 (251 + 5 Azure reserved addresses)'.
- Network security group:** 'None' with a right arrow.
- Route table:** 'None' with a right arrow.
- Service endpoints:**
 - Services:** '0 selected' with a dropdown arrow.
- Subnet delegation:**
 - Delegate subnet to a service:** 'None' with a dropdown arrow.
- Buttons:** 'OK' (blue).

Azure Administrator (AZ-104) | Lab Guide

Azure-FW-VN - Subnets
Virtual network

Search (Ctrl+F)

+ Subnet + Gateway subnet

Search subnets

NAME	ADDRESS RANGE	AVAILABLE ADDRESSES	SECURITY GROUP
AzureFirewallSubnet	10.0.0.0/24	251	-
Azure-FW-SN	10.0.1.0/24	251	-

Create virtual machines

Home > Resource groups > Azure-FW-RG > Everything > Windows Server > Create a virtual machine

Create a virtual machine

Subscription: MSDN Platforms (a7632955-7f0d-4342-adda-dbc286dbe45c)

Resource group: Azure-FW-RG

INSTANCE DETAILS

Virtual machine name: Azure-FW-VM

Region: Central US

Availability options: No infrastructure redundancy required

Image: Windows Server 2016 Datacenter

Size: Standard DS1 v2 (1 vcpu, 3.5 GB memory)

ADMINISTRATOR ACCOUNT

Username:

Password:

Review + create Previous Next: Disks >

Create a virtual machine

Basics Disks **Networking** Management Guest config Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution. [Learn more](#)

NETWORK INTERFACE

When creating a virtual machine, a network interface will be created for you.

CONFIGURE VIRTUAL NETWORKS

Virtual network: Azure-FW-VN

Subnet: Azure-FW-SN (10.0.1.0/24)

Public IP: (new) Azure-FW-VM-PIP

NIC network security group: ☐ None ☒ Basic ☐ Advanced

Public inbound ports: ☐ None ☒ Allow selected ports

Select inbound ports: RDP, HTTP, HTTPS

Review + create Previous Next: Management >

Note the private IP address. You'll use it later when you create the default route.

Deploy the firewall

Home > Resource groups > Azure-FW-RG > Everything > Firewall > Create a firewall

Create a firewall

Subscription: MSDN Platforms (a7632955-7f0d-4342-adda-dbc286dbe45c)

Resource group: Azure-FW-RG

INSTANCE DETAILS

Name: Azure-FW

Region: Central US

Choose a virtual network: ☐ Create new ☒ Use existing

Virtual network: Azure-FW-VN (Azure-FW-RG)

PUBLIC IP ADDRESS

Public IP address: ☒ Create new ☐ Use existing

Public IP address name: AzureFirewalls-PIP

Public IP address SKU: Standard

Review + create Previous Next: Tags > Download a template for automation

Create a firewall

Basics Tags **Review + create**

Summary

Subscription: MSDN Platforms

Resource group: Azure-FW-RG

Region: Central US

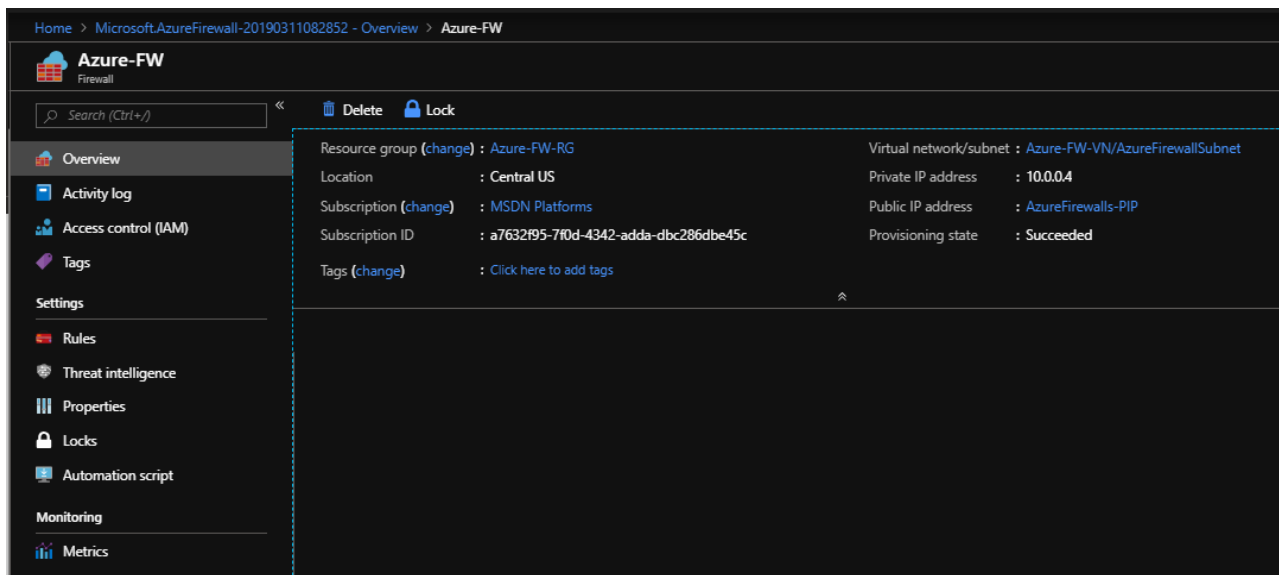
Virtual network: Azure-FW-VN

Address space: 10.0.0.0/16

Public IP address: AzureFirewalls-PIP

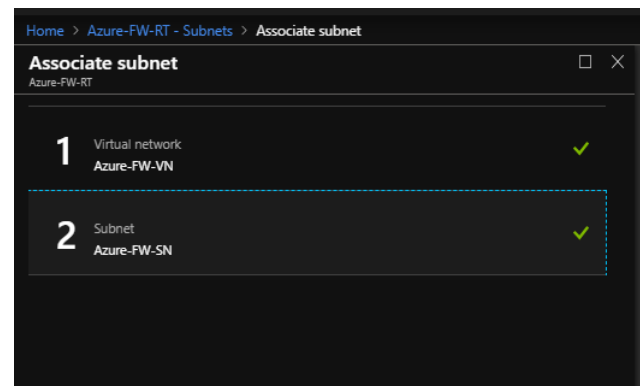
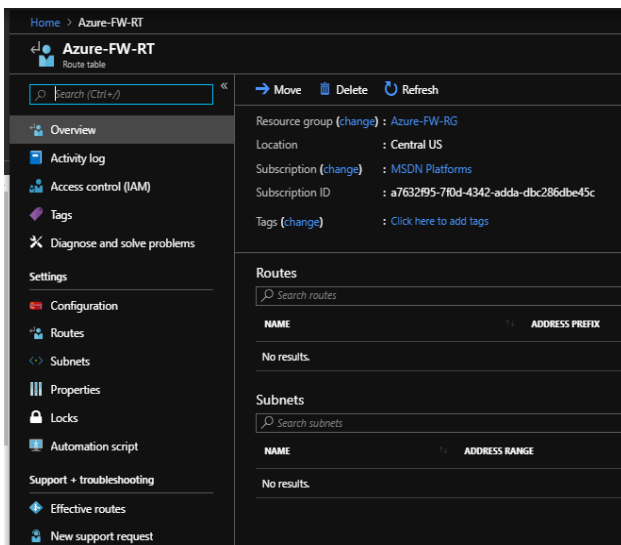
Create Previous Next Download a template for automation

Azure Administrator (AZ-104) | Lab Guide



Create a default route

From all services search for route table and after create Click **Refresh** then Click **Subnets** > **Associate**



Now all incoming traffic will be forwarded to the firewall, so now you can allow or deny network traffic coming to that subnet.

Azure Administrator (AZ-104) | Lab Guide

Click **Routes** > **Add**.

Configure an application rule

From firewall under setting select Rules, in this lab allows incoming RDP. Also change rdp port from 3389 to 5555

also
internet is
blocked,
so we
have to
allow
http and https
traffic.

www.abdelwahed.me

Network Routing and Endpoints

System Routes

Azure uses **system routes** to direct network traffic between virtual machines, on-premises networks, and the Internet. The following situations are managed by these system routes:

- Traffic between VMs in the same subnet.
- Between VMs in different subnets in the same virtual network.
- Data flow from VMs to the Internet.
- Site-to-Site and ExpressRoute communication through the VPN gateway.

Creating User defined Route

- 1- First create the routing table
- 2- Create routes

Home > Route tables >

Create Route table

Basics Tags Review + create

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ MSDN Platforms (6bde611b-...) ▼

Resource group * ⓘ (New) AZ-104-RG ▼
[Create new](#)

Instance details

Region * ⓘ West US 2 ▼

Name * ⓘ Test-RT ✓

Propagate gateway routes * ⓘ ☐ Yes ☒ No

[Review + create](#) [< Previous](#) [Next : Tags >](#)

Azure Administrator (AZ-104) | Lab Guide

Home > Microsoft.RouteTable-20201202170743 >

Test-RT
Route table

Search (Ctrl+/) << Move Delete Refresh

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- Configuration
- Routes**
- Subnets**
- Properties
- Locks

Automation

- Tasks (preview)

Essentials

Resource group (change) : AZ-104-RG Associations : 0 subnet associations

Location : West US 2

Subscription (change) : MSDN Platforms

Subscription ID : 6bde611b-7c73-4f7d-8942-8d3baa229533

Tags (change) : Click here to add tags

Routes

Search routes

Name	Address prefix	Next hop type
No results.		

Subnets

Search subnets

Name	Address range	Virtual network
No results.		

Home > Test-RT >

Add route

Test-RT

Route name *

Firewall-Route ✓

Address prefix * ⓘ

10.0.0.0/24 ✓

Next hop type ⓘ

Virtual appliance ^

Virtual network gateway

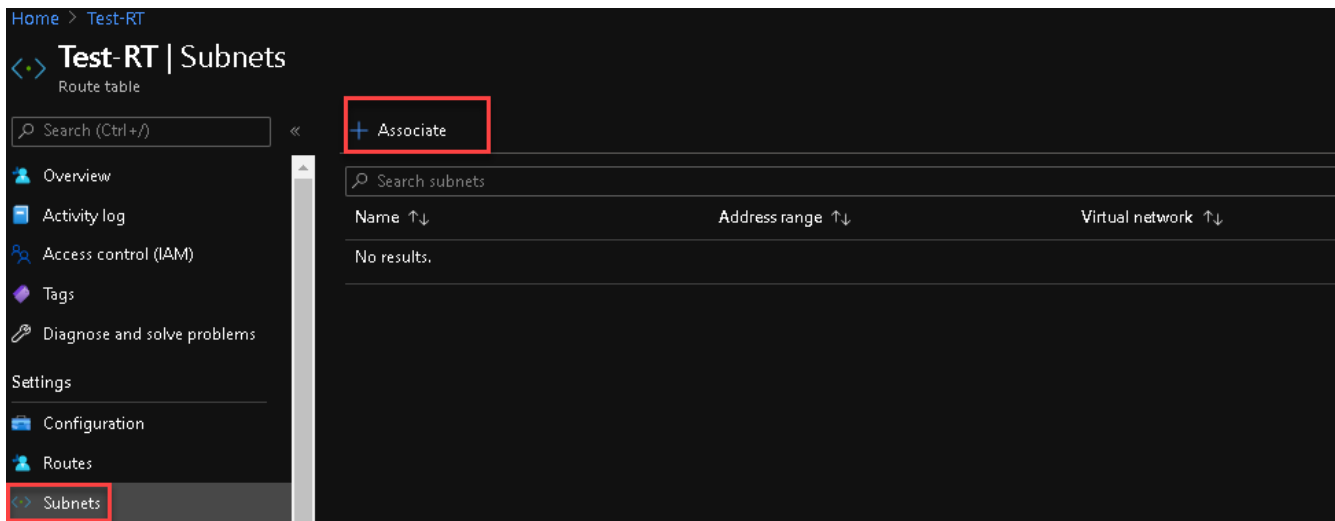
Virtual network

Internet

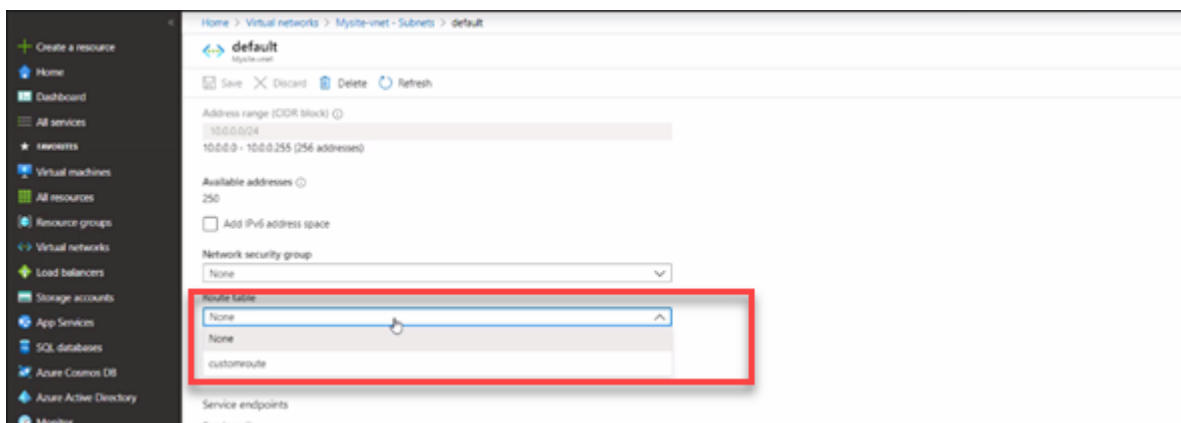
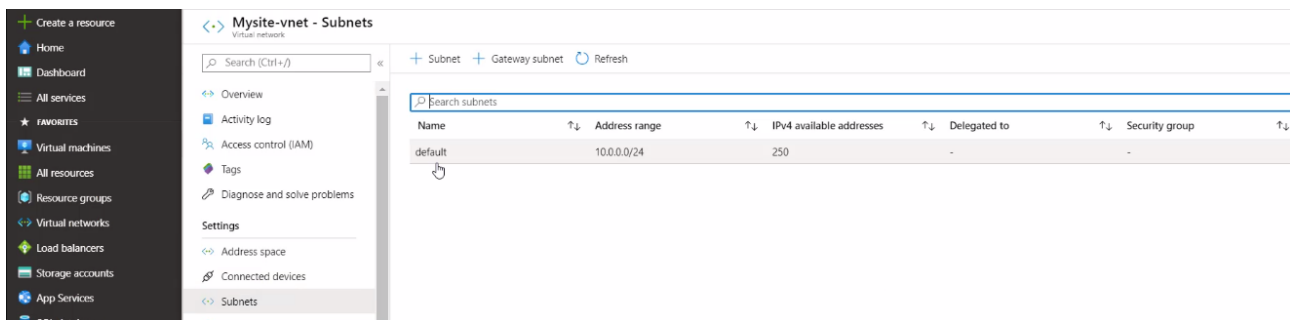
Virtual appliance

None

Azure Administrator (AZ-104) | Lab Guide



or from VNet select subnet you want to associate to the new custom route



Azure Load Balancers

Create 3 VMs to host web server

[Home](#) > [Virtual machines](#) >

Create a virtual machine ...

i Based on the number of availability zones selected, 3 virtual machines will be created. The following settings will be applied to each virtual machine unless specified otherwise.

Basics Disks Networking Management Monitoring Advanced Tags Review + create

Create a virtual machine that runs Linux or Windows. Select an image from Azure marketplace or use your own customized image. Complete the Basics tab then Review + create to provision a virtual machine with default parameters or review each tab for full customization. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ MSDN 5C
Resource group * ⓘ (New) LBLabs
[Create new](#)

Instance details

Virtual machine names ⓘ web-1, web-2, web-3
i 3 virtual machines will be created with the names shown above. [Edit names](#)
Region * ⓘ (US) Central US

Availability options ⓘ

Availability zone * ⓘ

Availability zone
Zones 1, 2, 3

i Based on your zone selection, we will place 3 virtual machines, one in each selected zone. It is highly recommended to place them behind a load balancing solution to make your configuration resilient. You can do this in the 'Networking' tab

Security type ⓘ

Standard

Image * ⓘ

Windows Server 2019 Datacenter - Gen2

[See all images](#) | [Configure VM generation](#)

VM architecture ⓘ

☐ Arm64
☒ x64

i Arm64 is not supported with the selected image.

Run with Azure Spot discount ⓘ

☐

Size * ⓘ

Standard_DS2_v2 - 2 vcpus, 7 GiB memory (SAR 399.67/month)

[See all sizes](#)

Azure Administrator (AZ-104) | Lab Guide

Administrator account

Username * ⓘ

ahmed

Password * ⓘ

.....

Confirm password * ⓘ

.....

Inbound port rules

Select which virtual machine network ports are accessible from the public internet. You can specify more limited or granular network access on the Networking tab.

Public inbound ports * ⓘ

☐ None

☒ Allow selected ports

Select inbound ports *

HTTP (80), RDP (3389)

⚠

This will allow all IP addresses to access your virtual machine. This is only recommended for testing. Use the Advanced controls in the Networking tab to create rules to limit inbound traffic to known IP addresses.

Basics

Disks

NetworkingManagementMonitoringAdvancedTagsReview + create

Azure VMs have one operating system disk and a temporary disk for short-term storage. You can attach additional data disks. The size of the VM determines the type of storage you can use and the number of data disks allowed. [Learn more](#)

Disk options

OS disk type * ⓘ

Standard SSD (locally-redundant storage)

If performance is critical for your workloads, choose Premium SSD disks for lower latency, higher IOPS and bandwidth, and bursting. [Learn more](#)

Delete with VM ⓘ

☒

Enable encryption at host ⓘ

☐

i

Encryption at host is not registered for the selected subscription. [Learn more about enabling this feature](#)

Encryption type *

(Default) Encryption at-rest with a platform-managed key

Enable Ultra Disk compatibility ⓘ

☐

Ultra disk is not supported for the selected VM size Standard_DS2_v2 in Central US.

Azure Administrator (AZ-104) | Lab Guide

Basics Disks **Networking** Management Monitoring Advanced Tags Review + create

Define network connectivity for your virtual machine by configuring network interface card (NIC) settings. You can control ports, inbound and outbound connectivity with security group rules, or place behind an existing load balancing solution.

[Learn more](#)

Network interface

When creating a virtual machine, a network interface will be created for you.

Virtual network * ⓘ

(new) LBLabs-vnet

[Create new](#)

Subnet * ⓘ

(new) default (10.2.0.0/24)

Public IP ⓘ

(new) web-1-ip, web-2-ip, web-3-ip

[Configure IP address](#)

3 public IPs will be created with the names shown above.

NIC network security group ⓘ

- ☐ None
- ☒ Basic
- ☐ Advanced

Delete public IP and NIC when VM is deleted ⓘ

☐

Enable accelerated networking ⓘ

☒

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#)

Load balancing options ⓘ

☐ None

☒ Azure load balancer

Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.

☐ Application gateway

Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

Select a load balancer * ⓘ

No existing load balancers in current subscription and location.

[Create a load balancer](#)

Create a load balancer



Load balancer name *

Ahmed-LB01

Type * ⓘ

☒ Public

Provides outbound connections for virtual machines inside your virtual network using public load balancers.

☐ Internal

Used to load balance traffic inside a virtual network. A load balancer frontend can be accessed from an on-premises network in a hybrid scenario.

Protocol * ⓘ

☒ TCP

☐ UDP

Rules

Rules

☒ Load balancer rule

☒ Inbound NAT rule

Load balancer rule

A load balancing rule distributes incoming traffic that is sent to a selected IP address and port combination across a group of backend pool instances. Only backend instances that the health probe considers healthy receive new traffic.

Port * ⓘ

80

Backend port * ⓘ

80

Inbound NAT rule

An inbound NAT rule forwards incoming traffic sent to a selected IP address and port combination to a specific virtual machine.

Frontend port range start * ⓘ

50000

Backend port * ⓘ

3389

Create

Cancel

Azure Administrator (AZ-104) | Lab Guide

Delete public IP and NIC when VM is deleted ⓘ

☐

Enable accelerated networking ⓘ

☒

Load balancing

You can place this virtual machine in the backend pool of an existing Azure load balancing solution. [Learn more](#) ⓘ

Load balancing options ⓘ

☐ None

☒

Azure load balancer

Supports all TCP/UDP network traffic, port-forwarding, and outbound flows.

☐

Application gateway

Web traffic load balancer for HTTP/HTTPS with URL-based routing, SSL termination, session persistence, and web application firewall.

Select a load balancer * ⓘ

(new) Ahmed-LB01

[Create a load balancer](#)



Review + create

< Previous

Next : Management >

Virtual machines



Default Directory (ahmedabdulwahedoutlook179.onmicrosoft.com)

+ Create ▾

↔ Switch to classic

🕒 Reservations ▾

⚙️ Manage view ▾

🔄 Refresh

⬇️ Export to CSV

🔗 Open query

🏷️ Assign tags

Filter for any field...

Subscription equals all

Type equals all

Resource group equals all ✕

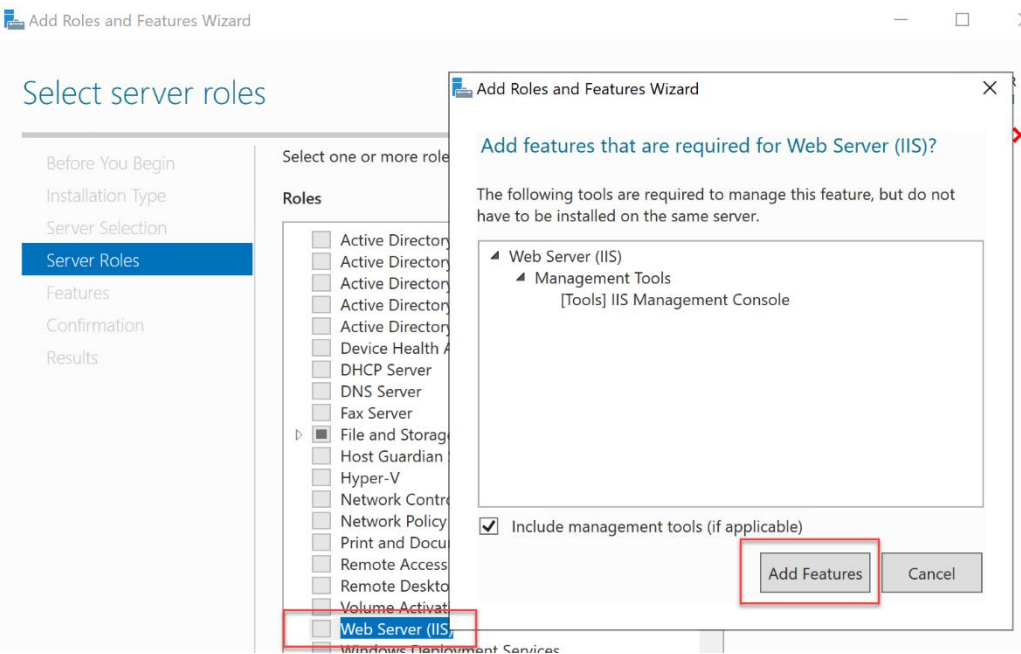
Location equals all ✕

+ Add filter

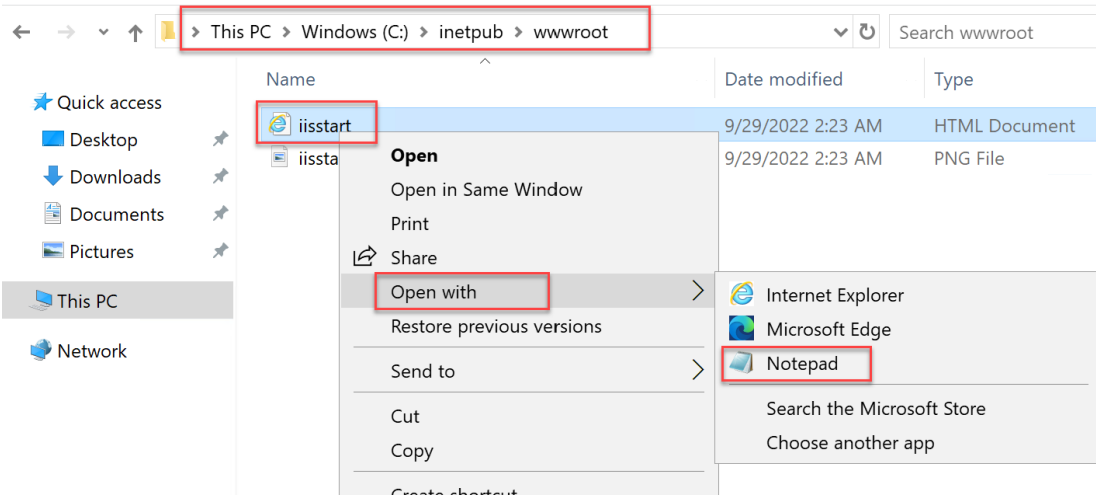
☐ Name ↑↓	Type ↑↓	Subscription ↑↓	Resource group ↑↓	Location ↑↓	Status ↑↓
☐  web-1	Virtual machine	MSDN 5C	LBLabs	Central US	Running
☐  web-2	Virtual machine	MSDN 5C	LBLabs	Central US	Running
☐  web-3	Virtual machine	MSDN 5C	LBLabs	Central US	Running

Azure Administrator (AZ-104) | Lab Guide

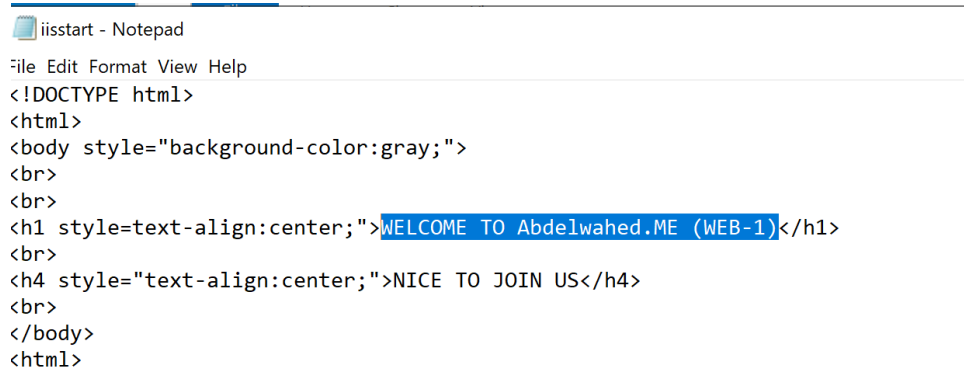
Now, access each server to add and configure web server (iis) using default installation settings.



The screenshot shows the 'Add Roles and Features Wizard' with the 'Web Server (IIS)' role selected. The 'Add Features' dialog box is open, showing the 'Web Server (IIS)' feature and its management tools. The 'Include management tools (if applicable)' checkbox is checked. The 'Add Features' button is highlighted.



The screenshot shows the Windows File Explorer with the path 'This PC > Windows (C:) > inetpub > wwwroot'. The 'iisstart' file is selected, and the context menu is open. The 'Open with' option is highlighted, and the 'Notepad' application is selected from the list of available applications.



The screenshot shows the Notepad application with the following HTML code:

```
File Edit Format View Help
<!DOCTYPE html>
<html>
<body style="background-color:gray;">
<br>
<br>
<h1 style=text-align:center;">WELCOME TO Abdelwahed.ME (WEB-1)</h1>
<br>
<h4 style="text-align:center;">NICE TO JOIN US</h4>
<br>
</body>
<html>
```

Azure Administrator (AZ-104) | Lab Guide

Using Load balancer public IP address, you can access web service in these 3 servers

Home > Load balancing | Load Balancer > Ahmed-LB01



Frontend IP configuration

Search

+ Add Refresh Give feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- Frontend IP configuration
- Backend pools

Filter by name...

Name ↑↓

IP address ↑↓

Ahmed-LB01-frontendconfig01

20.221.117.223 (Ahmed-LB01-publicip)

Home > Load balancing | Load Balancer > Ahmed-LB01



Backend pools

Search

+ Add Refresh Give feedback

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

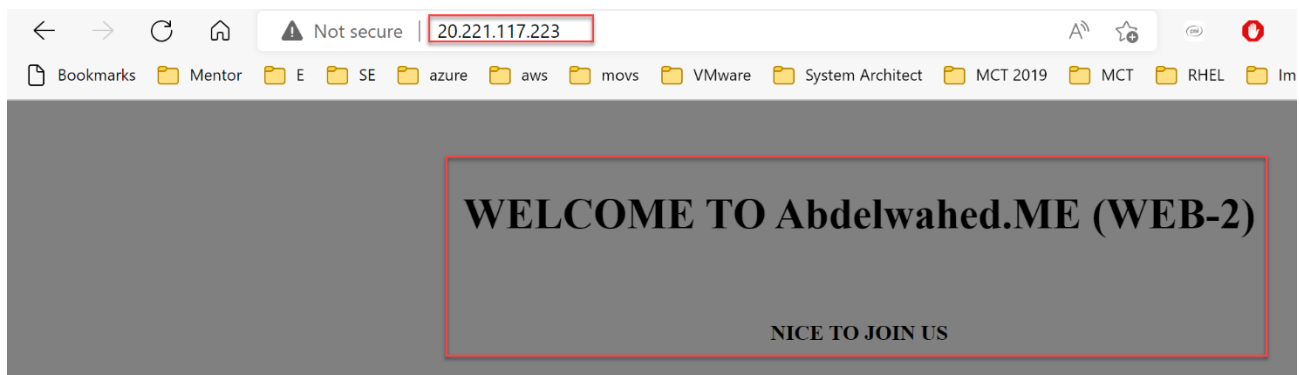
- Frontend IP configuration
- Backend pools
- Health probes
- Load balancing rules
- Inbound NAT rules

Filter by name...

Backend pool == all Resource Name == all Resource Status == all IP address == all Network interface == all Availability zone == all

Group by Backend pool

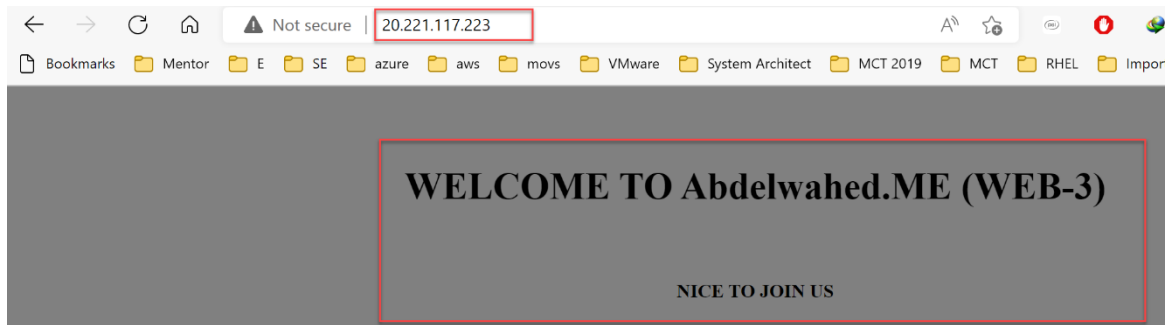
Backend pool	Resource Name	Resource Status	IP address	Network interface	Availability zone
Ahmed-LB01-backendpool01					
Ahmed-LB01-backendpool...	web-1	Running	10.2.0.4	web-117_z1	1
Ahmed-LB01-backendpool...	web-3	Running	10.2.0.5	web-1478_z3	3
Ahmed-LB01-backendpool...	web-2	Running	10.2.0.6	web-1779_z2	2



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Azure Administrator (AZ-104) | Lab Guide

To test load balancer refresh or shutdown web-2 and try to access using load balancer IP address



Azure Storage

Azure Storage is a service that you can use to store files, messages, tables, and other types of information. You can use Azure storage on its own—for example as a file share—but it is often used by developers as a store for working data. Such stores can be used by websites, mobile apps, desktop applications, and many other types of custom solutions. Azure storage is also used by IaaS virtual machines, and PaaS cloud services. You can generally think of Azure storage in three categories.

Storage for Virtual Machines

This includes disks and files. Disks are persistent block storage for Azure IaaS virtual machines. Files are fully managed file shares in the cloud.

Unstructured Data

This includes Blobs and Data Lake Store. Blobs are highly scalable, REST based cloud object store. Data Lake Store is Hadoop Distributed File System (HDFS) as a service.

Structured Data

This includes Tables, Cosmos DB, and Azure SQL DB. Tables are a key/value, auto-scaling NoSQL store. Cosmos DB is a globally distributed database service. Azure SQL DB is a fully managed database-as-a-service built on SQL.

Creating an Azure Storage account

[Home](#) > [Storage accounts](#) >

Create a storage account

Basics Advanced Networking Data protection Encryption Tags Review

Resource group * StorageRG ▼
[Create new](#)

Instance details

If you need to create a legacy storage account type, please click [here](#).

Storage account name * ohist02

Region * (US) East US ▼

Performance * ☒ **Standard:** Recommended for most scenarios (general-purpose v2 account)

☐ **Premium:** Recommended for scenarios that require low latency.

Redundancy * Geo-redundant storage (GRS) ▼

☒ Make read access to data available in the event of regional unavailability.

[Review](#)

[< Previous](#)

[Next : Advanced >](#)

[Home](#) > [Storage accounts](#) >

Create a storage account ...

Basics Advanced **Networking** Data protection Encryption Tags Review

Network connectivity

You can connect to your storage account either publicly, via public IP addresses or service endpoints, or privately, using a private endpoint.

Network access *

- ☒ Enable public access from all networks
- ☐ Enable public access from selected virtual networks and IP addresses
- ☐ Disable public access and use private access
- i** Enabling public access from all networks might make this resource available publicly. Unless public access is required, we recommend using a more restricted access type. [Learn more](#)

Create a storage account ...

Basics Advanced Networking **Data protection** Encryption Tags Review

- ☒ Enable soft delete for blobs
Soft delete enables you to recover blobs that were previously marked for deletion, including blobs that were overwritten. [Learn more](#)
Days to retain deleted blobs ⓘ
- ☒ Enable soft delete for containers
Soft delete enables you to recover containers that were previously marked for deletion. [Learn more](#)
Days to retain deleted containers ⓘ
- ☒ Enable soft delete for file shares
Soft delete enables you to recover file shares that were previously marked for deletion. [Learn more](#)
Days to retain deleted file shares ⓘ

Tracking

Manage versions and keep track of changes made to your blob data.

- ☐ Enable versioning for blobs
Use versioning to automatically maintain previous versions of your blobs. [Learn more](#)

Review

< Previous

Next : Encryption >

Azure Administrator (AZ-104) | Lab Guide

Home > Storage accounts >

ohist01
Storage account

Search

Upload Open in Explorer Delete Move Refresh Open in mobile Feedback

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Access Control (IAM)
- Data migration
- Events
- Storage browser

Data storage

- Containers
- File shares
- Queues
- Tables

Security + networking

- Networking
- Azure CDN

Essentials

Resource group (move) : StorageRG

Location : East US

Primary/Secondary Location : Primary: East US, Secondary: West US

Subscription (move) : MSDN SC

Subscription ID : a7632f95-7f0d-4342-adda-dbc286dbe45c

Disk state : Primary: Available, Secondary: Available

Performance : Standard

Replication : Read-access geo-redundant storage (RA-GRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 9/29/2022, 9:00:34 AM

Tags (edit) : Click here to add tags

Properties Monitoring Capabilities (7) Recommendations Tutorials Developer Tools

Blob service

Hierarchical namespaces Disabled

Default access tier Hot

Blob public access Enabled

Blob soft delete Disabled

Container soft delete Disabled

Versioning Disabled

Change feed Disabled

NFS v2 Disabled

Security

Require secure transfer for REST API operations Enabled

Storage account key access Enabled

Minimum TLS version Version 1.2

Infrastructure encryption Disabled

Networking

Allow access from All networks

Create Container and File Share

Home > ohist01

ohist01 | Containers

Storage account

Search

+ Container Change access level Restore containers Refresh Delete

Search containers by prefix

Show deleted containers

Name	Last modified	Public access level	Lease state
☐ \$logs	9/29/2022, 9:00:59 AM	Private	Available
☐ ahmed-container	9/29/2022, 9:03:56 AM	Container	Available

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Data storage

- Containers
- File shares
- Queues
- Tables

Azure Administrator (AZ-104) | Lab Guide

Home > ohist01

ohist01 | File shares

Storage account

Search

+ File share Refresh

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Data storage

Containers

File shares

Queues

File share settings

Active Directory: Not configured

Soft delete: Disabled

Maximum capacity: 5 TiB

Security: Maximum compatibility

Search file shares by prefix (case-sensitive)

Show deleted shares

Name	Modified	Tier	Quota
archive-data	9/29/2022, 9:12:50 AM	Cool	100 GiB
hr-share	9/29/2022, 9:11:25 AM	Hot	5 TiB
it-share	9/29/2022, 9:11:05 AM	Transaction optimized	5 TiB

Connect to file share

Home > Storage accounts > ohist01 | File shares >

archive-data

File share

Search

Connect Upload Add directory Refresh Delete share Change tier Edit quota

Search files by prefix

Name	Type
HR	Directory
IT	Directory
Sales	Directory

Connect

archive-data

authentication methods and run the PowerShell commands from a normal (not elevated) PowerShell terminal:

Drive letter

Z

Authentication method

Active Directory

Storage account key

Connecting to a share using the storage account key is only appropriate for admin access. Mounting the Azure file share with the Active Directory identity of the user is preferred.

Hide script

```
$connectTestResult = Test-NetConnection -ComputerName ohist01.file.core.windows.net -Port 445
if ($connectTestResult.TcpTestSucceeded) {
    # Save the password so the drive will persist on reboot
    cmd.exe /C "cmdkey /add:"ohist01.file.core.windows.net" /user:"localhost\ohist01" /pass:"lbnPFPzfhwvTRD6FKHOTnCioWwCNSQKd3vOORwVMhU7QOXABUvPv3XKMA04x+TNM2DM5UjW6+AS9vW5EA=="
    # Mount the drive
    New-PSDrive -Name Z -PSProvider FileSystem -Root "\\ohist01.file.core.windows.net\archive-data" -Persist
} else {
    Write-Error -Message "Unable to reach the Azure storage account via port 445. Check to make sure your organization or ISP is not blocking port 445, or use Azure P2S VPN, Azure S2S VPN, or Express Route to tunnel SMB traffic over a different port."
}
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Form PowerShell https://aka.ms/pscore6

PS C:\Windows\system32> netsh advfirewall set all

PS C:\Windows\system32> $connectTestResult = Test-NetConnection -ComputerName ohist01.file.core.windows.net -Port 445
if ($connectTestResult.TcpTestSucceeded) {
    # Save the password so the drive will persist on reboot
    cmd.exe /C "cmdkey /add:"ohist01.file.core.windows.net" /user:"localhost\ohist01" /pass:"lbnPFPzfhwvTRD6FKHOTnCioWwCNSQKd3vOORwVMhU7QOXABUvPv3XKMA04x+TNM2DM5UjW6+AS9vW5EA=="
    # Mount the drive
    New-PSDrive -Name Z -PSProvider FileSystem -Root "\\ohist01.file.core.windows.net\archive-data" -Persist
} else {
    Write-Error -Message "Unable to reach the Azure storage account via port 445. Check to make sure your organization or ISP is not blocking port 445, or use Azure P2S VPN, Azure S2S VPN, or Express Route to tunnel SMB traffic over a different port."
}

CMDKEY: Credential added successfully.

Name      Used (GB)  Free (GB) Provider      Root      CurrentLocation
-----
Z          0.00      100.00  FileSystem    \\ohist01.file.core.windows.net\...

```

Azure Administrator (AZ-104) | Lab Guide

Home > Storage accounts > ohist01 | File shares > archive-data

archive-data | Properties

Search «

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Properties

Operations

Snapshots

Backup

NAME

archive-data

URL

https://ohist01.file.core.windows.net/archive-data

LAST MODIFIED

9/29/2022, 9:12:50 AM

ETAG

0x8DAA1D94172E4DF

QUOTA

100 GiB

USAGE

472.61 KiB

TIER

Cool

Map Network Drive

What network folder would you like to map?

Specify the drive letter for the connection and the folder that you want to connect to:

Drive: Z: ▼

Folder: \\ohist01.file.core.windows.net\archive-data ▼

Browse...

Example: \\server\share

☒ Reconnect at sign-in

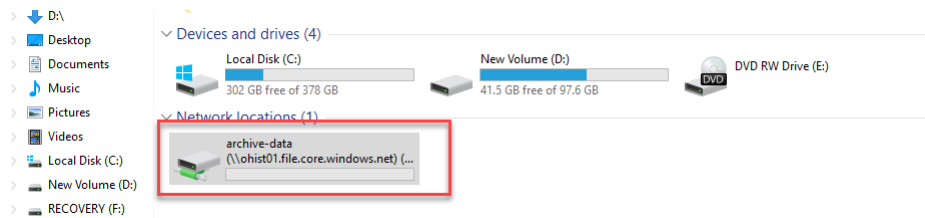
☐ Connect using different credentials

[Connect to a Web site that you can use to store your documents and pictures.](#)

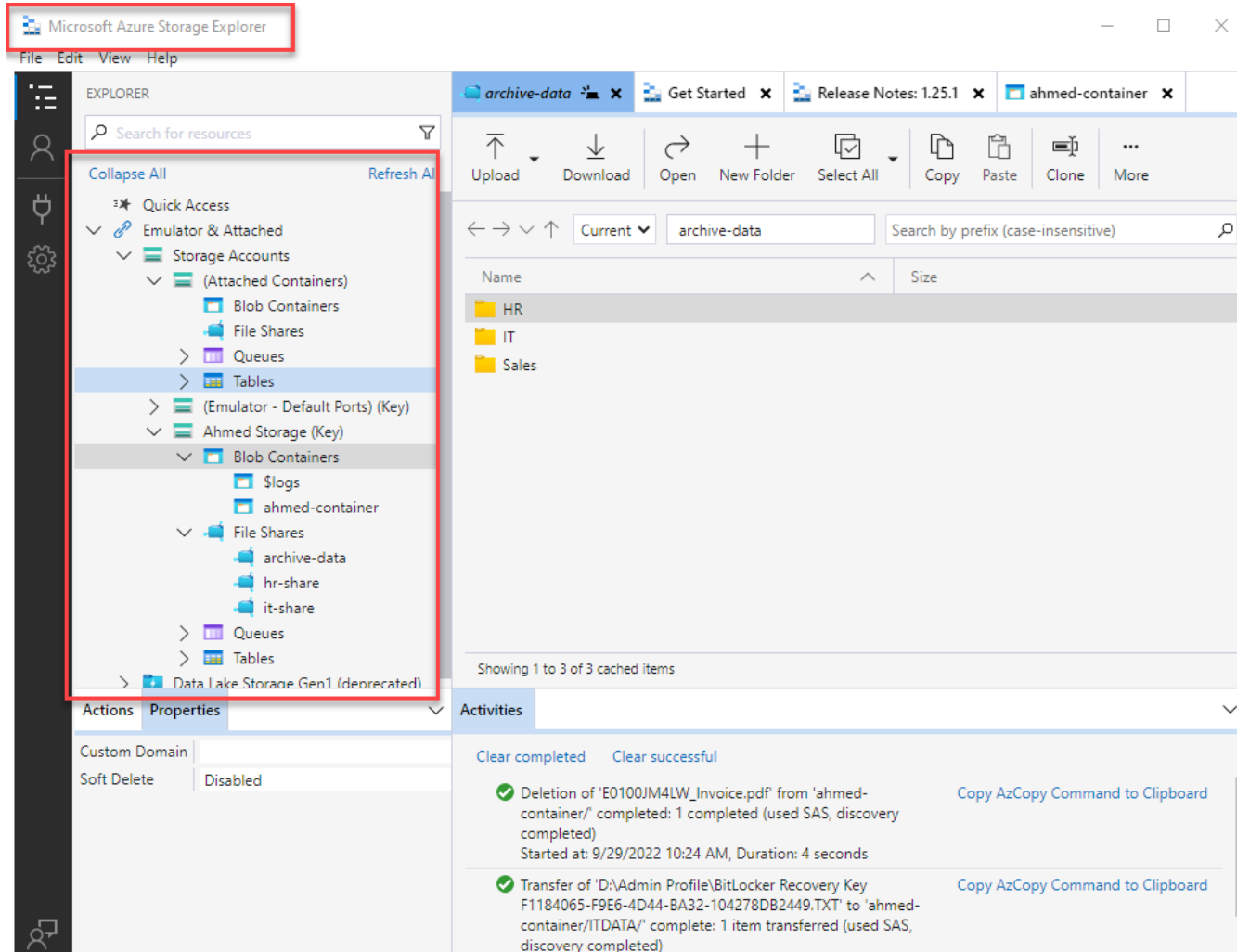
Finish

Cancel

Azure Administrator (AZ-104) | Lab Guide



[Access all storage account items using Microsoft storage explorer](#)



Data Protection

Backup Azure VMs

We have to create recovery service vault

[Home](#) > [Recovery Services vaults](#) >

Create Recovery Services vault ...

*** Basics** Tags Review + create

Project Details

Select the subscription and the resource group in which you want to create the vault.

Subscription * ⓘ	MSDN 5C
Resource group * ⓘ	LBLabs

[Create new](#)

Instance Details

Vault name * ⓘ	AhmedRSV
Region * ⓘ	Central US

Review + create

Next: Tags

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Go for properties to check vault configuration

AhmedRSV
Recovery Services vault

Search

Overview Activity log Access control (IAM) Tags Diagnose and solve problems

Settings

Identity Private endpoint connections

Properties

Locks

Getting started

Backup Site Recovery

Protected items

Backup items Replicated items

+ Backup + Enable Site Recovery Delete Refresh Feedback

For backups, try our new Backup Center. It offers Azure Backup customers a unified view of Recovery Services Vaults used for backup in Azure. It also has new governance capabilities. Click here to get the new experience.

Essentials

Overview Backup Site Recovery

What's new

Azure Backup Metrics are now in public preview →

Multi-user authorization for Azure Backup is now generally available →

Multiple backups a day for Azure Files is now in public preview →

Enhanced experience for archive is now generally available →

Backup for Azure Managed Disk is now Generally Available →

A new and improved way to enable Azure Site Recovery on your VMware machines is in public preview →

Backup

Site Recovery

Home > Microsoft.RecoveryServicesV2-1664425538893 | Overview > AhmedRSV

AhmedRSV | Properties

Recovery Services vault

Search

Overview Activity log Access control (IAM) Tags Diagnose and solve problems

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Backup Site Recovery

Protected items

Backup items Replicated items

Multi-User Authorization for Backup is now generally available. Click here to configure. →

Resource group

LB Labs

Diagnostics Settings

Update

Backup Configuration

Update

Encryption Settings

Update

Multi-User Authorization

Update

Security Settings

Update

Backup Configuration

AhmedRSV

Save Discard

Storage replication type

Locally-redundant Zone-redundant **Geo-redundant**

This option cannot be changed after protecting items. Geo-Redundant Storage (GRS) provides a higher level of data availability than Zonal-Redundant Storage and Local-Redundant Storage. Zonal-Redundant Storage helps to replicate the data in the availability zones of the same region. Review the trade-offs between lower cost and higher cost availability [here](#).

Cross Region Restore

Enable **Disable**

Note:

- This allows you to **restore in the secondary region** for both BCDR drills and outage scenarios.
- This is **available for Azure Virtual Machines** and SQL/SAP HANA databases running inside Azure VMs in this vault. No support for classic VMs.
- Cross Region Restore is currently **non-reversible** storage property.

Learn more about [Cross Region Restore](#) and [pricing impact](#).

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Home > Microsoft.RecoveryServicesV2-1664425538893 | Overview > AhmedRSV

AhmedRSV | Properties

Recovery Services vault

Search

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- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- Identity
- Private endpoint connections
- Properties
- Locks
- Getting started
 - Backup
 - Site Recovery
- Protected items

Multi-User Authorization for Backup is now generally available. Click here to configure. →

Resource group

LB Labs

Diagnostics Settings

Update

BACKUP

Backup Configuration

Update

Encryption Settings

Update

Multi-User Authorization

Update

Security Settings

Update

Security Settings

AhmedRSV

Save Discard

If you have enabled [Azure multi-factor authentication](#), you will be required to additionally authenticate using another device (for example, a mobile phone) while signing in to the Azure portal.

Soft Delete (For workloads running in Azure)

Enable this setting to protect backup data for Azure VM, SQL Server in Azure VM and SAP HANA in Azure VM from accidental deletes. [Learn More](#)

Enabled

Disable

Security Features (For workloads running on-premises)

Enable this setting to protect hybrid backups against accidental deletes and add additional layer of authentication for critical operations. Refer [this link](#) for minimum agent version requirement to enable this setting. [Learn more](#).

Enabled

Disable

Start VM backup

AhmedRSV | Backup

Recovery Services vault

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Settings

- Identity
- Private endpoint connections
- Properties
- Locks
- Getting started
 - Backup
 - Site Recovery

The storage replication is set to Geo-Redundant. This option cannot be changed later. Before proceeding further, click here. →

Where is your workload running?

Azure

What do you want to backup?

Virtual machine

Step: Configure Backup

Backup

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Home > Microsoft.RecoveryServicesV2-1664425538893 | Overview > AhmedRSV | Backup >

Configure Backup

AhmedRSV

Policy sub type *

☐ Standard

- Once a day backup
- 1-5 days operational tier
- Vault tier
- ZRS resilient snapshot tier
- Support for Trusted Azure VM

☒ Enhanced

- Multiple backups per day (Preview)
- 1-30 days operational tier
- Vault tier
- ZRS resilient snapshot tier
- Support for Trusted Azure VM

Backup policy * [Create a new policy](#)

Protection operations are based on the OS disk type selected for Azure VM. Enhanced Policy takes the first snapshot as configuration. Pricing of Standard policy and enhanced policy varies. [Learn more](#).

Policy Details

Full Backup

Backup Frequency
Every 4 hour(s) starting 8:00 AM UTC for 12 Hour(s)

Instant Restore
Retain instant recovery snapshot(s) for 2 day(s)

Retention of daily backup point
Retain backup taken every day for 30 Day(s)

Virtual Machines

Name

No Virtual machines selected.

[Add](#)

[Enable backup](#) [Download a template for automation](#)

Create policy

Azure Virtual Machine

Policy name

Backup schedule

Frequency * Start time * Schedule * Duration * Timezone *

Instant Restore

Retain instant recovery snapshot(s) for Day(s)

Retention range

Retention of daily backup point
☒ Retention of daily backup point
For Day(s)

☐ Retention of weekly backup point
Not Configured

☐ Retention of monthly backup point
Not Configured

☐ Retention of yearly backup point

[OK](#)

Home > Microsoft.RecoveryServicesV2-1664425538893 | Overview > AhmedRSV | Backup >

Configure Backup

AhmedRSV

Backup policy * [Edit this policy](#)

Protection operations are based on the OS disk type selected for Azure VM. Enhanced Policy takes the first snapshot as configuration. Pricing of Standard policy and enhanced policy varies. [Learn more](#).

Policy Details

Full Backup

Backup Frequency
Every 4 hour(s) starting 8:00 AM UTC for 12 Hour(s)

Instant Restore
Retain instant recovery snapshot(s) for 7 day(s)

Retention of daily backup point
Retain backup taken every day for 30 Day(s)

Virtual Machines

Name

No Virtual machines selected.

[Add](#)

OS Disk only backup option allows you to backup Azure Virtual Machine with only OS disk and exclude all the data disks. You can learn more about this feature, its limitation and pricing - [Learn more](#).

[Enable backup](#) [Download a template for automation](#)

Select virtual machines

Discovering virtual machines that can be backed up, are in the same region as vault and not protected by another vault.

Filter items by name

Virtual machine name	Resource Group
☒ web-1	LBLabs
☐ web-2	LBLabs
☐ web-3	LBLabs

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[OK](#) [Cancel](#)

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AhmedRSV | Backup items

Recovery Services vault

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<<

Refresh

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Replicated items

For backups, try our new Backup Center. It offers Azure Backup customers a unified view of Recovery Services Vaults used fo
Click here to get the new experience.

Primary RegionSecondary Region

BACKUP MANAGEMENT TYPE

BACKUP ITEM COUNT

Azure Virtual Machine1

Azure Backup Agent0

Azure Backup Server0

DPM0

Azure Storage (Azure Files)0

SQL in Azure VM0

SAP HANA in Azure VM0

Home > ConfigureProtection-1664426939480 | Overview > AhmedRSV | Backup items >

Backup Items (Azure Virtual Machine)

AhmedRSV

RefreshAddFilter

With Backup center, you can view all your laasVM items across vaults, subscriptions and regions in a single pane of glass. Click here to use the new experience. →

All data fetched from the service.

Filter items ...

Name ↑↓

Resource Group ↑↓

Backup Pre-Check

Last Backup Status

Latest restore point ↑↓

web-1

LB Labs

Passed

Warning (Initial backup pending)

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web-1

Backup Item

Backup now

Restore VM

File Recovery

Stop backup

Resume backup

Delete backup data

Restore to Secondary Region

Undelete

For backups, try our new Backup Center. It offers Azure Backup customers a unified view of Recovery Services Vaults used for backup in Azure. It also provides improved sorting and filtering along with new governance capabilities.

Essentials

Recovery services vault : [AhmedRSV](#)

Subscription (move) : [MSDN 5C](#)

Subscription ID : a7632f95-7f0d-4342-adda-dbc286dbe45c

Alerts (in last 24 hours) : [View alerts](#)

Jobs (in last 24 hours) : [View jobs](#)

Backup Pre-Check : Passed

Last backup status : Warning (Initial backup pending)

Backup policy : [AhmedBKPolicy](#)

Oldest restore point : -

Recovery points

This list is filtered for last 30 days of recovery points. To recover from recovery point older than 30 days, as well as vault-archive, [click here](#).

Long term recovery points can be moved to vault-archive. To move all 'recommended recovery points' to vault-archive tier, [click here](#).

CRASH CONSISTENT

APPLICATION CONSISTENT

FILE-SYSTEM CONSISTENT

0

0

0

Time ↑↓

Consistency

Recovery type

No restore points available.

Home > [ConfigureProtection-1664426939480](#) | Overview > [AhmedRSV](#)

AhmedRSV | Backup Jobs

Recovery Services vault

Search

Choose columns

Filter

Export jobs

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Metrics

Diagnostic settings

Logs

Advisor recommendations

Backup Jobs

Site Recovery jobs

Filtered by: Item Type - All, Operation - All, Status - All, Start Time - 9/28/2022, 10:52:43 AM, End Time - 9/29/2022, 10:52:43 AM

For backups, try our new Backup Center. It offers Azure Backup customers a unified view of Recovery Services Vaults used for backup in Azure. It also provides improved sorting and filtering along with new governance capabilities. [Click here to get the new experience.](#)

All data fetched from the service.

Filter items ...

Workload name ↑↓	Operation	Status	Type	Start time ↑↓	Total Duration ↑↓	Details
web-1	Backup	In progress	Azure Virtual Machine	9/29/2022, 10:52:30 AM	00:00:11	View details
web-1	Backup	In progress	Azure Virtual Machine	9/29/2022, 10:31:53 AM	00:20:48	View details
web-1	Configure backup	Completed	Azure Virtual Machine	9/29/2022, 8:49:11 AM	00:00:30	View details

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Now you can restore and stop the backup

Home > AhmedRSV | Backup items > Backup Items (Azure Virtual Machine) >

web-1

Backup Item

Backup now

Restore VM

File Recovery

Stop backup

Resume backup

Delete backup data

Restore to Secondary Region

Undelete

Backup pre-checks for this item point to issues that may lead to backup failures. Click here to resolve them →

Essentials

Recovery services vault : AhmedRSV

Subscription (move) : MSDN_SC

Subscription ID : a7632f95-7f0d-4342-adda-dbc286dbe45c

Alerts (in last 24 hours) : View alerts

Jobs (in last 24 hours) : View jobs

Backup Pre-Check : Warning

Last backup status : Success 9/29/2022, 10:52:30 AM

Backup policy : AhmedBKPolicy

Oldest restore point : 9/29/2022, 10:31:57 AM (37 minute(s) ago)

Recovery points

This list is filtered for last 30 days of recovery points. To recover from recovery point older than 30 days, as well as vault-archive, click here.

Long term recovery points can be moved to vault-archive. To move all 'recommended recovery points' to vault-archive tier, click here.

CRASH CONSISTENT

0

APPLICATION CONSISTENT

2

FILE-SYSTEM CONSISTENT

0

Time ↑↓	Consistency	Recovery type
9/29/2022, 10:52:33 AM	Application Consistent	Snapshot
9/29/2022, 10:31:57 AM	Application Consistent	Snapshot

Home > AhmedRSV | Backup items > Backup Items (Azure Virtual Machine) > web-1 >

Restore Virtual Machine

web-1

Restore allows you to restore VM/disks from a selected Restore Point.

Restore point *

9/29/2022, 10:31:57 AM

Select

Data Store

Snapshot

Restore Configuration

Create new

Replace existing

To create an alternate configuration when restoring your VM (from the following menus), use PowerShell cmdlets.

Restore Type *

Virtual machine name *

Resource group *

Virtual network *

Subnet *

Staging Location *

Create new virtual machine

Create new virtual machine

Restore disks

LBLABS

Select an option

Select an option

Select an option

Can't find your storage account?

Restore

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Backup files and folders

Home > Recovery Services vaults > AhmedRSV

AhmedRSV | Backup
Recovery Services vault

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Where is your workload running?
On-Premises

What do you want to backup?
Files and folders
☒ Files and folders
☐ Hyper-V Virtual Machine
☐ VMWare Virtual Machine
☐ Microsoft SQL Server
☐ Microsoft SharePoint
☐ Microsoft Exchange
☐ System State
☐ Bare Metal Recovery

AhmedRSV | Backup
Recovery Services vault

Search

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems

Settings

Where is your workload running?
On-Premises

What do you want to backup?
Files and folders

Step: Prepare Infrastructure

Prepare Infrastructure

Download both of backup agent and vault credentials to the server you want to backup (I'll use web-2 VM)

Home > Recovery Services vaults > AhmedRSV | Backup >

Prepare infrastructure ...

Recovery Services Agent

Please follow the steps mentioned below.

1. Install Recovery Services agent
[Download Agent for Windows Server or Windows Client](#)
2. Download vault credentials to register the server to the vault. Vault credentials will expire after 10 days.

☒ Already downloaded or using the latest Recovery Services Agent

Download

3. Schedule backup using Recovery Services Agent UI. [Learn More](#)
4. Once the backups are scheduled, you can use backup jobs page to monitor the backups. [Browse jobs page](#)
5. You can also Configure Notifications from alerts page to receive email alerts for backup failures. [Browse alerts page](#)

 Microsoft Azure Recovery Services Agent Setup Wizard

✕



Installation Settings

Installation Stages

- Installation Settings
- Proxy Configuration
- Microsoft Update Opt-In
- Installation

Installation Folder

Microsoft Azure Recovery Services Agent will be installed in the following folder. To choose a different installation folder, click Browse. The location specified must have at least 1 GB of free space.

C:\Program Files\Microsoft Azure Recovery Services Agent

Browse

Cache Location

Microsoft Azure Recovery Services Agent can use this to keep track of files being backed up from your computer. The location specified must have free space which is atleast 5% of the backup data.

C:\Program Files\Microsoft Azure Recovery Services Agent\Scratch

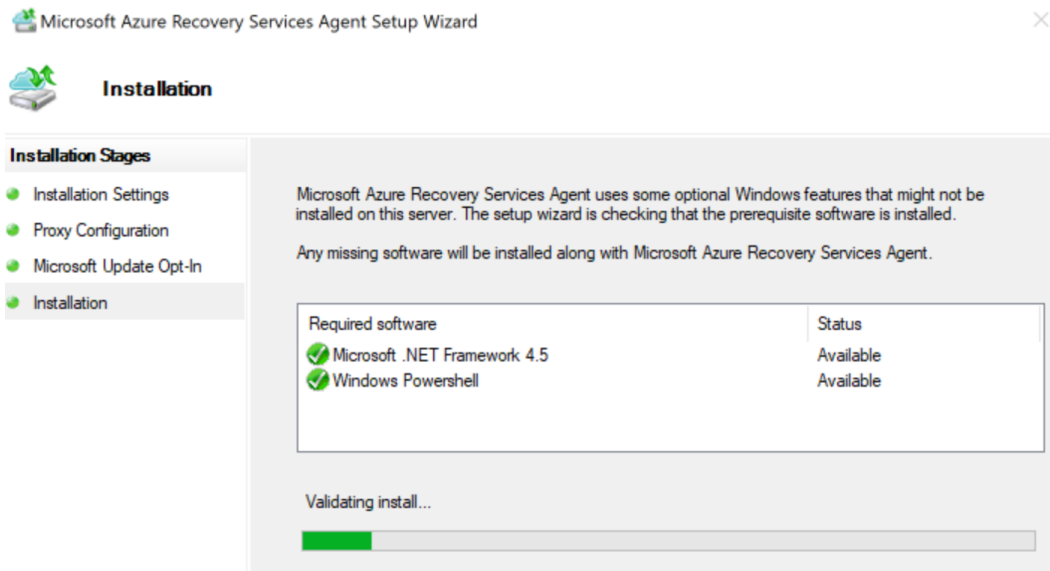
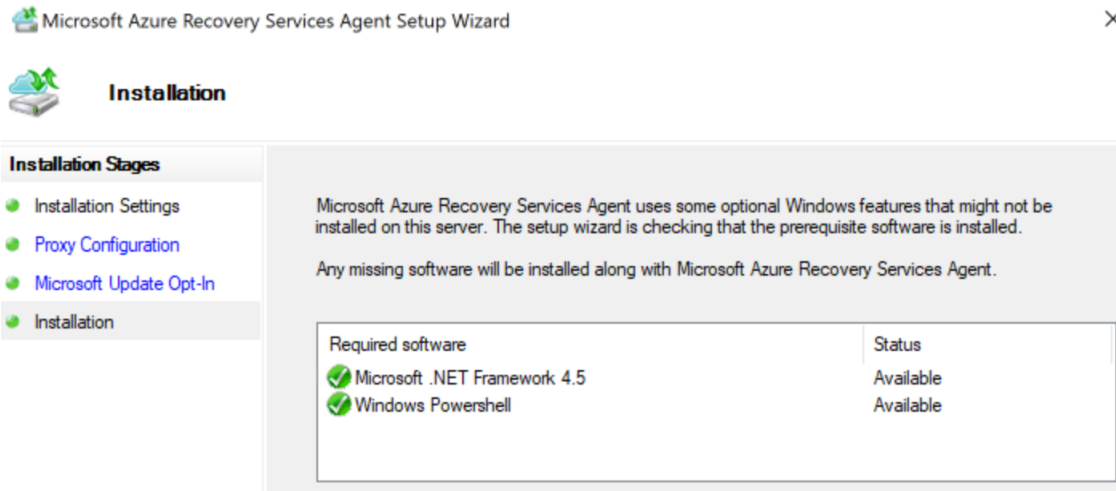
Browse

< Back

Next >

Cancel

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Microsoft Azure Recovery Services Agent Setup Wizard



Installation

Installation Stages

- Installation Settings
- Proxy Configuration
- Microsoft Update Opt-In
- Installation

Microsoft Azure Recovery Services Agent uses some optional Windows features that might not be installed on this server. The setup wizard is checking that the prerequisite software is installed.

Any missing software will be installed along with Microsoft Azure Recovery Services Agent.

Required software	Status
✓ Microsoft .NET Framework 4.5	Available
✓ Windows Powershell	Available

✓ Microsoft Azure Recovery Services Agent installation has completed successfully.

Click proceed to registration to register the server to a backup vault.

Proceed to Registration

Close

Register Server Wizard



Vault Identification

Vault Identification

Encryption Setting

Server Registration

Select the vault credentials downloaded from the quick start page in the Microsoft Azure Backup Vault.

Vault Credentials:

Backup Vault:

Region:

Subscription Identifier:

Vault Credential expiry:

Register Server Wizard ✕

Encryption Setting

Vault Identification

Encryption Setting

Server Registration

Backups are encrypted to protect the confidentiality of your data.

Generate or type a passphrase to encrypt and decrypt backups from this server.

Enter Passphrase (minimum of 16 characters)

***** (36)

Generate Passphrase

Confirm Passphrase

***** (36)

Enter a location to save the passphrase

C:\Users\ahmed\Desktop

Browse

Saving the passphrase locally does not protect it from corruptions and disasters. Microsoft cannot recover data, if the passphrase is lost or forgotten. It is strongly advised to store a copy of the passphrase in a secure 'external' location, like the Azure Key Vault. [Learn to store the passphrase in an Azure Key Vault](#)

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Next >

Finish

Cancel

Now server registered and we can back up our files

Register Server Wizard

Server Registration

Vault Identification

Encryption Setting

Server Registration

Microsoft Azure Backup is now available for this server.

The passphrase was saved to the following file :
[C:\Users\ahmed\Desktop\9_29_2022_08_03_30_AhmedRSV_web-2.txt](#)

WARNING! Storing the passphrase locally can lead to passphrase loss due to deletions and ransomware attacks. Microsoft cannot help to recover your data if the passphrase is lost. Save a copy of the passphrase to another computer or Azure Key Vault immediately.
[Learn how you can store the passphrase in Azure Key Vault.](#)

Before your server is backed up you must configure and schedule backup options.

☒ Launch Microsoft Azure Recovery Services Agent

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Now going to back up user documents

The screenshot shows the Microsoft Azure Backup portal. The main heading is "Microsoft Azure Backup" with a sub-header stating it supports scheduled backups of files and folders, and Windows System State (New) to an online location. Below this, there's a section for "Jobs (Activity in the past 7 days, double click on the message to see details)" with tabs for "Jobs" and "Alerts". A table with columns "Status", "Time", "Message", and "Description" is present but empty. To the right, an "Actions" menu is open, with "Schedule Backup" highlighted. Other options include "Register Server", "Recover Data", "Change Properties", "Open Portal", "About Microsoft Azure Reco", "Privacy & Cookies", "View", and "Help".

Microsoft Azure Backup

Microsoft Azure Backup supports scheduled backups of files and folders, and Windows System State (New) to an online location

Now you can also backup Windows System State and protect your File Servers, Domain Controllers and IIS Web-Servers from corruptions, attacks and disasters. [Learn more](#)

You can also Configure Notifications from Alerts blade to receive email alerts for backup failures. [Learn More](#)

Jobs (Activity in the past 7 days, double click on the message to see details)

Jobs Alerts

Status	Time	Message	Description
--------	------	---------	-------------

Status

Last Backup	Next Backup	Available Recovery Points	Last Recovery
Status: - Time: - View details	Status: Not Scheduled Time: -	Total backups: None Latest copy: - Oldest copy: - View details	Status: - Time: - View details

Actions

- Backup
 - Register Server
 - Schedule Backup**
 - Recover Data
 - Change Properties
 - Open Portal
 - About Microsoft Azure Reco
- Privacy & Cookies
- View
- Help

Schedule Backup Wizard



Select Items to Backup

The screenshot shows the "Select Items to Backup" wizard. The left sidebar has a "Getting started" section with options: "Select Items to Backup", "Specify Backup Schedule...", "Select Retention Policy (F...", "Choose Initial Backup Ty...", "Confirmation", and "Modify Backup Progress". The main area has a heading "Click Add Items to select the files and folders you want to backup." Below this is a text box with "Name" and "C:\Users\ahmed\Documents\". A file explorer window is open, showing the "Documents" folder with two items: "9_29_2022_08_03_30_AhmedRSV_web-2" and "AhmedRSV_Thu Sep 29 2022.VaultCredentials". The "Add Items" button is highlighted in the bottom right.

Getting started

- Select Items to Backup
- Specify Backup Schedule ...
- Select Retention Policy (F...
- Choose Initial Backup Ty...
- Confirmation
- Modify Backup Progress

Click Add Items to select the files and folders you want to backup.

Name
C:\Users\ahmed\Documents\

Documents

Name	Date modified	Type
9_29_2022_08_03_30_AhmedRSV_web-2	9/29/2022 8:03 AM	Text
AhmedRSV_Thu Sep 29 2022.VaultCredentials	9/29/2022 7:59 AM	VAUI

Add Items **Remove Items** **Exclusion Settings**

Schedule Backup Wizard



Specify Backup Schedule (Files and Folders)

Getting started

Select Items to Backup

Specify Backup Schedule ...

Select Retention Policy (F...

Choose Initial Backup Ty...

Confirmation

Modify Backup Progress

Define a schedule when you want to create a backup copy for selected files and folders

Schedule a backup every

☒ Day

☐ Week

At following times (Maximum allowed is three times a day)

11:30 PM

4:00 PM

8:30 PM

Schedule Backup Wizard



Select Retention Policy (Files and Folders)

Getting started

Select Items to Backup

Specify Backup Schedule ...

Select Retention Policy (F...

Choose Initial Backup Ty...

Confirmation

Modify Backup Progress

Specify the retention policy for the backup copy of files and folders

☒ Daily Retention Policy

Retain backup copies taken

At 11:30 PM

4:00 PM

for 180 Days

8:30 PM

☒ Weekly Retention Policy

Retain backup copies taken on

Saturday

Modify

At ☒ 11:30 PM

☐ 4:00 PM

for 104 Weeks

☒ Monthly Retention Policy

Retain backup copies taken on

☒ Saturday of Last Week

Modify

At ☒ 11:30 PM

☐ 4:00 PM

for 60 Months

☐ On day(s) 1

Modify

☒ Yearly Retention Policy

Retain backup copies taken on

☒ Saturday of Last Week of March

Modify

At ☒ 11:30 PM

☐ 4:00 PM

for 10 Years

☐ March 1

Modify

Schedule Backup Wizard



Choose Initial Backup Type (Files and Folders)

Getting started

Select Items to Backup

Specify Backup Schedule ...

Select Retention Policy (F...

Choose Initial Backup Ty...

Confirmation

Modify Backup Progress

Initial Backup transfers a full backup of your data to Azure. Subsequent backups transfer only changes in your data.

Specify the option for transferring the Initial Backup. [Help me choose.](#)

Online

☒ Transfer over the network

Uses your network to move the initial backup data to Azure

Offline

 Seed initial backup using storage devices without using network. Recommended for limited network bandwidth environments. Subsequent backups are done over the network.

☐ Transfer using Microsoft Azure Data Box disks

Order an Azure Data Box (up to 100TB) or a single Data Box disk (up to 8TB) to transfer data from this server to Azure. Saves effort to procure and ship disks. Ensure you have already ordered your Data Box device. To order go to Azure > [Data Box](#). [Learn more.](#)

☐ Transfer using my own disks

To transfer initial backup using this option, you must use Azure-compatible disks, connectors and shipping partners. Refer to our Offline Backup documentation for more details. [Learn more.](#)

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Schedule Backup Wizard



Confirmation

Getting started

Select Items to Backup

Specify Backup Schedule ...

Select Retention Policy (F...

Choose Initial Backup Ty...

Confirmation

Modify Backup Progress

You are about to save the following backup schedule and initiate initial backup.

Policy Type

Files and Folders

Files and Folders

Backup Items Selected items (C:\)

File (s) Excluded: None

Backup Time: 11:30 PM, 4:00 PM, 8:30 PM

Backup Days: Everyday

Weekly frequency: Every 1 week(s)

Retention Days: 180

Initial Backup Creation Network

Schedule Backup Wizard



Modify Backup Progress

Getting started

Select Items to Backup

Specify Backup Schedule ...

Select Retention Policy (F...

Choose Initial Backup Ty...

Confirmation

Modify Backup Progress

Status:

✓ You have successfully created a backup schedule.

Action

✓ Create backup schedule for Files and Folders

Status

Success

You can also backup now

Back Up Now Wizard

Confirmation

Select Backup Item

Retain Backup Till

Confirmation

Backup progress

Back Up Now will backup this server using the following settings.

Backup Items:

C:\Users\ahmed\Documents\

Files excluded:None

Retain Backup Till:10/29/2022

Server Settings

Network throttling settings:Not Configured

Proxy server settings:Not Configured

[Change Properties](#)

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Next >

Back Up

Cancel

Backup

Register Server

Schedule Backup

Back Up Now

Recover Data

Change Properties

Open Portal

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Home > Recovery Services vaults > AhmedRSV

AhmedRSV | Backup items ...

Recovery Services vault

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For backups, try our new Backup Center. It offers Azure Backup customers a unified view of Recovery Services V along with new governance capabilities. Click here to get the new experience.

Primary Region Secondary Region

BACKUP MANAGEMENT TYPE	BACKUP ITEM COUNT
Azure Virtual Machine	1
Azure Backup Agent	1

Home > Recovery Services vaults > AhmedRSV | Backup items > Backup Items (Azure Backup Agent) >

C:\ on web-2. ...

Backup Item

Essentials

Recovery services vault : [AhmedRSV](#)

Computer name : [web-2.](#)

Item Type : File-Folders

Last backup status : ✔ Success

Last refreshed at : 9/29/2022, 1:24:05 PM

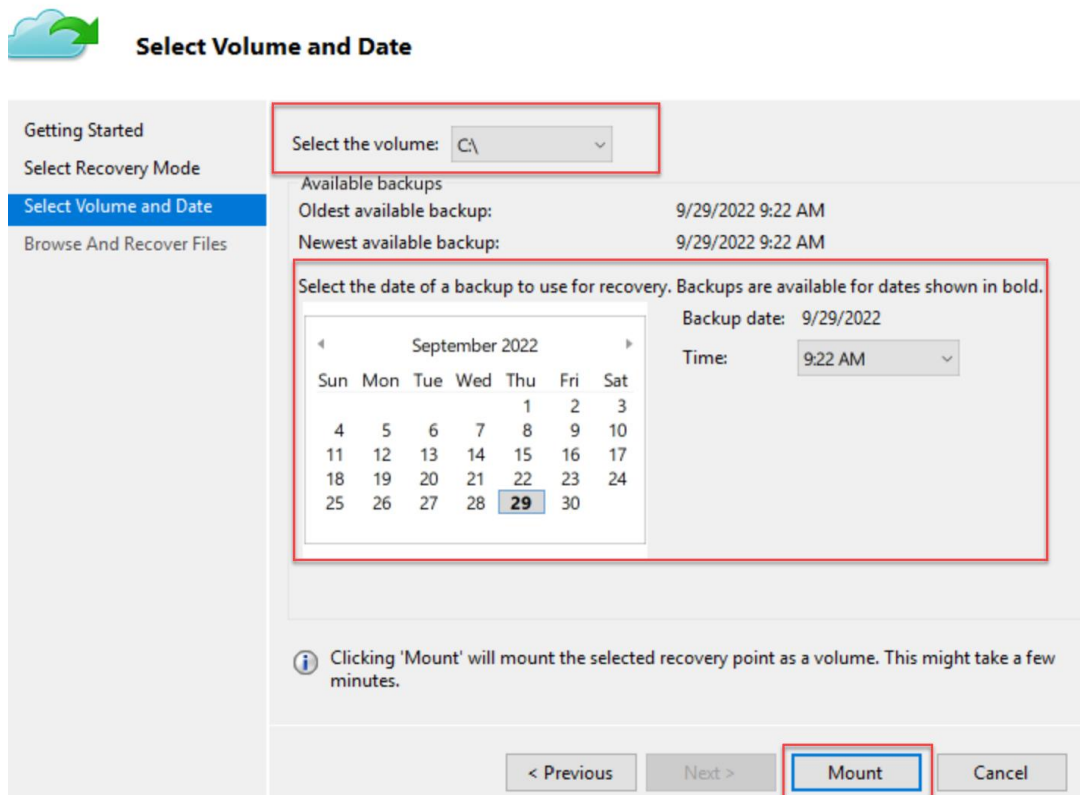
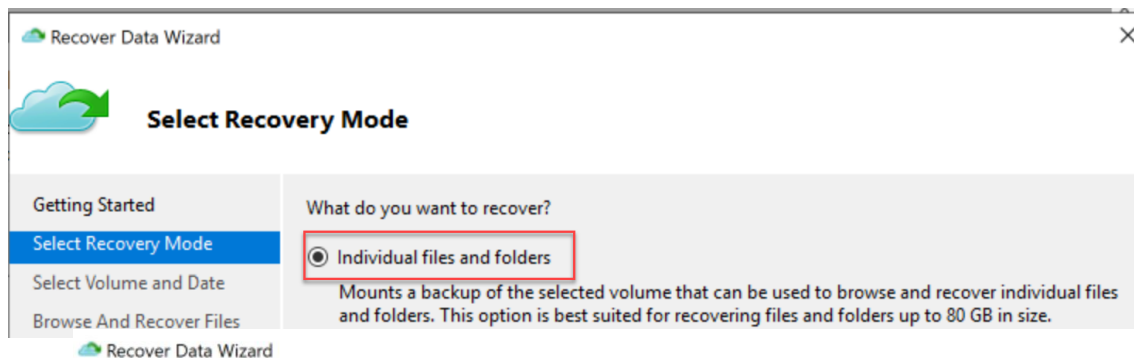
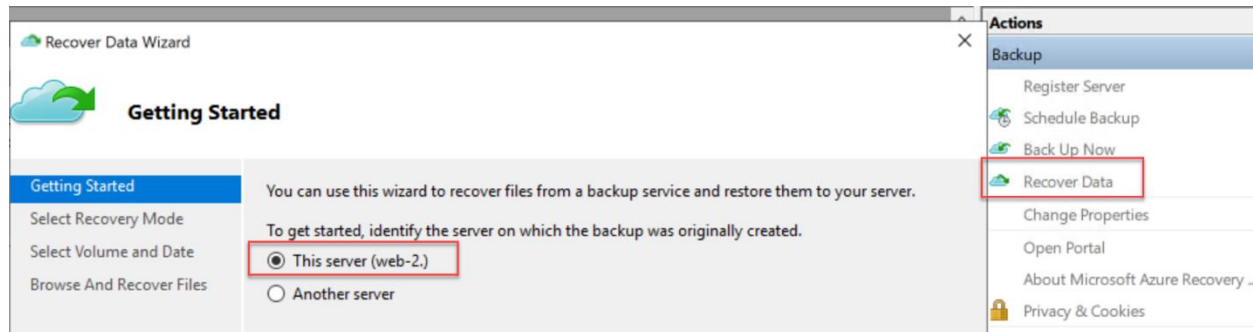
Monitoring

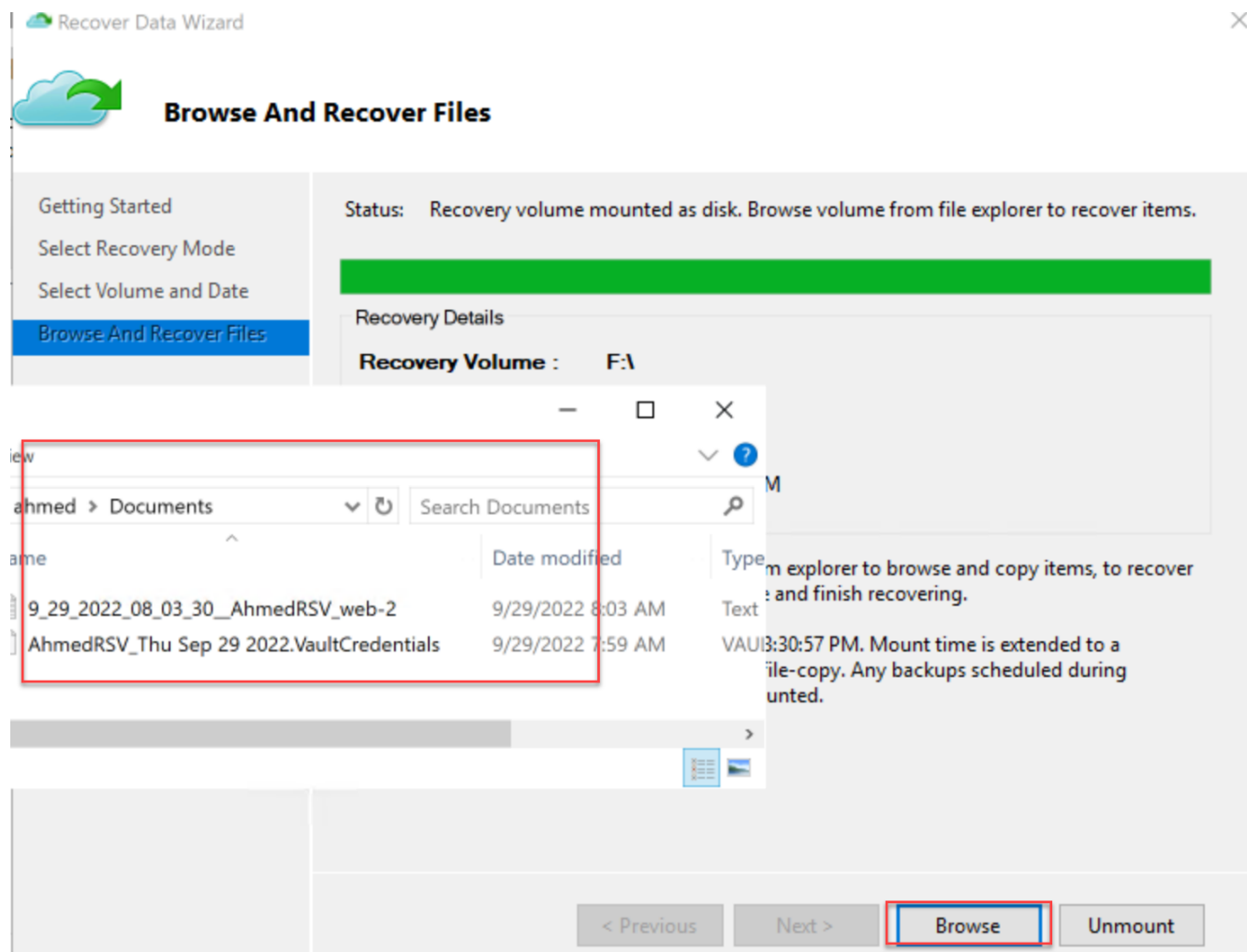
Recovery points

Latest	9/29/2022, 1:22:12 PM
Oldest	9/29/2022, 1:22:12 PM
Total	1

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To recover these data, you have to recover from azure backup agent

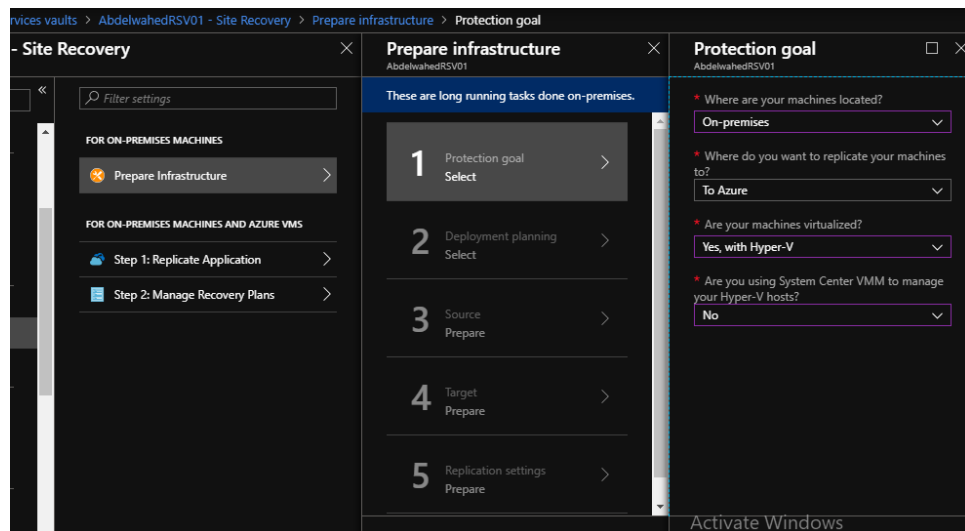
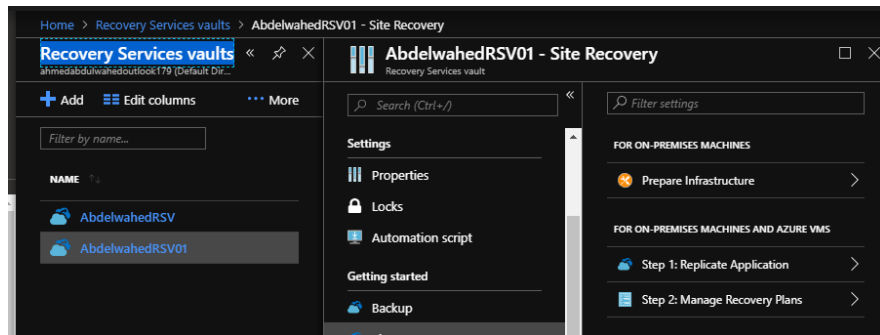


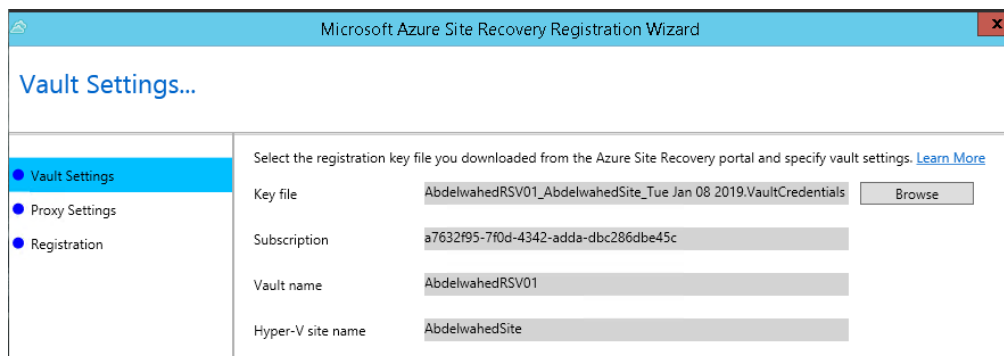
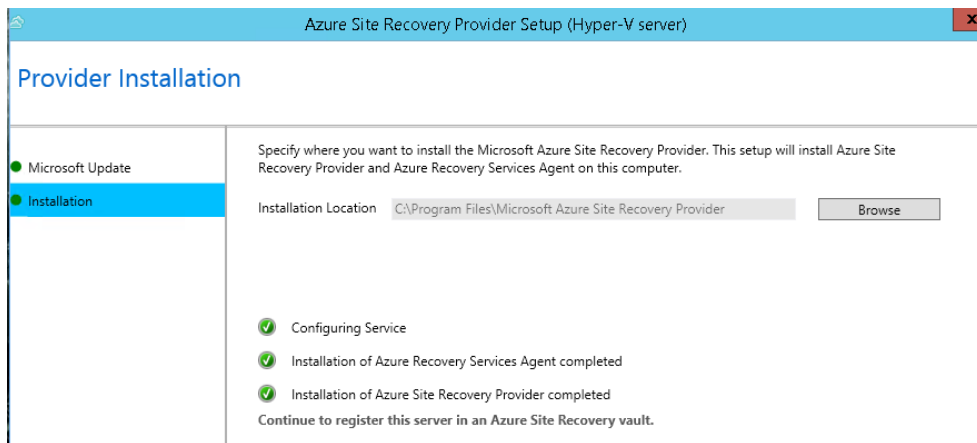
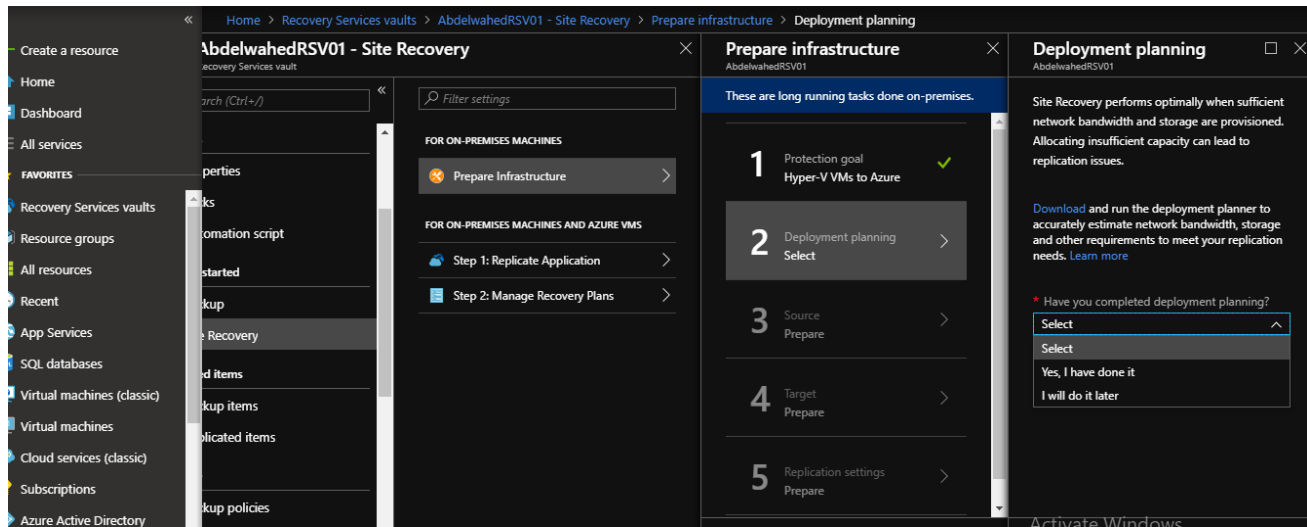


Azure Site Recovery

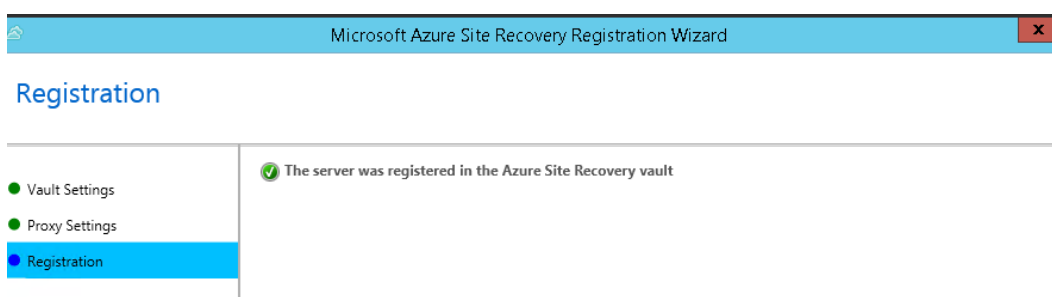
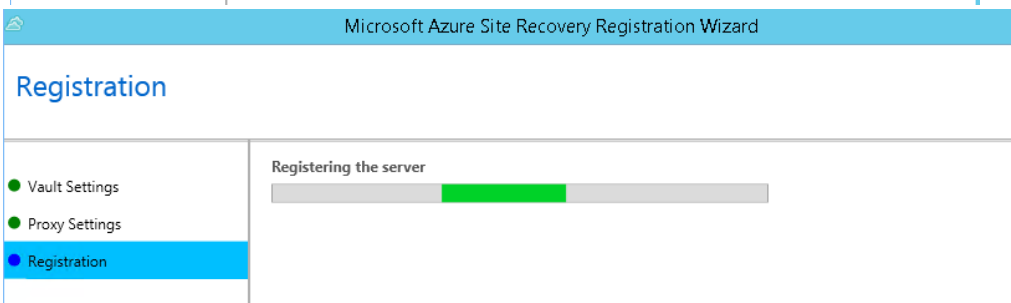
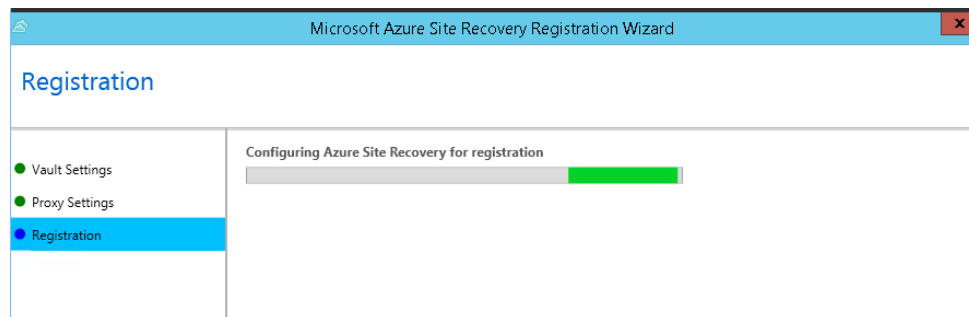
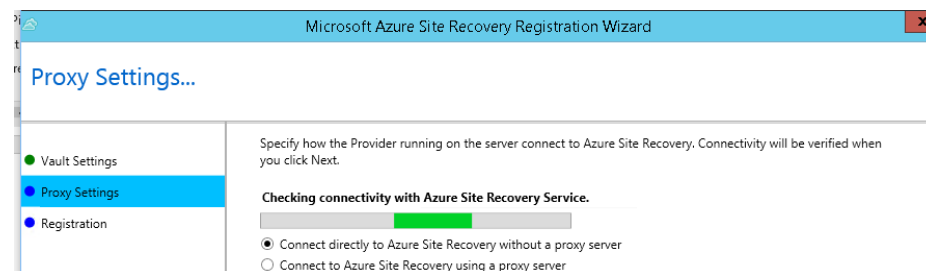
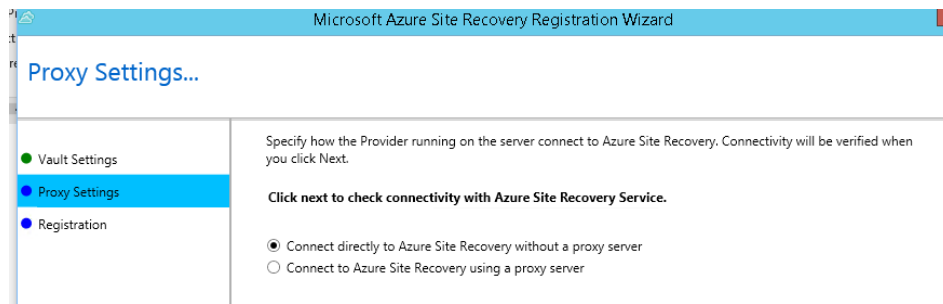
Configuring ASR for on-premises resources

1. Prepare Infrastructure
2. Replicate Application
3. Manage Recovery Plans





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Home > AbdelwahedRSV01 - Site Recovery > Prepare infrastructure > Prepare source > Add Server

Recovery

AbdelwahedRSV01

Filter settings

FOR ON-PREMISES MACHINES

Prepare Infrastructure

FOR ON-PREMISES MACHINES AND AZURE VMs

Step 1: Replicate Application

Step 2: Manage Recovery Plans

Prepare infrastructure

AbdelwahedRSV01

These are long running tasks done on-premises.

- 1 Protection goal
Hyper-V VMs to Azure
- 2 Deployment planning
I have done it
- 3 Source
Prepare
- 4 Target
Prepare
- 5 Replication settings
Prepare

OK

Prepare source

AbdelwahedRSV01

+ Hyper-V Site + Hyper-V Server

✓ Step 1: Select Hyper-V site

* Hyper-V Site

AbdelwahedSite

✓ Step 2: Ensure Hyper-V servers are added

BNZShare.Binzafrah.com

OK

Add Server

AbdelwahedRSV01

Server type

Hyper-V server

Adding Hyper-V server may take 15 minutes to 30 minutes

Register your Hyper-V host(s)
On-premises

1. Make sure the host is running Windows Server 2012 R2 or above. [Learn more.](#)
2. Configure Proxy setting and ensure each host can access the [Service URLs](#)
3. Download the installer for the Microsoft Azure Site Recovery Provider.
4. Download the vault registration key to register the host in a Hyper-V site
5. Install the Provider on the Hyper-V host and use the registration key to register

AbdelwahedSite

Download

Home > AbdelwahedRSV01 - Site Recovery > Prepare infrastructure > Target

AbdelwahedRSV01 - Site Recovery

Recovery Services vault

Search (Ctrl+/)

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Settings

Properties

Locks

Automation script

Getting started

Backup

Site Recovery

Protected items

Prepare infrastructure

AbdelwahedRSV01

These are long running tasks done on-premises.

- 1 Protection goal
Hyper-V VMs to Azure
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I have done it
- 3 Source
AbdelwahedSite
- 4 Target
Prepare
- 5 Replication settings
Prepare

Target

AbdelwahedRSV01

+ Storage account + Network

* Subscription

MSDN Platforms (a7632f95-7f0d-43...

* Select the deployment model used after failover

Resource Manager

✓ Step 2: Ensure that at least one compatible Azure storage account exist

Storage account(s)

Found abdelwahedstrg01diag (Standard) storage account.

✓ Step 3: Ensure that at least one compatible Azure virtual network exist

Network(s)

abdelwahedstrg01-vnet

Home > AbdelwahedRSV01 - Site Recovery > Prepare infrastructure > Replication policy > Create and associate policy

Prepare infrastructure

AbdelwahedRSV01

These are long running tasks done on-premises.

1

Protection goal

Hyper-V VMs to Azure

✓

2

Deployment planning

I have done it

✓

3

Source

AbdelwahedSite

✓

4

Target

Azure

✓

5

Replication settings

Prepare

>

Replication policy

AbdelwahedRSV01

+ Create and Associate Associate Dissociate

→ Step 1: Ensure 'AbdelwahedSite' is associated to at least one replication policy

No replication policies exist. Click '+Create and Associate' button on top to create a replication policy and associate 'AbdelwahedSite' to it.

Create and associate policy

AbdelwahedRSV01

* Name AbdelwahedPolicy ✓

Source type Hyper-V

Target type Azure

Copy frequency 5 Minutes

* Recovery point retention in hours 2

* App-consistent snapshot frequency in hours 1

Initial replication start time Immediately

Associated Hyper-V site AbdelwahedSite

AbdelwahedRSV01 - Site Recovery > Prepare infrastructure > Replication policy

Recovery

Filter settings

FOR ON-PREMISES MACHINES

Prepare Infrastructure >

FOR ON-PREMISES MACHINES AND AZURE VMs

Step 1: Replicate Application >

Step 2: Manage Recovery Plans >

Prepare infrastructure

AbdelwahedRSV01

These are long running tasks done on-premises.

1

Protection goal

Hyper-V VMs to Azure

✓

2

Deployment planning

I have done it

✓

3

Source

AbdelwahedSite

✓

4

Target

Azure

✓

5

Replication settings

Prepare

>

Replication policy

AbdelwahedRSV01

+ Create and Associate Associate Dissociate

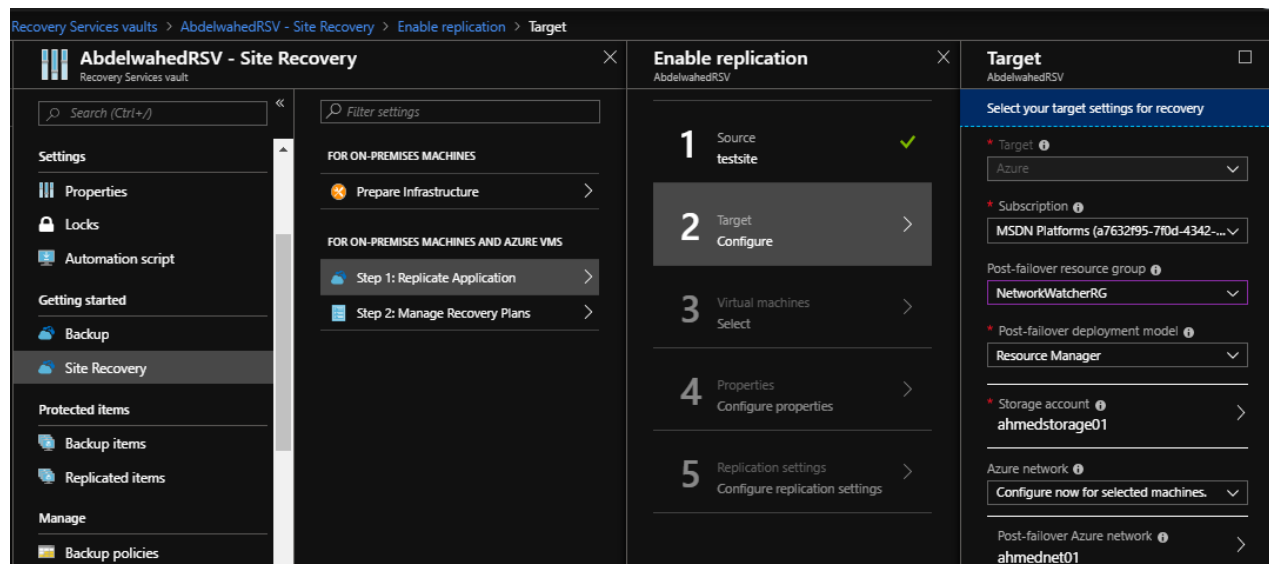
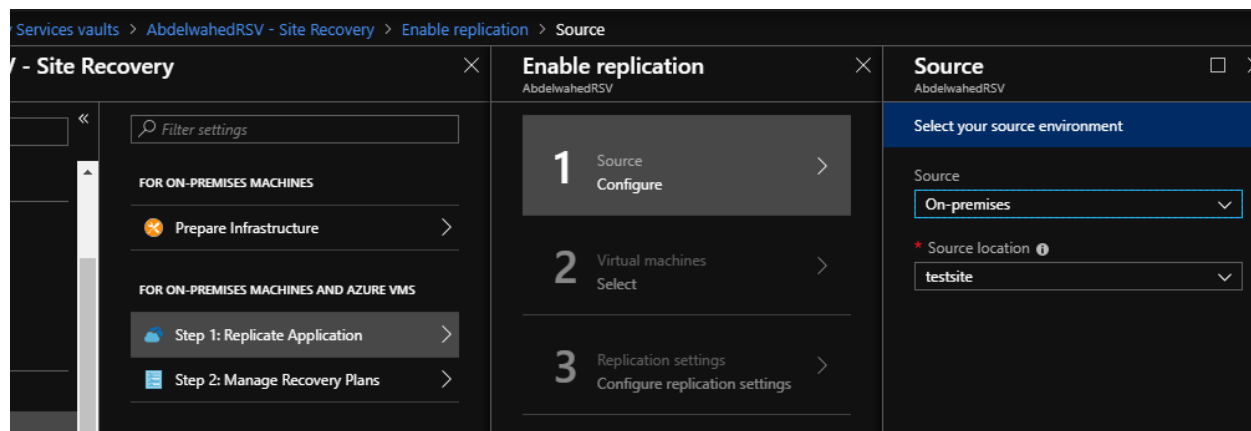
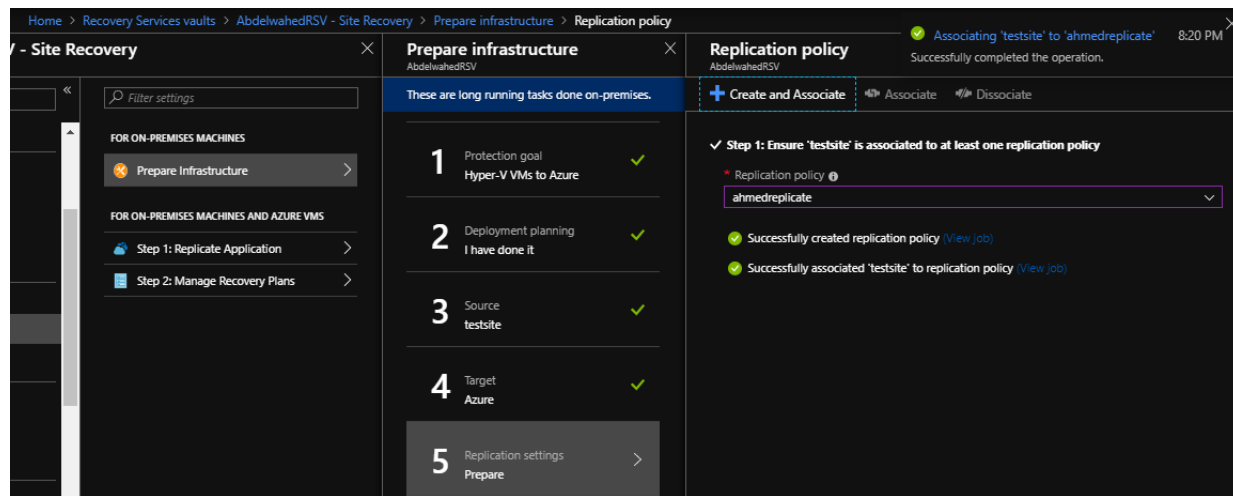
→ Step 1: Ensure 'AbdelwahedSite' is associated to at least one replication policy

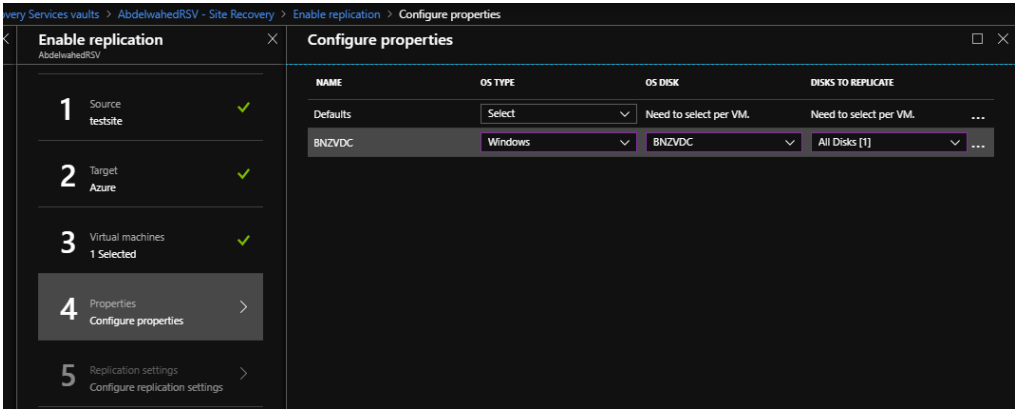
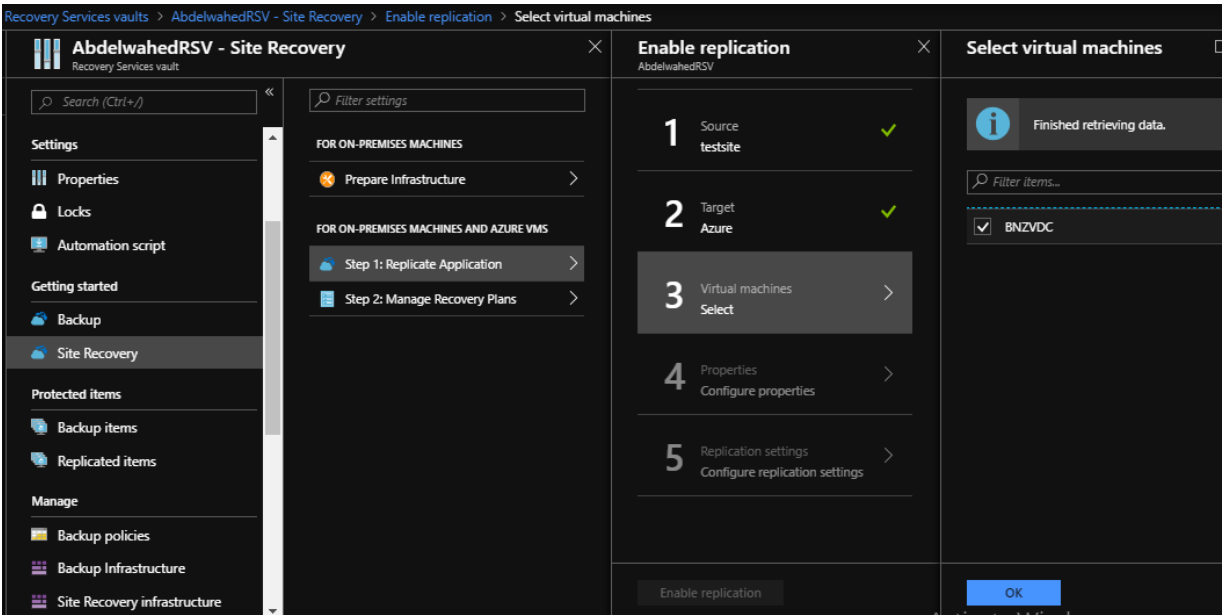
* Replication policy AbdelwahedPolicy

✓ Successfully created replication policy (View job)

... Associating 'AbdelwahedSite' to replication policy (View job in progress)

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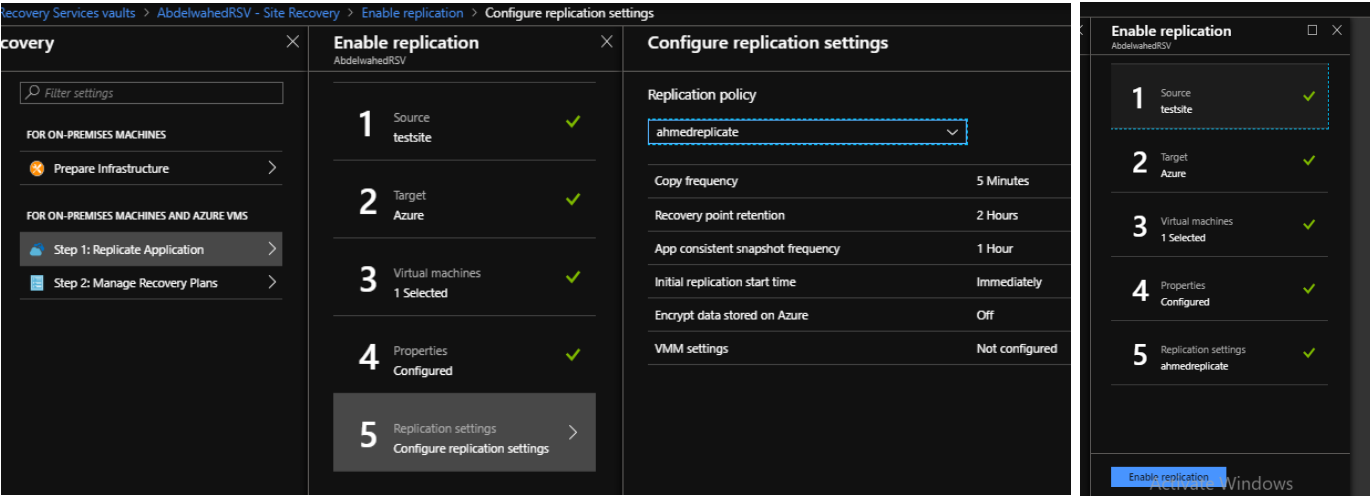




set

policy

replication



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Home > Recovery Services vaults > AbdelwahedRSV - Replicated items > BNZVDC

BNZVDC
Replicated items

Search (Ctrl+/)

Overview

General

Properties

Compute and Network

Disks

Essentials

Health and status

Replication Health: Healthy

Status: 1% synchronized

RPO: -

Failover readiness

Last successful Test Failover: -

Configuration issues: No issues

Latest recovery points
Click above to see the latest recovery points.

Errors(0) [Open in new page](#)

No errors

Events - Last 72 hours(0) [Open in new page](#)

TIME	EVENT NAME	SEVERITY
No events		

Home > Recovery Services vaults > AbdelwahedRSV - Replicated items

Recovery Services vaults
ahmedabdulwahedoutlook179 (Default Dir...

+ Add Edit columns More

Filter by name...

NAME

- AbdelwahedRSV
- AbdelwahedRSV01

AbdelwahedRSV - Replicated items
Recovery Services vault

Search (Ctrl+/)

Refresh Replicate Columns Filter

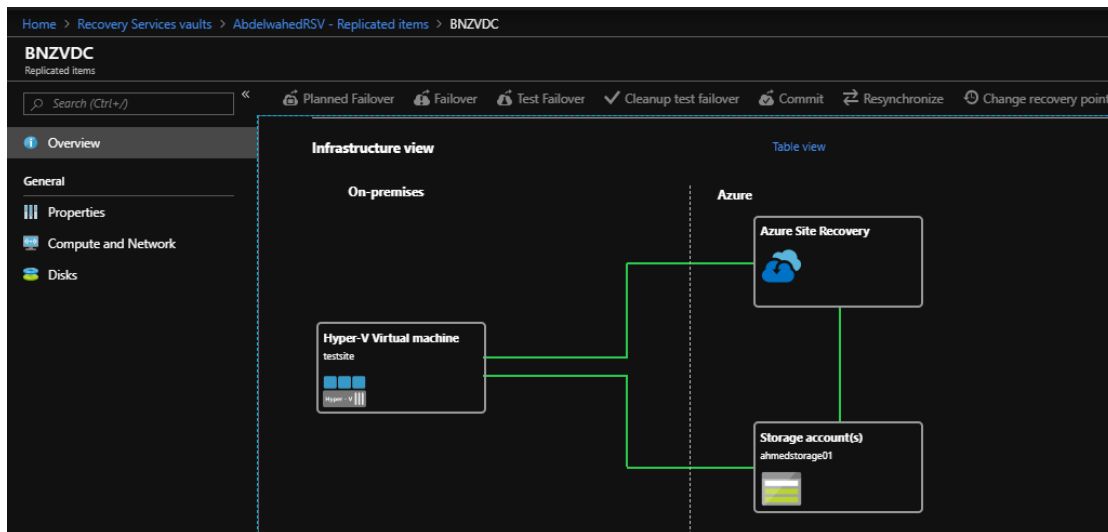
You can run your machines on managed disks after a failover or migration from on-premises to Azure. Set the option to use managed disks in Replicated item -> Settings -> Compute and Network.

Last refreshed at: 1/8/2019, 9:52:05 PM

Finished loading data from service.

Filter items...

NAME	REPLICATION HEALTH	STATUS	ACTIVE LOCATION
BNZVDC	Healthy	1% synchronized	testsite



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Failover and migrating the VM

